A First-Principles Dynamic Model of a Rubber V-Belt CVT for General Transient Simulation

With Mechanical Actuation and Traction-Limited Torque Transfer

Kai Arseneau

July 20, 2026

Abstract

This work develops a first-principles dynamic model for a mechanically actuated dry rubber V-belt continuously variable transmission (CVT) and its coupling to the vehicle drivetrain. The model is motivated by a persistent gap in the available literature: while quasi-static Baja-oriented models are common and detailed dynamic models exist for hydraulically actuated metal-belt CVTs, publicly available formulations for mechanically actuated rubber belt CVTs do not generally combine sheave inertia, flyweight actuation, torque-reactive secondary clamping, belt geometric closure, traction-limited torque transmission, and vehicle longitudinal dynamics within a single unified framework.

Starting from pulley geometry and belt-length closure, the derivation develops the CVT kinematics, the rotational dynamics of the primary and secondary pulleys, the axial equation of motion governing shift, and the torque-transfer closure required to distinguish sticking from traction-limited slip. The primary clutch model includes centrifugal flyweight actuation and spring preload, while the secondary model includes helix-generated torque feedback together with spring forces. A centrifugal belt correction is also derived to account for the reduction in effective normal force at speed. These elements are assembled into a closed nonlinear state-space model suitable for direct numerical time integration.

The resulting formulation is intended both as a simulation model and as a derivation document in which each major modeling step remains physically interpretable. Simulation results are used to examine whether the completed model behaves in mechanically sensible ways under transient launch conditions. Comparisons of launch shift trajectories, pulley axial-force balance, and torque admissibility across multiple tuning cases show a consistent mechanism chain from tuning changes, to clamping-force evolution, to stick–slip behavior, to overall launch outcome. The default tuning produces the most balanced launch response, combining strong early admissible coupling torque, controlled low-ratio retention, and a smoother main shift. More aggressive cases reveal renewed slip and visibly underdamped oscillatory transients during shift onset, highlighting both the usefulness of the model and the limitations introduced by idealized loss assumptions.

Overall, the formulation provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behavior during launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behavior remains traceable back to clear mechanical causes.

Contents

Nomenclature and Notation

All quantities are expressed in SI units unless otherwise stated. Angles are expressed in radians. Angular velocities are expressed in rad/s. Torques are expressed in N m, forces in N, lengths in m, and masses in kg.

Symbol Summary

Symbols used throughout the derivation are summarized in Table 1.

Symbol	Meaning
State and kinematics	
$\boldsymbol{x}(t)$	state vector
t	time
$\omega_p,\ \omega_s$	primary and secondary angular speed
$s,\ \dot{s},\ \ddot{s}$	shift position, velocity, and acceleration
$R,\ \dot{R}$	CVT ratio and ratio rate
v_b	belt transport speed
$v_b^*,\ T_b$	slip-branch target belt speed and belt-speed relaxation time constant
v_Δ	slip metric: primary-imposed minus secondary-imposed belt-line speed
Geometry	
$r_{p,\text{outer}}(s),\ r_{s,\text{outer}}(s)$	primary and secondary outer radius
$r_{p,\text{eff}}(s),\ r_{s,\text{eff}}(s)$	primary and secondary effective torque radius
$r_{p,\text{outer,min}}$	minimum primary outer radius
$r_{s,\text{outer,max}}$	maximum secondary outer radius
$s_{\text{dz}},\ s_{\max}$	deadzone threshold and maximum shift
β	sheave face half-angle (radial reference)
C	pulley center distance
L_b	total belt length
ϕ	belt wrap angle used in distributed contact model
$r,\ \dot{r}$	local belt path radius and its time derivative
r_{cm}	belt element center-of-mass radius
Constraint and mapping functions	
$f(r_p, r_s, C)$	belt-length compatibility root function ($f = 0$)
$\text{root}_{r_s}(\cdot)$	numerical root solve for secondary radius
$r_f(s),\ \dfrac{dr_f}{ds}$	primary flyweight radius map and gradient
$\theta_s(s),\ \dfrac{d\theta_s}{ds}$	secondary helix angle map and gradient

Symbol	Meaning
Forces and torques	
$F_{p,\mathrm{ax}},\ F_{s,\mathrm{ax}}$	primary and secondary axial mechanism force
$F_{c,p},\ F_{c,s}$	primary and secondary belt centrifugal axial contribution
F_{ax}	generic axial clamp force at one pulley contact
$\tau_{\mathrm{eng}},\ \tau_{\mathrm{load}}$	engine input torque and reflected load torque
$\tau_p,\ \tau_s$	coupling torque seen at primary and secondary
$\tau,\ \tau_c$	generic interface torque and applied coupling/friction torque
τ_{ns}	no-slip requested coupling torque
$\tau_-^{\mathrm{stick}},\ \tau_+^{\mathrm{stick}}$	aggregate lower and upper coupling-torque bounds for stick admissibility
$\tau_-^{\mathrm{slip}},\ \tau_+^{\mathrm{slip}}$	aggregate lower and upper coupling-torque bounds on slip branches
$\tau_{c,p,\pm}^{\mathrm{stick}},\ \tau_{c,s,\pm}^{\mathrm{stick}}$	primary/secondary pulley-wise stick-branch lower/upper bounds
$\tau_{c,p,\pm}^{\mathrm{slip}},\ \tau_{c,s,\pm}^{\mathrm{slip}}$	primary/secondary pulley-wise slip-branch lower/upper bounds
Mass/inertia and material parameters	
$I_p,\ I_s$	primary and secondary rotational inertia
m_f	effective flyweight mass
$m_{p,m},\ m_{s,m}$	equivalent primary and secondary translating mechanism mass
ρ_b	belt material density
A_b	belt cross-sectional area
μ	belt–pulley traction coefficient
$k_p,\ x_{p,0}$	primary spring stiffness and preload displacement
$k_{s,\theta},\ \theta_{s,0}$	secondary torsional stiffness and preload angle
$k_{s,x},\ x_{s,0}$	secondary axial stiffness and preload displacement

Table 1: Core symbols used throughout the derivation.

Constants and Parameters

All parameters in Table 2 are treated as constant values that do not vary through the model. They are either physical constants, fixed by the physical design of the CVT and vehicle or else are tuning parameters that are set and held constant during a given simulation run.

Symbol	Name	Unit	Description
Geometry and belt			
L_b	Belt length	m	Fixed total belt length
β	Sheave half-angle	rad	Cone half-angle used for radial/axial conversion
h_b	Belt thickness	m	Radial belt thickness
b_{out}	Belt outer width	m	Belt width at outer face
b_{in}	Belt inner width	m	Belt width at inner face
$r_{p,\mathrm{outer,min}}$	Primary minimum outer radius	m	Radius at fully open primary
$r_{s,\mathrm{outer,max}}$	Secondary maximum outer radius	m	Radius at fully closed secondary
s_{dz}	Shift deadzone threshold	m	Onset of effective radial motion

Symbol	Name	Unit	Description
$s_{\max}$	Maximum shift	m	Physical upper travel bound
Primary/secondary actuation			
m_f	Flyweight mass	kg	Effective flyweight mass
$r_{f,0}$	Initial flyweight radius	m	Flyweight radius offset at $s = 0$
k_p	Primary spring stiffness	N/m	Primary compression spring rate
$x_{p,0}$	Primary spring preload	m	Initial compression of primary spring
$k_{s,\theta}$	Secondary torsional stiffness	N m/rad	Torsional spring rate in helix assembly
$\theta_{s,0}$	Secondary torsional preload	rad	Initial torsional deflection preload
r_h	Helix reference radius	m	Effective cylindrical radius in helix kinematics
$k_{s,x}$	Secondary axial stiffness	N/m	Secondary axial compression spring rate
$x_{s,0}$	Secondary axial preload	m	Initial compression of secondary axial spring
Inertia and drivetrain			
I_{eng}	Engine inertia	$\text{kg}\,\text{m}^2$	Engine rotational inertia
I_{prim}	Primary pulley inertia	$\text{kg}\,\text{m}^2$	Primary CVT pulley inertia
I_p	Effective primary inertia	$\text{kg}\,\text{m}^2$	Implemented primary-side inertia in current setup
I_{sec}	Secondary pulley inertia	$\text{kg}\,\text{m}^2$	Secondary CVT pulley inertia
I_{gb}	Gearbox inertia	$\text{kg}\,\text{m}^2$	Gearbox/intermediate shaft inertia
I_{wheel}	Wheel inertia	$\text{kg}\,\text{m}^2$	Wheel/hub inertia about wheel axis
I_s	Effective secondary inertia	$\text{kg}\,\text{m}^2$	Lumped secondary-side inertia about secondary axis
G	Final drive ratio	1	Fixed ratio defined by $\omega_s = G\,\omega_w$
m	Vehicle mass	kg	Total vehicle mass
r_w	Wheel rolling radius	m	Effective tire rolling radius
Contact and environment			
ρ_b	Belt density	kg/m^3	Belt material density
μ_s	Belt–pulley static friction coefficient	1	Static traction coefficient used at impending slip
μ_k	Belt–pulley kinetic friction coefficient	1	Kinetic traction coefficient used on slip branches
C_{rr}	Rolling resistance coefficient	1	Tire-road rolling resistance coefficient
C_d	Drag coefficient	1	Aerodynamic drag coefficient
A_f	Frontal area	m^2	Vehicle aerodynamic reference area
ρ	Air density	kg/m^3	Ambient air density used in drag model
g	Gravitational acceleration	m/s^2	Gravity constant

Table 2: Canonical non-varying parameters used by the CVT model equations.

Chapter 1

Introduction & Background

1.1 Problem and Motivation

Mechanically actuated rubber V-belt continuously variable transmissions (CVTs) are widely used in compact powersport drivetrains because they can vary the speed ratio automatically while transmitting power through a relatively simple mechanical assembly. Their visible layout is straightforward: two variable-radius pulleys are connected by a belt. Their transient behaviour, however, is not. The operating state of the transmission emerges from the interaction of its internal mechanisms, the belt-pulley contacts, and the vehicle drivetrain to which it is attached.

Several coupled effects determine that response. Engine speed influences the centrifugal actuation of the primary pulley, while transmitted torque influences the torque-reactive response of the secondary pulley. Together, these mechanisms generate the clamping forces that press the belt into the sheave grooves. Clamping force determines the traction available at the belt–pulley contacts, which in turn determines how torque is carried and whether the belt and pulleys can remain coupled without gross slip. At the same time, pulley motion changes the belt radii and therefore the transmission ratio, while drivetrain inertia, road grade, and external vehicle load react back through the secondary side. These effects do not occur independently; each changes the conditions under which the others act.

This coupling makes the behaviour of a mechanically actuated CVT difficult to interpret from vehicle response alone. A shift curve, engine-speed trace, or loss of acceleration may reflect the combined effect of primary engagement, secondary reaction, belt traction, shift motion, and changing drivetrain load. The underlying states responsible for that behaviour are largely internal to the transmission and are not usually available directly during vehicle testing. This is especially consequential in racing-oriented applications such as Baja SAE, where launch behaviour, shift response, and torque transfer strongly affect performance, while changes to springs, flyweights, ramps, helices, or other hardware may require fabrication and repeated vehicle testing before their effect can be assessed.

Addressing this problem requires a physically interpretable dynamic framework that connects the internal CVT mechanisms to the vehicle-level response they produce. The model should be able to begin from varied initial conditions and respond to changing engine torque, road grade, external load, and other time-varying inputs, so that launch, clutching, shifting, backshift, traction loss, and vehicle acceleration emerge from the same coupled system response.

The remainder of this chapter develops the context for that framework. Section 1.2 introduces the physical layout and operating principles of the belt-sheave CVT, including ratio change, traction, mechanical actuation, and the surrounding drivetrain. Section 1.3 contains a literature review that organizes prior work by modelling emphasis and identifies the foundations available for mechanical actuation, belt contact, shift dynamics, and drivetrain coupling. Given this context, Section 1.4 then states the aim, scope, assumptions, and contributions of the present formulation.

1.2 Conceptual Overview of a Continuously Variable Transmission

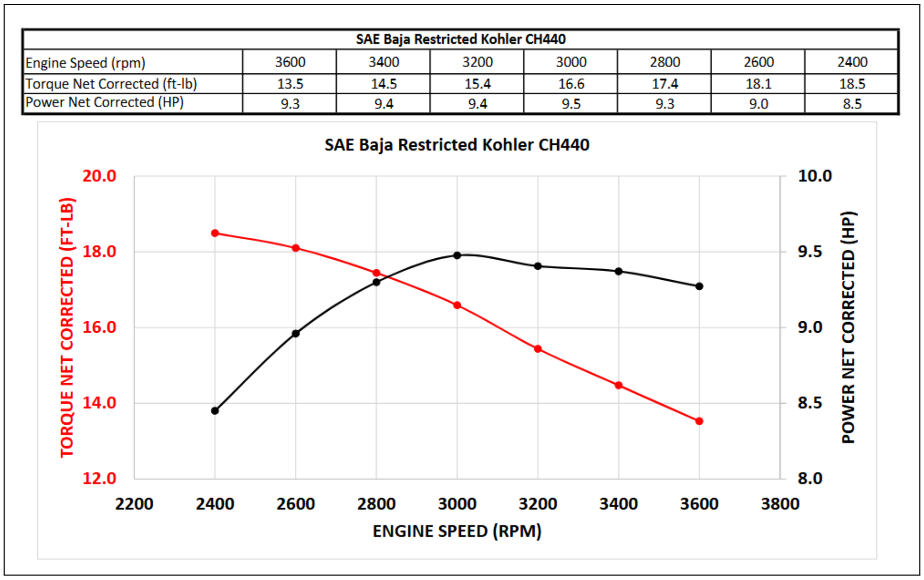
This section introduces the operating principles of the continuously variable transmission (CVT) considered in this work. The goal is not yet to derive the model equations, but to build the mechanical intuition needed to understand them. In particular, this section explains why a CVT is useful, how a rubber V-belt CVT changes ratio, why clamping and traction matter, how the primary and secondary pulleys generate their forces, and why the CVT must be understood as part of a larger drivetrain system.

1.2.1 Why Use a Continuously Variable Transmission?

A transmission is responsible for matching the operating speed of an engine to the speed required at the vehicle wheels. In a conventional stepped gearbox, this is achieved using a fixed set of discrete gear ratios. Each gear imposes a fixed relationship between engine speed and wheel speed. As the vehicle accelerates, the engine speed rises through one gear, drops when the transmission shifts, and then rises again in the next gear. This repeated sweep through the engine speed range is a direct consequence of having only a small number of available ratios.

This matters because an internal combustion engine (ICE) does not produce the same torque and power at every speed. Instead, the engine has a useful operating region where it can produce power more effectively, as shown conceptually in Figure 1.1. A stepped gearbox can pass through this region, but it cannot generally stay there while vehicle speed continues to increase. Figure 1.2 contrasts this stepped behavior with the smoother operating path made possible by a continuously variable transmission.

A continuously variable transmission (CVT) addresses this limitation by allowing the transmission ratio to vary smoothly within a finite range rather than changing through a fixed set of gear steps. Since the ratio can change continuously, the engine can remain closer to a desired operating region while the vehicle accelerates. This is the main benefit of a CVT: it gives the drivetrain more freedom to match engine behavior to vehicle load.



SAE Baja Restricted Kohler CH440							
Engine Speed (rpm)	3600	3400	3200	3000	2800	2600	2400
Torque Net Corrected (ft-lb)	13.5	14.5	15.4	16.6	17.4	18.1	18.5
Power Net Corrected (HP)	9.3	9.4	9.4	9.5	9.3	9.0	8.5

Figure 1.1: Torque and power of the Kohler CH440 engine as functions of engine speed. Both quantities vary substantially across the operating range, with peak torque occurring at a lower engine speed than peak power. (Baja SAE, 2022)

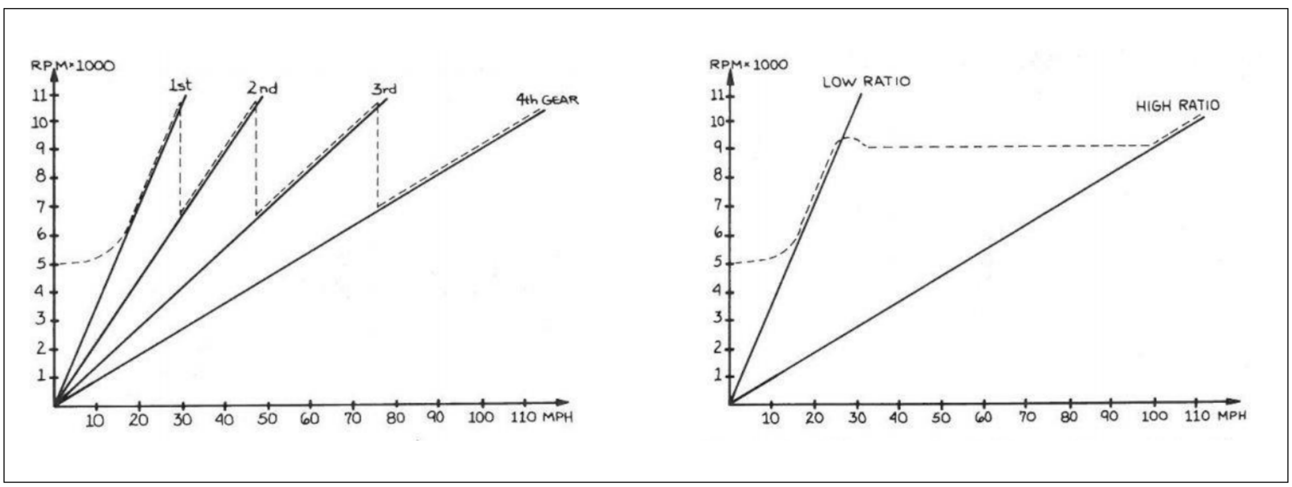


Figure 1.2: Conceptual comparison of engine-speed behavior during acceleration. A stepped gearbox requires repeated engine-speed sweeps and drops between gears, whereas a CVT can vary its ratio continuously to keep the engine closer to a desired operating speed as vehicle speed increases. (Aaen, 2007)

1.2.2 Physical Layout and Single-Pulley Geometry

There are several forms of continuously variable transmission, including different mechanical layouts and different ways of producing the variable speed relationship. This work focuses on a belt-sheave CVT, shown conceptually in Figure 1.3. In this arrangement, two variable-radius pulleys are connected by a rubber V-belt. The pulley connected to the engine side is called the primary (or driving) pulley. The pulley connected to the downstream drivetrain is called the secondary (or driven) pulley. The belt transmits motion and torque between these two pulleys.

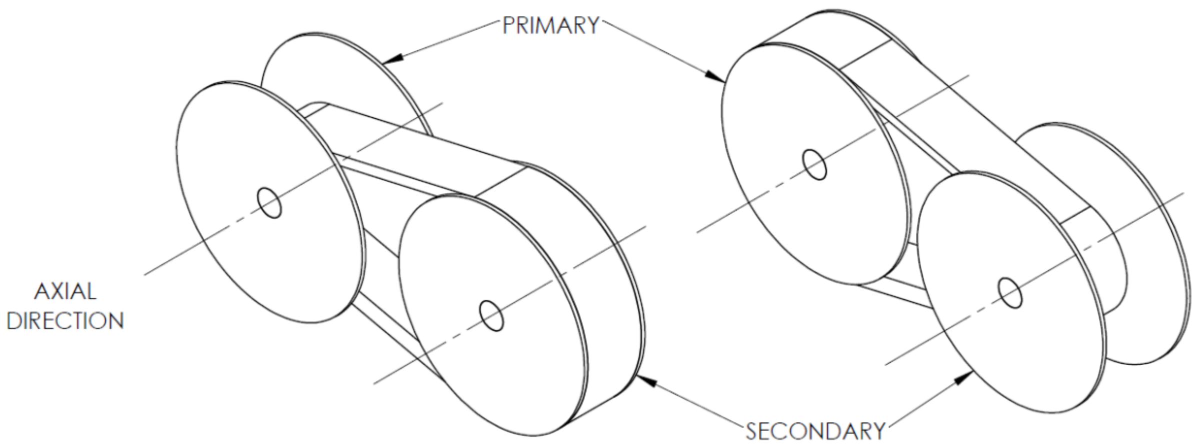


Figure 1.3: Two sample configurations of the belt-sheave CVT considered in this work. The primary pulley is connected to the engine side, the secondary pulley is connected to the downstream drivetrain, and the rubber V-belt transmits motion and torque between them. (Messick, 2018)

Before considering the full two-pulley system, it is helpful to look at the geometry of a single pulley. Both the primary and secondary pulleys share the same basic anatomy. Each pulley is made from two opposing conical sheaves that form a V-shaped groove. One sheave is fixed axially to the shaft, while the other can move along the shaft. The rubber V-belt sits between the sheave faces. This basic layout is shown in Figure 1.4.

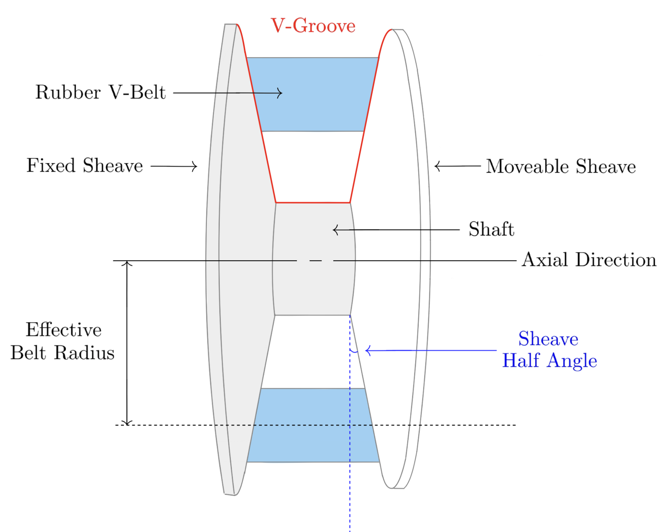


Figure 1.4: Generic anatomy of a belt-sheave CVT pulley. Each pulley consists of opposing conical sheaves that form a V-groove. Motion of the movable sheave changes the groove width and therefore changes the radius at which the belt rides.

The key geometric idea is simple: changing the axial separation between the sheaves changes the effective radius of the belt. If the movable sheave moves toward the fixed sheave, the groove becomes narrower. Since the belt has a finite width, it cannot occupy the same radial position in the narrower groove, so it is pushed outward along the conical sheave faces. The effective belt radius therefore increases.

If the movable sheave moves away from the fixed sheave, the groove becomes wider and the belt can fall deeper between the sheaves. This reduces the effective belt radius. At this level, the mechanism is just wedge geometry: axial motion of the sheave becomes radial motion of the belt. Closing the sheaves makes the belt radius larger, while opening the sheaves makes the belt radius smaller, as shown in Figure 1.5.

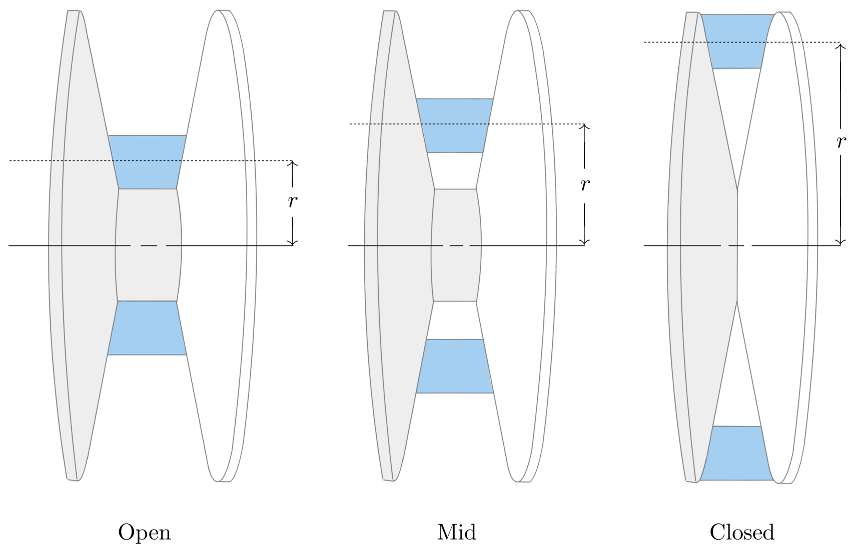


Figure 1.5: Single-pulley radius change. As the sheaves close, the V-groove narrows and the belt is pushed outward to a larger effective radius. As the sheaves open, the belt can fall deeper in the groove and the effective radius decreases.

1.2.3 Two-Pulley Coupling and Ratio Change

The full CVT contains two of the variable-radius pulleys described above. The important difference is that the two pulleys are connected by the same belt. Because the belt length is approximately fixed, the effective radii of the two pulleys cannot change independently.

If the belt rides outward on the primary pulley, it must ride inward on the secondary pulley. Likewise, if the primary radius decreases, the secondary radius must increase. This opposite motion is the core geometric behavior that allows the CVT ratio to change. In the physical mechanism, the actual motion of the sheaves occurs through contact forces, springs, actuator mechanisms, and the axial force balance of the pulleys.

The CVT begins launch in a low-speed, high-reduction configuration. In this state, the primary pulley has a small effective radius and the secondary pulley has a large effective radius. This behaves like a low gear in a conventional drivetrain: the engine-side pulley turns relatively quickly while the downstream side turns more slowly, amplifying torque through the drivetrain.

As the CVT upshifts, the primary sheaves close and the primary effective radius increases. At the same time, the secondary effective radius decreases. The transmission therefore moves from its launch configuration toward a higher-speed configuration. By the end of the shift, the primary radius is larger and the secondary radius is smaller. This coupled motion is shown in Figure 1.6.

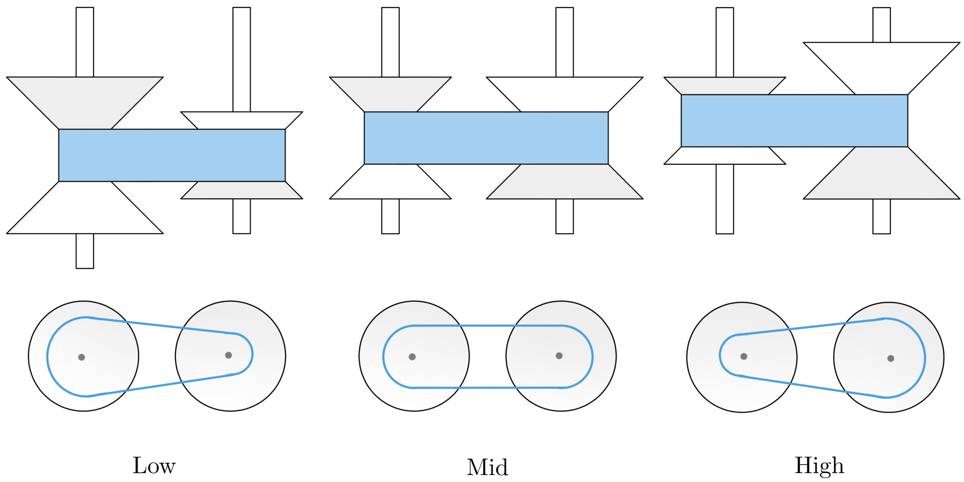


Figure 1.6: Low-, intermediate-, and high-ratio states of a belt–sheave CVT. As the belt moves outward on the primary pulley and inward on the secondary pulley, the effective radii vary in opposite directions while the belt length remains approximately constant. The lower views highlight the corresponding change in transmission ratio.

1.2.4 Torque Transfer, Clamping Force, and Slip

The geometry described so far determines the speed relationship between the primary and secondary pulleys. However, a speed relationship alone does not guarantee that torque can be transmitted. Unlike a gear pair, the belt and pulley are not mechanically locked together by teeth. Torque is transmitted through frictional traction between the belt and the sheave faces.

For traction to develop, the belt must be physically squeezed between the sheave faces. Figure 1.7 shows how this happens at a small portion of the belt–pulley contact. The circled region on the pulley wrap is enlarged to show the belt cross-section within the pulley V-groove. At that location, the movable sheave presses the belt toward the fixed sheave. This clamping action produces contact normal forces on the two sides of the belt, creating the contact pressure needed for frictional traction.

The same local interaction occurs continuously around the wrapped portion of the belt. At each position along the wrap, the normal forces between the belt and sheave faces allow a tangential friction force to act. Together, these small tangential traction forces transfer torque between the pulley and the belt. In practical terms, transferring more torque through the CVT requires enough clamping force to create enough available traction at the belt–pulley interface. More clamping force increases the normal contact force, which in turn increases the tangential force that friction can support.

If the available traction is not sufficient for the torque being transferred, the belt slips relative to the pulley. Slip reduces useful torque transfer and can produce heat, wear, and efficiency loss. A working CVT must therefore do two things at once: it must allow the pulley radii to change so the ratio can shift, and it must maintain enough clamping force for the belt to transmit torque without excessive slip.

This distinction is central to the rest of the model. The pulley geometry determines how the speed ratio changes. The contact mechanics determine how much torque can actually be carried through the belt–pulley contacts. A CVT is therefore not only a variable-geometry mechanism; it is also a clamping and traction problem.

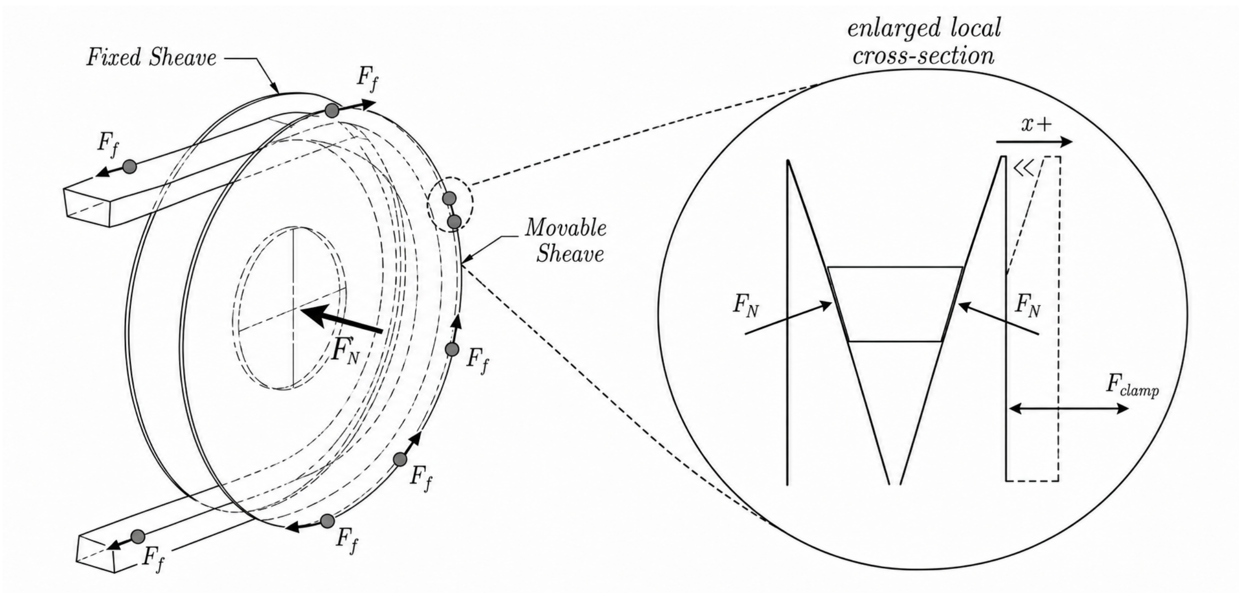


Figure 1.7: Clamping and traction in a rubber V-belt CVT. A representative location on the wrapped belt–pulley contact is circled and enlarged to show the local belt cross-section within the V-groove. Motion of the movable sheave toward the fixed sheave creates normal contact forces on the belt faces. These normal forces allow tangential frictional traction to act continuously along the wrap, collectively transmitting torque between the pulley and belt. The indicated tangential forces are representative local contributions to this distributed traction.

1.2.5 Mechanical Actuation in the Present CVT

Different CVTs generate radius change and clamping force in different ways. Some systems use electronic or hydraulic actuation to command pulley motion or clamping pressure directly. The system studied here is mechanically actuated: its pulley forces arise from mechanisms inside the primary and secondary pulley assemblies rather than from an externally commanded ratio.

These mechanisms are important because they determine both how the belt moves through the ratio range and how much clamping force is available for torque transfer. The primary pulley is mainly speed-sensitive, while the secondary pulley is mainly torque-reactive.

The primary pulley uses flyweights, ramps, and a spring. As engine speed increases, the flyweights move outward under centrifugal effects. The ramp surfaces convert this outward flyweight motion into a closing force on the primary sheaves. The spring resists this motion and helps set the speed at which the primary begins to close. As the primary closes, the belt is pushed outward, the primary effective radius increases, and the CVT shifts in the upshift direction introduced in Figure 1.6. The primary actuation mechanism is shown conceptually on the left side of Figure 1.8.

The secondary pulley uses a helix mechanism and spring loading. The helix can be understood like a screw-like interface: when torque is transmitted through the secondary, the angled helix surfaces convert part of that torque reaction into a clamping force between the sheaves. In this way, torque through the secondary helps create the clamping needed to carry torque through the belt. The spring provides preload and helps set the baseline clamping behavior.

The helix also means that secondary shift is not purely axial. As the secondary moves through its shift range, axial motion is coupled with relative rotation between secondary sheave components. This detail becomes important later when the secondary force model is developed, but the basic intuition is simply that the helix converts torque and relative rotation into axial clamping. The secondary actuation mechanism is shown conceptually on the right side of Figure 1.8.

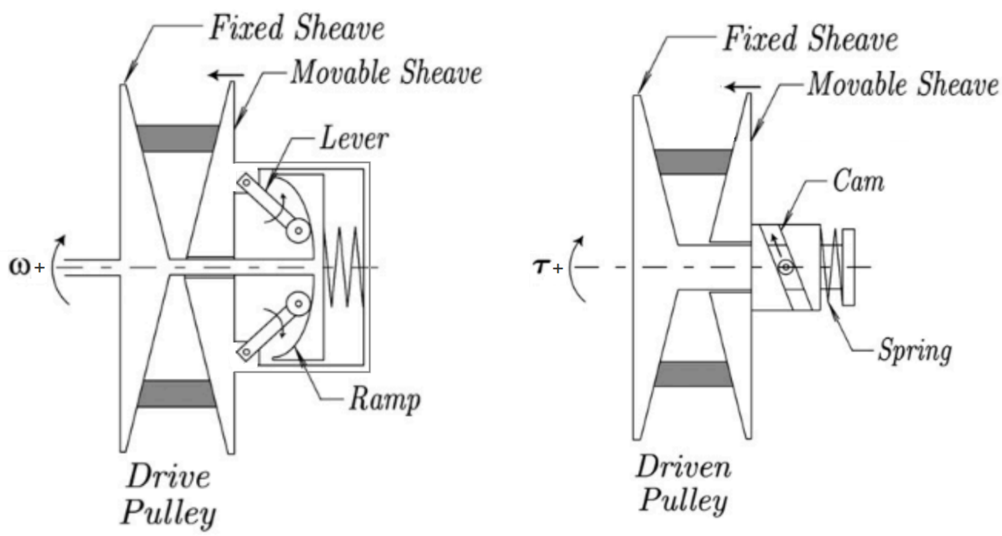


Figure 1.8: Conceptual illustration of the mechanical actuation mechanisms in the present CVT. Left: the primary (drive) pulley, where increasing rotational speed moves the flyweights outward and the ramp geometry converts that motion into a closing force on the sheaves, opposed by a spring. Right: the secondary (driven) pulley, where transmitted torque loads a cam or helix mechanism that converts part of the torque reaction into sheave clamping force, together with spring preload. In this way, the primary is mainly speed-sensitive and the secondary is mainly torque-reactive. (Juli'o and Plante, 2011)

Although the primary and secondary mechanisms respond most strongly to different physical quantities, they act on the same belt and the same shift process. The primary is mainly driven by speed, the secondary is mainly driven by transmitted torque, and the resulting CVT behavior depends on their combined interaction with the belt, pulley geometry, drivetrain inertia, external load, and available clamping force.

1.2.6 Full Drivetrain Architecture

The CVT does not operate in isolation. The physical drivetrain architecture considered in this work is shown in Figure 1.9. The figure shows the CVT within the complete vehicle context, including the downstream gearbox, disconnect, shafts, bevel units, axles, and all four wheels.

Engine torque enters the primary pulley, passes through the belt to the secondary pulley, and then enters the downstream driveline. This driveline contains a fixed-ratio gearbox and the remaining shaft, bevel, and axle components that transmit torque to the wheels. At the tire–ground interface, tractive force accelerates the vehicle, while vehicle inertia, grade resistance, rolling resistance, and aerodynamic drag produce the load that feeds back into the drivetrain through the secondary side.

Although the full physical architecture is shown in Figure 1.9, the dynamic model does not resolve the individual motion of every U-joint, bevel unit, axle, or wheel. Instead, their combined effect is represented through the downstream fixed reduction, equivalent drivetrain inertia, and vehicle longitudinal load. This preserves the coupling that matters to the CVT while avoiding unnecessary detail outside the scope of the present model.

Many drivetrain architectures are possible, but the engine–CVT–downstream driveline–wheel arrangement shown here is the system modeled in this work. This system-level view is important because the CVT ratio is not simply chosen independently of the rest of the vehicle. Engine speed affects primary actuation. Transmitted torque affects secondary clamping. Vehicle load affects the torque carried by the secondary. Belt traction determines whether that torque can be carried without slip. All of these quantities evolve together during a launch.

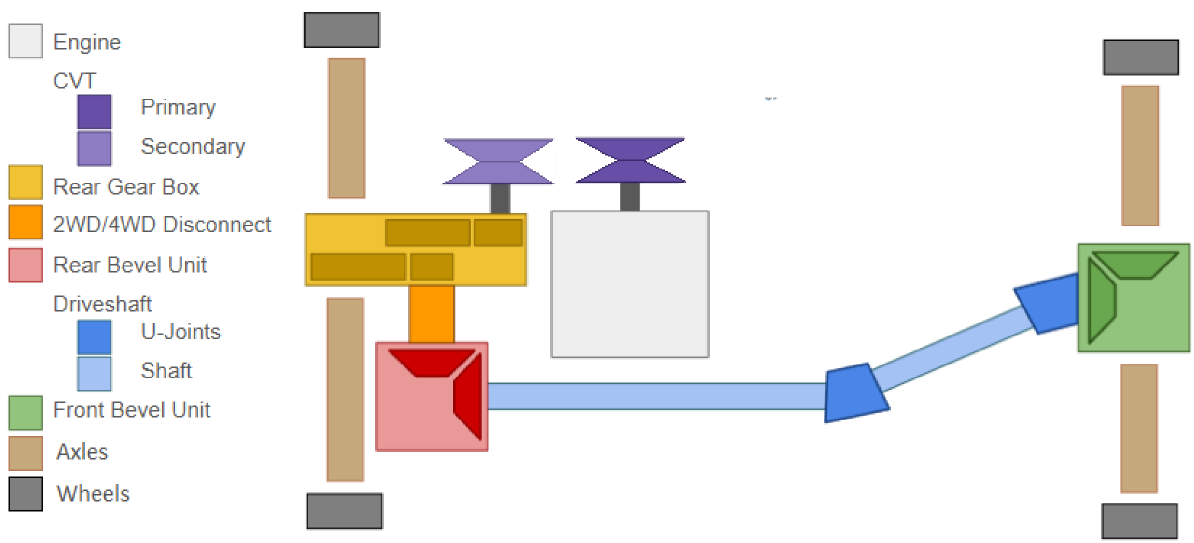


Figure 1.9: Physical drivetrain architecture considered in this work. The figure shows the CVT within the complete four-wheel vehicle layout, including the downstream gearbox, disconnect, shaft, bevel, and axle components. In the dynamic model, these downstream elements are represented by an equivalent fixed reduction, reflected inertia, and vehicle load acting through the secondary side of the CVT.

The rest of this document turns this physical picture into a dynamic model. The geometry describes how pulley radii change during shift. The force models describe how the primary and secondary sheaves are driven. The traction model describes when torque transfer can stick and when slip occurs. The drivetrain model closes the loop between engine torque, CVT behavior, vehicle load, and vehicle acceleration. Together, these pieces explain how the CVT behaves as a coupled mechanical system during launch.

1.3 Motivation and Literature Review

CVT modelling is not a single, uniform problem. A model developed to size a variator, estimate belt efficiency, design a control strategy, or predict launch behaviour may involve the same physical components, but it will usually make different choices about which quantities are prescribed and which are solved. Some models are primarily design tools. Some focus on mechanical actuation and force balance. Some study belt forces, axial thrust, slip, or efficiency at fixed operating points. Others attempt to resolve transient shift behaviour or detailed belt-pulley contact mechanics.

The progression followed here is from design intuition, to component force balance, to mechanical actuation, to belt contact, to ratio change, to full transient launch behaviour. This ordering mirrors the way the modelling problem becomes difficult. A belt-sheave CVT can first be understood as a variable-radius transmission, then as a collection of mechanical actuators that generate axial force, then as a friction-limited belt contact problem, and finally as a coupled dynamic system whose engine speed, belt state, shift position, and vehicle motion evolve together. The literature contains strong foundations for each of these pieces, but those foundations are distributed across several modelling traditions.

1.3.1 Framework for Classifying CVT Models

A useful way to compare CVT models is to ask what part of the system is being allowed to move, and what part is being held fixed. The same physical transmission can be treated as a geometry problem, a force-balance problem, a belt-contact problem, a shift-dynamics problem, or a drivetrain problem. Each viewpoint can be useful, but each one answers a different question. This is why the literature is best read by tracking the modelling choices that define each formulation, rather than by treating all CVT models as attempts at the same final system.

One of the first choices is how the ratio is represented. In the simplest case, the transmission is evaluated at a fixed ratio. A slightly broader approach computes the ratio from a quasi-static force balance at each operating point. Other models prescribe a ratio trajectory, or use an empirical relationship for the rate of ratio change. More detailed shift models treat ratio change as a mechanical process in its own right, involving radial belt motion, pulley motion, and contact forces (Kim et al., 2002; Sorge, 2008; Cammalleri and Sorge, 2009; Sorge and Cammalleri, 2011). These approaches are not interchangeable. A fixed-ratio model can explain forces at a known condition, while a shift model tries to explain how the system moves from one condition to another.

A second choice is how axial force is produced. Some models take pulley clamping forces as known inputs. This is natural when the goal is to study belt response under specified loading, or when the variator is externally controlled. Other models include the mechanism that generates those forces, such as centrifugal weights, springs, torque-sensitive cams, hydraulic pressure, or electronic actuation (Cammalleri, 2005; Kim et al., 2002; Skinner, 2020; Sorge and Cammalleri, 2011). The important distinction is not only the actuator type, but whether clamping force is prescribed from outside the model or emerges from the same system being simulated.

A third choice is how much of the belt-pulley contact is represented. At low fidelity, the belt may be treated through a capstan-style tension relation, often adjusted for the wedge action of the V-groove. More detailed models include radial and tangential friction, sliding angle, belt penetration, bending stiffness, local normal pressure, tension variation, slip mechanisms, and efficiency losses (Gerbert, 1996; Kim and Kim, 1989; Sorge, 1996; Kong and Parker, 2005, 2006; Chen et al., 1998; Bertini et al., 2014; Messick, 2018). At still higher fidelity, the belt may be discretized into nodes or elements so that local belt motion and contact behaviour are solved directly (Juli'o and Plante, 2011; Ballew, 2015). Increasing contact fidelity can make the local mechanics more realistic, but it also changes the purpose of the model: a model built for efficiency or local belt force prediction may not be the most transparent way to study the full drivetrain response.

A fourth choice is how the rotating system is coupled. Many models prescribe one or both pulley speeds, or impose the speed ratio algebraically. This is often appropriate for steady contact analysis, efficiency prediction, or controlled operating-point studies. Other models solve rotational equations of motion so that input speed, output speed, and transmitted torque evolve together (Srivastava and Haque, 2006; Ballew, 2015; Duan et al., 2016). Extending this further, the input may be coupled to an engine model and the output may be coupled to a dynamometer or vehicle load. This determines whether the CVT is being studied as an isolated variator or as one part of a larger powertrain.

These choices build on one another. A model can have detailed belt contact but prescribed pulley speeds. Another can include vehicle dynamics but use a simplified belt law. Another can model mechanical actuation carefully but treat shifting as quasi-static. None of these choices is inherently wrong; they reflect the question being asked. The important point is that the assumptions define the model's domain of usefulness.

Launch and clutching sit near the intersection of these choices. In that regime, ratio change, actuation, belt contact, rotational dynamics, and external loading can all matter at once. This does not make launch the only goal of CVT modelling, but it does make launch a useful stress test of model closure: a formulation that prescribes too many of these quantities may still be valuable, but it is not resolving the same coupled problem. The following review uses this framework to move from practical and quasi-static design models, to mechanical actuation and shift mechanics, to belt contact and efficiency models, to transient rubber-belt formulations, and finally to related dynamic models in metal-belt and chain CVTs.

1.3.2 Practical Design and Tuning Foundations

The practical understanding of mechanically actuated rubber V-belt CVTs developed largely through machine design and tuning practice. General machine design references such as Budynas and Nisbett (2015) provide the background needed to treat the CVT as a real mechanical assembly involving belts, springs, clutches, shafts, friction interfaces, uncertainty, and design iteration. Not all of these effects are included in a compact model, but they provide useful context for deciding which behaviours are retained and which are idealized.

Practical tuning literature provides a more application-specific foundation. Aaen (1979) describes powersport CVTs using the language of engagement speed, shift speed, flyweights, primary springs, secondary springs, torque sensing, backshift, and speed variance. Its value is not that it provides a complete dynamic model, but that it builds component-level intuition from the perspective of a tuner or racer. It explains how changes to weights, springs, ramps, and helices are expected to affect behaviour, while leaving much of the coupled system response to experience and testing.

These sources are valuable because they describe the physical machine before it is reduced to equations. They also show why prediction is difficult: the same hardware change can affect engagement, shift rate, belt grip, backshift, engine speed, and vehicle acceleration at the same time. Qualitative tuning rules are therefore useful starting points, but they are not enough to determine the full system response under changing operating conditions.

1.3.3 Design-Oriented and Quasi-Static CVT Models

Design-oriented CVT models attempt to convert practical component knowledge into quantitative design procedures. Rather than asking how every transient state evolves in time, these models usually ask which geometry, actuator, belt, or tuning parameters should be selected to satisfy a desired operating requirement. They commonly relate vehicle requirements, pulley geometry, belt length, wrap angle, speed ratio, flyweight force, spring preload, helix force, belt tension, and torque capacity. Their main value is that they connect tunable hardware to expected operating behaviour.

A useful starting point is the numerical design method of Arango and Mu noz Alzate (2020), which begins from vehicle requirements, road specifications, and desired performance before working toward the CVT design. This is valuable because it places component selection in its proper context: the belt, pulley geometry, speed-ratio range, engine or motor selection, and mechanical centrifugal actuation are not independent choices, but responses to the requirements imposed by the vehicle and its operating route. In this sense, the work provides a design-level framework where system requirements drive CVT component choices.

Other design-oriented works focus more narrowly on particular parts of the variator design. Cammalleri (2005) develops a speed-torque-controlled rubber V-belt variator design using equilibrium relationships for a centrifugal driver and torque-sensitive driven pulley. La Battaglia et al. (2022) focuses on the kinematic design of motorcycle rubber V-belt CVTs, emphasizing the roller sliding profile of the front pulley as a critical design variable. Both works show how simplified mathematical models can be used to choose actuator or geometry parameters without requiring a full transient simulation of the drivetrain.

Within Baja-style applications, Skinner (2020) provides a practical force-balance and tuning-oriented model. The work emphasizes quantifiable tuning objectives such as engagement speed, shift-out speed, static RPM shifting, and efficient power transfer, and relates those objectives to the primary and secondary subsystems. It is useful as an applied tuning reference because it connects the variables adjusted by teams to predicted operating behaviour, while remaining primarily an operating-point and force-balance model.

The common strength of these works is that they make CVT design less arbitrary. They identify the parameters that matter and show how geometry, springs, flyweights, helices, belt forces, and vehicle requirements can be related. They also highlight a recurring tension in CVT modelling: models that are simple enough to support early design may omit important coupled behaviour, while models that resolve more detailed mechanics can become difficult to apply directly as design tools. In this sense, design-oriented and quasi-static models occupy an important middle ground. They provide usable relationships for component selection and tuning, but they do not generally describe the complete time history of clutching, slip, shift motion, and vehicle acceleration.

1.3.4 Mechanical Actuation and Clamping Force Generation

After the basic design variables are identified, the next modelling question is how the CVT generates the axial forces that clamp the belt and drive ratio change. In externally controlled CVTs, these forces may be treated as prescribed hydraulic or electronic inputs. In mechanically actuated rubber V-belt CVTs, however, the forces are produced by the hardware itself. The primary force is typically speed-sensitive, while the secondary force is torque-sensitive.

Cammalleri (2005) provides an important design treatment of this problem through a speed-torque-controlled rubber V-belt variator. The centrifugal driver and torque-sensitive driven pulley are treated as part of the variator design, rather than as external force inputs. This helps establish the link between actuator geometry, spring loading, pulley torque, and the intended operating behaviour of the CVT.

Kim et al. (2002) also study a rubber belt CVT with mechanical actuators, including both steady-state and transient characteristics. Their work is useful because it connects the mechanically generated driver and driven pulley forces to the resulting CVT behaviour. Although these earlier mechanical-actuator models include torque-reactive secondary behaviour, they simplify how torque is shared and reacted through the driven sheave assembly.

Sorge and Cammalleri (2011) address this secondary-side issue in more detail by studying the helical shift mechanics of rubber V-belt variators. Their work shows that the relative rotation between the driven sheaves is important to the secondary response, and that the helix should not be treated only as a simple torque-to-axial-force conversion. This makes the secondary a coupled torque-reactive mechanism whose axial force depends on helix geometry, transmitted torque, spring loading, and the internal motion of the driven pulley.

Together, these works show that mechanical actuation is more than a source of prescribed clamp force. The primary and secondary mechanisms are part of the dynamics of the variator: speed, torque, spring preload, ramp or helix geometry, and pulley motion all influence the axial forces applied to the belt. This establishes the actuator-side foundation needed before considering the belt contact and transient shift response.

1.3.5 Rubber V-Belt Contact, Slip, and Efficiency Models

Once axial force is generated by the actuation mechanisms, the belt-pulley interface determines what that force actually does. The sheaves may clamp the belt, but the resulting torque capacity, axial reaction, radial belt motion, slip, and loss all depend on how the rubber belt interacts with the pulley groove. A rubber V-belt therefore cannot be treated as an ideal kinematic constraint. It wedges into the sheaves, develops a tension distribution around the wrap, experiences radial and tangential friction, deforms under load, and dissipates energy through several mechanisms.

Classical belt theory gives the starting point: friction and wrap angle determine the maximum tension ratio that can be supported before gross sliding occurs. For V-belts, this picture is modified by the groove angle and the associated wedge effect. However, real belt slip is more complicated than a single capstan-style traction limit. Gerbert (1996) presents a unified treatment of belt slip in which creep, radial compliance, shear deflection, and seating/unseating effects all contribute to speed loss. The important point is not that every model must include all of these effects, but that "slip" is not one physical mechanism. A model should be clear about which slip and loss mechanisms it represents and which it leaves outside its scope.

The contact problem also determines the axial reaction felt by the sheaves. Kim and Kim (1989) developed a theoretical analysis of axial forces in a rubber V-belt CVT using simplified assumptions about the contact regions and belt force distribution. Sorge (1996) developed a compact model for axial thrust in V-belt drives, emphasizing radial penetration and the connection between belt forces and pulley thrust. These works are useful because they connect the belt contact state back to the sheave force balance: axial force is not only generated by the actuator, but also shaped by how the belt seats, slides, and transmits load in the groove.

A more detailed treatment is given by Kong and Parker (2005), who formulate the steady mechanics of belt-pulley systems using distributed belt equations. Kong and Parker (2006) extend this framework to pulley grooves and sliding friction. This line of work resolves local belt mechanics at much greater fidelity than simple force-balance models, including the distribution of forces and sliding behaviour around the contact arc.

Messick (2018) extends this contact-mechanics tradition for rubber V-belt CVTs by building on the Kong and Parker framework and validating axial-force and efficiency predictions experimentally. Messick's work is valuable for its contact-region formulation, belt property measurement, operating-range analysis, and dynamometer validation. It is also clear about its modelling scope: shifting is treated as a quasi-equilibrium process rather than as a transient response-time problem. In this sense, it is a strong reference for belt force and efficiency modelling, while leaving the time evolution of the complete drivetrain to a different class of model.

Efficiency-focused studies provide another view of the same interface. Chen et al. (1998) experimentally investigated speed and torque losses in a rubber V-belt CVT, while Bertini et al. (2014) developed an analytical model for rubber V-belt CVT power losses including sliding, hysteresis, and entry/exit effects. These works show that efficiency depends on multiple interacting mechanisms, not just on whether the belt remains below a gross slip threshold.

The contact and efficiency literature therefore provides the foundation for understanding how belt force, axial thrust, slip, and loss arise. It also shows why contact fidelity must be chosen carefully. A high-fidelity contact model may be necessary for efficiency prediction, axial-force prediction, or local belt mechanics. A lower-dimensional contact model may be more appropriate when the goal is to preserve physical intuition while coupling the belt to actuator dynamics, rotational dynamics, and drivetrain response.

1.3.6 Transient Rubber-Belt Models and the Launch Problem

Transient rubber-belt models address time-dependent behaviour more directly, but "transient" can mean several different levels of closure. A model may study how the belt migrates radially during shift, how local belt elements move through the pulley contact, how the variator responds to prescribed axial forces, or how the CVT interacts with an engine and vehicle load. Launch lies near the most coupled end of this spectrum. During launch, the belt may be slipping, the engine may be accelerating, clamping forces may be changing, the secondary torque reaction depends on transmitted torque, and the vehicle load evolves as the vehicle begins to move. A launch model therefore has to manage torque transfer and ratio change while the drivetrain state is still developing.

A useful bridge between steady force-balance models and fully transient belt simulations is the shift-mechanics work of Sorge (2008). Sorge shows that ratio change in a rubber belt variator is not simply a sequence of steady operating points. During shift, the belt moves radially through the sheave grooves while also travelling circumferentially and transmitting torque, so belt mass conservation, radial motion, sliding, and circumferential transport interact. Cammalleri and Sorge (2009) later develop approximate closed-form solutions for this shift-mechanics problem, making the theory more accessible for engineering calculations. These works isolate the mechanics of ratio change itself, without attempting to close the full engine–vehicle launch problem.

Juli'o and Plante (2011) developed an experimentally validated transient model of rubber-belt CVT mechanics using a discretized belt formulation. This work represents the belt as a distributed mechanical body rather than relying only on lumped contact assumptions. Its strength is that local belt motion and contact response are resolved directly. The tradeoff is that the global mechanism chain becomes less compact, and the relationship between model behaviour and a small set of practical tuning variables becomes less direct.

Ballew (2015) continues in the discretized-belt direction and expands the model to include engine torque output and vehicle loads such as rolling resistance and aerodynamic drag. Ballew also argues against prescribing input pulley speed, since doing so removes an important rotational degree of freedom. This makes the work especially relevant to drivetrain-level rubber-belt CVT simulation. At the same time, the reported simulations use a PI controller with feed-forward gain to generate primary axial force, fix the secondary axial force to simplify the simulation, and introduce damping between belt nodes to manage numerical behaviour. Thus, the model includes important engine and vehicle coupling, but it does not close the launch problem through a mechanically self-actuated flyweight and helix model.

The transient rubber-belt literature therefore provides important precedents, while also showing why the problem remains difficult. Shift-mechanics models clarify how ratio change occurs, discretized belt models resolve local belt behaviour, and drivetrain-coupled models show how engine and vehicle dynamics can be included. The remaining challenge is not simply to make the belt model more detailed, but to choose a level of fidelity that preserves the coupling between actuation, torque transfer, ratio evolution, and vehicle response without losing physical interpretability.

1.3.7 Adjacent Metal-Belt and Chain CVT Dynamic Models

The broader CVT literature contains an extensive dynamic modelling tradition for metal push-belt and chain CVTs, especially in automotive applications. These systems are not direct equivalents of dry rubber V-belt CVTs, but they are useful adjacent references because they show how other CVT architectures have handled shift dynamics, slip, clamping force, deformation, and validation.

Srivastava and Haque (2009) review the dynamics and control of belt and chain CVTs, covering modelling and control strategies for friction-limited transmissions. Their review is useful not only because it summarizes the field, but because it shows that CVT modelling has developed unevenly across architectures. Different CVT types are modelled with different assumptions, and fully dynamic, experimentally validated models remain less common than steady-state or control-oriented formulations. The review also reinforces an important distinction: rubber V-belts, metal push belts, and chains should not be treated interchangeably because their contact mechanics, actuation methods, deformation behaviour, torque capacity, and control objectives differ.

For metal belt CVTs, Carbone et al. (2005) study the influence of pulley deformation on shifting behaviour. Their model explains the distinction between creep-mode and slip-mode shifting and shows that pulley deformation can be essential to reproducing slow-shift behaviour under Coulomb-friction assumptions. Yildiz et al. (2017) later validate the Carbone–Mangialardi–Mantriota model under steady-state conditions and shift manoeuvres with torque load, supporting the usefulness of this modelling framework for controlled metal-belt or chain CVTs.

Other metal-belt studies provide useful analogues for traction-limited behaviour and ratio-change dynamics. Srivastava and Haque (2006) model the dynamics of metal V-belt CVTs, while Bonsen et al. (2004) analyze slip behaviour and the relationship between slip and transmitted torque. These works are relevant because they treat slip and dynamic ratio change as part of the variator mechanics, even though the transmission architecture differs from a dry rubber V-belt system.

Chain CVTs provide another adjacent comparison. Duan et al. (2016) develop a physics-based chain CVT model beginning from the kinematics and dynamics of an infinitesimal chain segment. Their model includes speed-dependent friction, pulley axial dynamics coupled with chain radial movement, and pulley deformation. This makes the work a useful example of a physically coupled CVT formulation in which rotational dynamics, axial dynamics, contact mechanics, and friction state are treated together.

The important distinction is that these models are usually developed for lubricated automotive metal-belt or chain systems with hydraulic or electronic clamping. Their methods are therefore valuable, but their assumptions cannot be imported directly into dry rubber V-belt CVTs with mechanical flyweights, springs, and torque-sensitive helices. They show what is possible in adjacent CVT architectures, while also reinforcing the need for a formulation matched to rubber-belt contact and mechanical actuation.

1.3.8 Synthesis of Prior Work

The literature establishes a complementary set of foundations rather than a single preferred model family. Practical and design references identify the hardware variables and force relationships that govern rubber V-belt CVTs. Mechanical-actuation studies show how primary and secondary clamping forces can arise from the pulley mechanisms themselves. Contact and efficiency models resolve the belt–pulley interface in increasing detail, while transient formulations address ratio change, belt motion, and, in some cases, drivetrain response. Adjacent metal-belt and chain formulations further demonstrate how axial, rotational, and traction dynamics can be coupled when the transmission architecture and actuation system are different.

The remaining difficulty is not any single missing equation, but the closure of the whole system under assumptions appropriate to a mechanically actuated dry rubber V-belt CVT. Launch and clutching sit at the intersection of engine acceleration, primary engagement, secondary torque reaction, belt traction, shift motion, and vehicle load. Treating one or more of these quantities as a prescribed input can be appropriate for a focused study, but it changes the problem being solved. The central modelling challenge is therefore to retain the dominant couplings required for transient vehicle response without obscuring the system behind unnecessary local detail.

Table 1.1 compares selected formulations that most directly bracket the present modelling objective. The preceding sections discuss the individual models in detail; the table brings together their principal modelling choices and contributions. It therefore provides a comparison of modelling scope across the selected formulations.

Selected formulation	**Ratio / shift**	**Actuation closure**	**Belt / contact treatment**	**Rotational vehicle coupling**	**Principal contribution**
Cammalleri (2005)	Quasi-static equilibrium	Mechanical primary and secondary	Analytical belt-force model	Steady imposed conditions	Mechanical variator design with centrifugal and torque-sensitive actuation
Kim et al. (2002)	Steady plus empirical transient response	Mechanical primary and secondary	Simplified belt-force model	Externally supplied conditions	Empirical transient response of a mechanically actuated rubber-belt CVT
Sorge (2008)	Derived transient shift mechanics	Pulley loading prescribed	Radial belt motion and sliding during shift	No engine–vehicle model	Derived mechanics showing that ratio change is inherently transient
Sorge and Cammalleri (2011)	Secondary helix motion	Torque-reactive helix and spring	Belt loads supplied to secondary	No complete drivetrain model	Secondary helix resolved as a coupled torque–rotation–axial mechanism
Messick (2018)	Quasi-equilibrium operating states	Axial loading prescribed	Detailed distributed contact, thrust, and efficiency model	Dynamometer operating points	Experimentally validated rubber-belt contact, axial-thrust, and efficiency formulation
Juli'o and Plante (2011)	Transient discretized-belt response	Axial and kinematic inputs prescribed	Local belt inertia, elasticity, and contact	External speed and load conditions	Experimentally validated transient rubber-belt mechanics
Ballew (2015)	Transient discretized-belt response	Primary controlled; secondary force fixed	Nodal stick–sliding contact with damping	Engine torque and vehicle road loads included	Engine–vehicle coupling with controlled or prescribed pulley clamping
Duan et al. (2016)	Dynamic axial and radial shift	Applied clamping and axial dynamics	Chain elasticity, slip, and pulley deformation	Coupled pulley and chain dynamics	Coupled axial, rotational, contact, and friction dynamics in a chain CVT
Present formulation	Dynamic shift coordinate	Mechanical flyweight primary and helix secondary	Lumped geometry and gross-slip traction model	Engine, belt, pulleys, and vehicle coupled in time	Mechanically actuated, vehicle-coupled transient rubber-belt framework

Table 1.1: Selected CVT formulations most relevant to the present modelling objective. The table summarizes the principal closure choices and contribution of each formulation.

1.4 Aim, Scope, and Contributions

The literature review in Section 1.3 motivates a specific modelling objective: to develop a physically interpretable dynamic formulation for a mechanically actuated dry rubber V-belt CVT coupled to a vehicle drivetrain. The formulation is intended to close the dominant system-level interactions within a single time-domain model, allowing the transmission and vehicle response to evolve together under changing operating conditions. It is not intended to provide the highest-fidelity representation of local rubber-belt contact or a complete efficiency-loss model; instead, it focuses on the coupled mechanisms needed to understand how actuation, shift behaviour, torque transfer, and vehicle loading produce transient CVT behaviour.

1.4.1 Aim

The aim of this work is to derive a coupled nonlinear dynamic model of a mechanically actuated rubber V-belt CVT and its longitudinal vehicle drivetrain. The model extends from the engine-side pulley, through the belt and downstream driveline, to the vehicle response. Pulley speeds, belt transport, torque transfer, axial shift position, and vehicle motion evolve from the governing equations rather than being prescribed independently.

The formulation is intended to describe how the drivetrain responds from specified admissible initial conditions under changing engine torque, road grade, external load, and other time-varying inputs. Primary actuation, secondary torque reaction, belt traction, ratio change, rotational inertia, and vehicle loading are treated as interacting parts of the same system: a change in one mechanism alters the conditions under which the others act. This allows launch, clutching, shifting, backshift, gross slip, engine braking, and vehicle acceleration to be studied as outcomes of a common coupled response.

A further aim is to preserve the physical interpretation of that response. The derivation is organized so that the role of pulley geometry, primary speed-sensitive actuation, secondary torque-reactive actuation, belt traction, rotational inertia, and vehicle loading remains visible throughout the model. Its assumptions are stated explicitly, together with the mechanisms they exclude, so that simulation results can be interpreted in terms of both the physics retained and the limits of the formulation. The goal is therefore not only to predict system behaviour, but also to clarify why a particular response occurs and how a mechanical or operating change propagates through the drivetrain.

1.4.2 Scope

The system considered in this work is a mechanically actuated dry rubber V-belt CVT coupled to a simplified longitudinal vehicle drivetrain. The CVT consists of a speed-sensitive primary pulley, a torque-sensitive secondary pulley, a rubber V-belt, and a downstream fixed-ratio driveline that transmits power to the vehicle wheels. The vehicle is represented through one-dimensional longitudinal motion.

The model includes the following effects:

- variable pulley geometry, belt-length closure, and ratio kinematics;
- a dynamic shift coordinate relating sheave motion to pulley radii, wrap angles, and transmission ratio;
- mechanically generated primary and secondary axial forces;

- flyweight, ramp, spring, helix, and torque-reactive effects within the pulley actuation mechanisms;
- effective inertia associated with axial shift motion;
- explicit rotational dynamics of the primary pulley, belt, and secondary pulley;
- belt–pulley torque transfer subject to traction limits and gross-slip conditions;
- engine torque and engine-side inertia;
- downstream fixed reduction, reflected driveline inertia, and longitudinal vehicle road load; and
- transient response to specified initial conditions and time-varying input histories.

The belt is represented through a lumped geometric and traction model rather than a fully distributed contact formulation. The model determines whether the belt–pulley contacts can support the torque required by the coupled system, but it does not separately resolve detailed belt-slip and loss mechanisms such as creep, radial compliance, shear deformation, seating and unseating, local rubber hysteresis, or thermal effects. Local belt deformation, detailed radial penetration, belt wear, sheave compliance, detailed tribology, and three-dimensional vehicle dynamics are also outside the present scope. Tire–ground contact is assumed to provide sufficient traction for the longitudinal force demanded by the drivetrain; wheel slip and tire-force limits are not represented.

The formulation retains these assumptions explicitly so that their consequences remain clear when interpreting a simulation result. Its structure also permits individual closures to be refined in later work. For example, the traction relation could be replaced by a distributed contact model, belt elasticity or pulley deformation could be introduced, or thermal, efficiency-loss, and tire-traction models could be incorporated while preserving the surrounding drivetrain framework.

1.4.3 Contributions

The primary contribution of this work is the closure of a coupled mechanism chain for a mechanically actuated dry rubber V-belt CVT operating within a vehicle drivetrain. Individual elements of the formulation build on established ideas in CVT and belt-drive modelling. The contribution lies in combining these elements into a physically interpretable time-domain framework in which the response of the transmission and vehicle emerges from their interaction.

The main contributions are as follows:

1. A vehicle-coupled dynamic formulation that closes the speed-sensitive primary mechanism, torque-reactive secondary mechanism, belt-mediated torque transfer, ratio change, pulley dynamics, and longitudinal vehicle response within a single system of governing equations.

2. A dynamic ratio-change and belt-transport treatment in which axial sheave motion, pulley radii, belt geometry, and belt speed evolve as physical states of the system rather than being imposed through a prescribed ratio map or instantaneous equilibrium condition.

3. A traction-limited torque-transfer formulation that compares the torque required for no-slip motion with the torque supportable by the belt–pulley contacts. This permits gross-slip behaviour to be represented when the sticking response is not admissible.

4. A transient drivetrain framework that evolves from specified admissible initial conditions under changing engine torque, road grade, external load, and other time-varying inputs. Launch, clutching, shifting, backshift, engine braking, traction loss, and vehicle acceleration can therefore be studied as responses of the same coupled system.

5. A derivation structure that preserves the physical role of the major mechanisms and states the modelling assumptions explicitly. This supports interpretation of the predicted response and provides a modular basis for future extensions, including distributed belt contact, belt elasticity, pulley deformation, thermal effects, efficiency losses, and tire-traction limits.

Together, these contributions provide a framework for tracing how a change in component geometry, spring loading, actuator parameters, operating conditions, or vehicle load propagates through the CVT. The resulting effect on clamping, traction state, shift behaviour, torque transfer, engine speed, and vehicle acceleration can then be examined within the assumptions of the model.

Chapter 2

Model Setup

2.1 System Definition

Placeholder

2.2 Modeling Assumptions

The following assumptions define the modeling scope and physical idealizations used in the CVT dynamic formulation. These assumptions establish the limits within which the derived equations remain valid.

2.2.1 Material and Structural Idealizations

Assumption 1 (Constant Material Properties)

Material properties of structural components are treated as constant. No temperature-dependent material variation is modeled. Thermal effects are not solved as part of this system.

Assumption 2 (Rigid Structural Components)

All structural components (pulleys, shafts, housing) are modeled as rigid bodies. Elastic deformation under operational loading is neglected. The belt is assumed axially rigid (no longitudinal stretch), while still conforming geometrically around pulleys.

Assumption 3 (Negligible Wear and Surface Evolution)

Component wear, belt degradation, and pulley surface evolution are neglected. The model represents short-term operational behavior.

2.2.2 Contact and Friction Modeling

Assumption 4 (Idealized Belt–Pulley Friction Law)

Belt traction is modeled using a simplified dry-friction law with an explicit stick–slip transition rule. The model distinguishes between gross adhered contact and gross slipping contact at each pulley. Microscopic contact mechanics, rubber hysteresis, local creep, partial slip within the contact patch, lubrication effects, and detailed Stribeck behavior are not explicitly modeled.

Assumption 5 (Constant Effective Friction Coefficient)

The belt–pulley coefficient of friction is treated as constant. It does not evolve with temperature, wear, belt conditioning, surface contamination, normal pressure, or slip speed. Separate static and kinetic friction coefficients may be used, but each is treated as constant during a given simulation.

Assumption 6 (Averaged Contact Distribution)

Contact pressure and traction over the pulley wrap angle are treated as effectively uniform or averaged. Localized edge loading and non-uniform stress distributions are neglected.

Assumption 7 (Gross Tangential Traction Approximation)

Belt–pulley traction is resolved only in the gross tangential direction around each pulley wrap. Radial belt migration through the sheave groove during shift is not modeled as a separate friction demand. Local radial sliding, local creep, and contact redistribution associated with radial belt motion are not resolved.

Assumption 8 (Equal Secondary Sheave-Face Torque Partition)

The torque transmitted from the belt to the secondary pulley is treated as equally shared between the fixed and movable secondary sheave faces. Face-level torque redistribution between the fixed and movable sheaves is not resolved. Changes in torque share due to helix-induced relative rotation, local belt deformation, contact pressure redistribution, and face-level microslip are neglected.

2.2.3 Geometric and Kinematic Idealizations

Assumption 9 (Perfect Axis Alignment)

Primary and secondary pulley axes are assumed perfectly aligned. Misalignment, shaft eccentricity, and geometric runout are neglected.

Assumption 10 (Constant Belt Length)

The belt length is treated as constant. Longitudinal elasticity, transient stretch, permanent creep, and progressive elongation are not modeled. Effective pulley radii are determined solely from geometry and belt-length constraints.

Assumption 11 (Single Belt Transport Speed)

The belt is represented by a single lumped transport speed. Internal belt shear, belt span vibration, local belt stretch, and separate belt speeds on the two sheave faces are not resolved.

Assumption 12 (Uniform Contact Radius Around Wrap)

At any given instant, the belt contacts each pulley at a single, uniform effective radius throughout the entire wrap angle. The contact radius is constant around the circumference of contact, with no spiral or progressive radial variation along the wrap. While the effective radius may evolve with axial shift position, it remains spatially uniform at each time instant.

2.2.4 Dynamic and Inertial Modeling

Assumption 13 (Lumped Inertia Representation)

Rotational inertias of the engine, pulleys, drivetrain, and wheels are represented as lumped equivalent inertias. Distributed shaft flex and torsional compliance are neglected.

Assumption 14 (Negligible Vibrational Effects)

High-frequency vibrational effects, structural resonances, and drivetrain oscillations are neglected. The model captures dominant rigid-body dynamics only.

Assumption 15 (Negligible Parasitic Friction)

Frictional losses in non-critical components (e.g., bearings, seals, auxiliary shafts) are neglected. Only friction directly relevant to belt traction and CVT torque transfer is modeled.

Assumption 16 (No Explicit CVT Loss or Thermal Model)

The model resolves torque transfer, traction admissibility, and slip state. Heat generation, temperature rise, wear, and efficiency loss are not solved. Power dissipated during slip is not propagated into a thermal, wear, or degradation model.

2.2.5 Environmental and Operational Scope

Assumption 17 (Negligible Gravitational Influence on CVT Internals)

Gravitational forces acting on CVT internal components are negligible relative to centrifugal and contact forces at operating speeds.

Assumption 18 (Full-Throttle Engine Model)

The engine is modeled using a prescribed full-throttle torque curve. Load feedback modifies engine speed dynamically, but part-throttle dynamics and throttle transients are not explicitly modeled.

2.3 Reference Material

This section records any information for easy reference.

2.3.1 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions.

Vehicle Longitudinal Coordinate and Incline Angle

Vehicle motion is modeled along a single longitudinal coordinate that follows the surface of the road.

The coordinate x is defined locally along the direction of travel, tangent to the road surface. The positive direction, $+x$, corresponds to the forward direction of vehicle motion.

Because the road may be inclined relative to horizontal, the local direction of the x axis depends on the road slope. The incline angle is denoted by α and is measured between the road surface and a horizontal reference line.

$$\alpha > 0 \quad : \text{uphill in the } +x \text{ direction}$$

$$\alpha < 0 \quad : \text{downhill in the } +x \text{ direction}$$

As illustrated in Figure 2.1, the x direction always lies tangent to the road surface. Consequently, the orientation of the longitudinal axis varies with the local road incline α.

Vehicle velocity is defined along this coordinate. A velocity $v > 0$ corresponds to motion in the forward $+x$ direction along the road, while $v < 0$ corresponds to motion in the opposite direction.

These definitions establish the geometric orientation of vehicle motion and the sign convention for road incline used throughout the model.

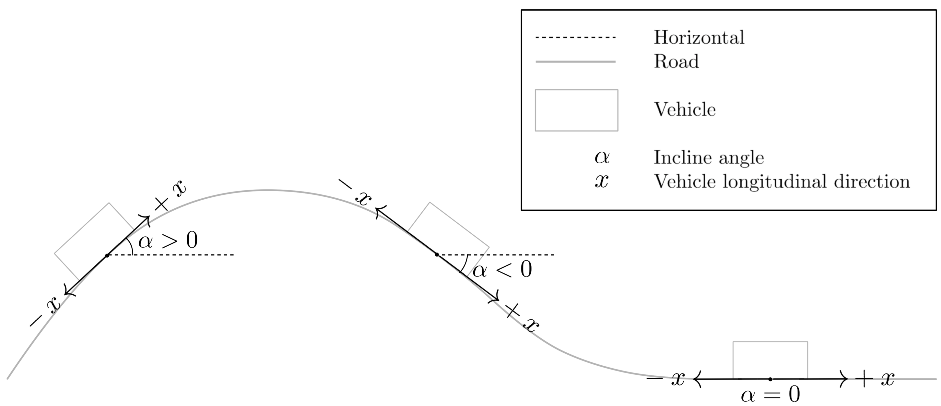


Figure 2.1: Definition of the longitudinal vehicle coordinate and incline angle. The coordinate x is defined tangent to the road surface and points in the forward direction of travel. The road incline α is measured relative to the horizontal reference line. Positive α corresponds to uphill motion in the $+x$ direction, while negative α corresponds to downhill slope.

Rotational Sign Convention

All rotating components in the drivetrain follow a consistent angular sign convention defined relative to forward vehicle motion.

The positive longitudinal direction $+x$ was defined previously as the forward direction of travel along the road surface. The rotational directions of all drivetrain components are chosen such that positive rotation produces vehicle motion in this $+x$ direction.

As illustrated in Figure 2.2, positive wheel angular velocity $\omega_w > 0$ corresponds to wheel rotation that drives the vehicle forward along $+x$.

The same convention is propagated upstream through the drivetrain. Positive rotation of the secondary pulley, $\omega_s > 0$, corresponds to the direction of rotation that produces this forward wheel motion through the final reduction gearbox and axle.

Likewise, positive rotation of the primary pulley, $\omega_p > 0$, corresponds to the direction that drives the secondary pulley in this same sense through the CVT belt.

Engine angular velocity follows the same convention. The engine rotates positively when it drives the primary pulley in the defined positive direction.

Torque follows the same sign convention. A torque is considered positive when it acts in the defined positive rotational direction and therefore tends to increase the corresponding angular velocity.

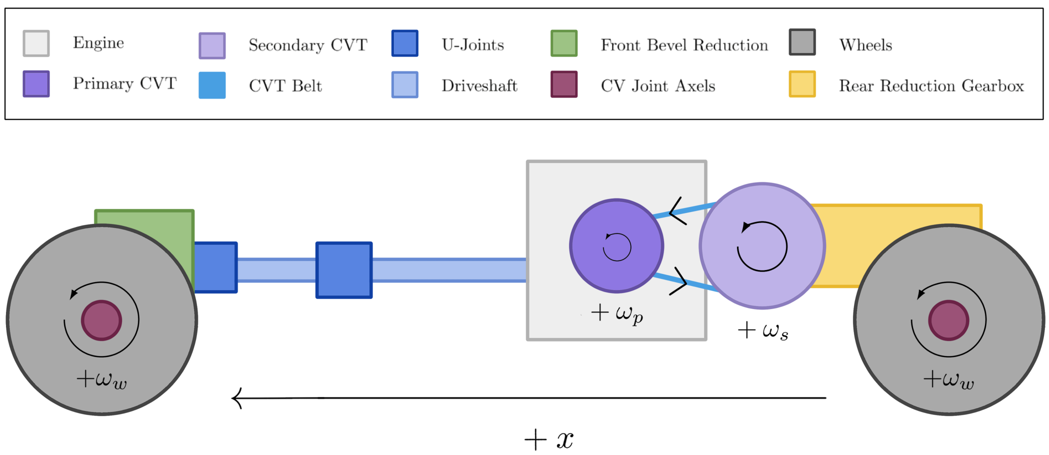


Figure 2.2: Rotational sign convention for the drivetrain. Positive angular velocities for the wheels (ω_w), secondary pulley (ω_s), and primary pulley (ω_p) are defined such that drivetrain rotation produces forward vehicle motion along the $+x$ direction.

CVT Axial Direction

The axial coordinate of the primary pulley is defined along the axis of the pulley shaft, parallel to the pulley centerline and perpendicular to the plane of belt rotation.

In the reference configuration shown in Figure 2.3 (left), the primary pulley is fully open. The movable sheave is separated from the fixed sheave by the initial gap, and this configuration defines the origin of the axial coordinate.

The axial variable s measures the translation of the movable sheave relative to this open configuration.

$$s = 0 \quad : \text{fully open configuration}$$

A positive value of s corresponds to the movable sheave translating toward the fixed sheave along the pulley axis.

$$s > 0 \quad : \text{sheave closing motion}$$

As the movable sheave closes, the belt is squeezed between the conical faces and is forced outward along the pulley ramps. This increases the effective belt radius on the primary pulley and corresponds to an upshift of the transmission.

The maximum displacement $s = S_{\max}$ corresponds to the fully closed configuration shown in Figure 2.3 (right).

Although s is defined using the primary pulley, the motion cannot occur independently. Because the belt length remains constrained, closure of the primary pulley forces the secondary pulley to open. The precise geometric relationship between these motions will be derived later.

Axial velocity follows the same sign convention: $\dot{s} > 0$ corresponds to the movable sheave translating toward the fixed sheave.

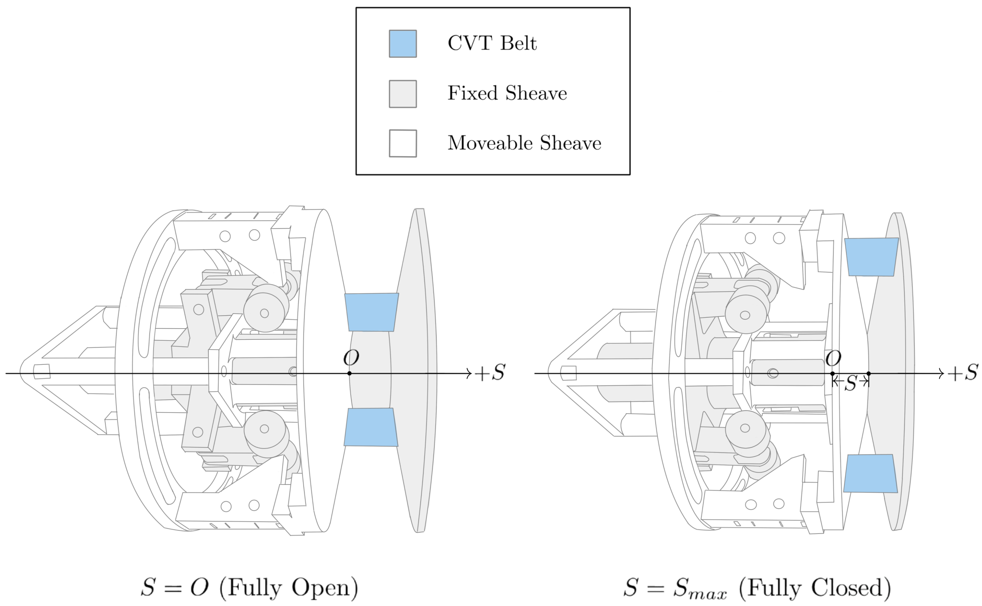


Figure 2.3: Definition of the primary axial coordinate. The system begins in an open configuration. The axial displacement s measures how much the movable sheave has closed from this initial position. Increasing s increases primary effective radius and is coupled to secondary axial motion.

Radial and Tangential Directions

Within the plane of pulley rotation, directions are defined using the standard polar coordinate convention commonly used in rotational mechanics.

At any point in the plane, two orthogonal directions are defined: the radial direction $\hat{e}_r$ and the tangential direction $\hat{e}_\theta$. These directions form a local coordinate basis centered on the pulley axis.

The radial direction lies in the plane of rotation and points directly outward from the center of the pulley. A radial force is defined as positive when it acts away from the center of rotation. Thus, $F_r > 0$ corresponds to a force directed outward along $\hat{e}_r$.

The tangential direction lies in the plane of rotation and is perpendicular to the radial direction. The positive tangential direction $\hat{e}_\theta$ is defined to be consistent with the positive rotational convention introduced in Section 2.3.1. Tangential motion in this direction corresponds to motion produced by positive angular velocity.

Figure 2.4 illustrates these directions using physical examples from the primary pulley. The centrifugal flyweights generate outward radial forces F_r, while the belt moves along the tangential direction with velocity v_t. These quantities serve as representative examples of forces and motion that act along the radial and tangential directions.

The previously defined axial direction is perpendicular to this plane of rotation and points along the pulley shaft, into the page.

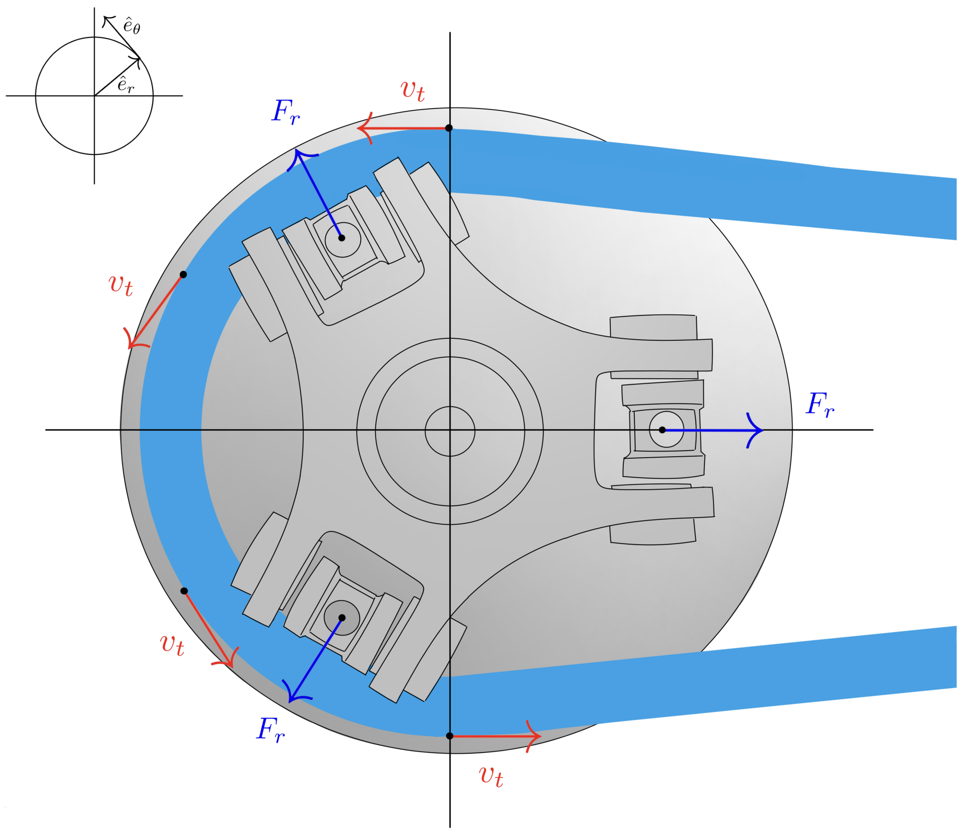


Figure 2.4: Radial and tangential directions in the plane of pulley rotation. The radial direction $\hat{e}_r$ points outward from the center of the pulley, while the tangential direction $\hat{e}_\theta$ is perpendicular to $\hat{e}_r$ and aligned with positive rotation. Example quantities illustrated include outward radial forces F_r generated by flyweights and tangential belt velocity v_t. For reference, the previously defined axial coordinate s is perpendicular to the page in this view and points along the pulley shaft, into the page.

2.4 Governing Principles

Theory 1: Newton's Second Law (Translation)

Equation

$$\sum F = ma$$

Description

- $\sum F$ is the net force (N)
- m is mass (kg)
- a is acceleration (m/s^2)

Source

Standard mechanics identity.

Derivation

From the definition of acceleration and empirical laws of motion; used here as a governing relationship.

Theory 2: Newton's Second Law (Rotation)

Equation

$$\sum \tau = I\alpha$$

Description

- $\sum \tau$ is net torque (N m)
- I is moment of inertia (kg m^2)
- α is angular acceleration (rad/s^2)

Source

Standard rigid-body dynamics identity.

Derivation

Standard rigid-body rotational dynamics relationship.

Theory 3: Torque Definition

Equation

$$\tau = Fr$$

Description

- τ is torque (N m)
- F is tangential force (N)
- r is moment arm radius (m)

Source

Standard mechanics identity.

Derivation

Defines torque as the product of tangential force and moment arm. Equivalently, $F = \tau/r$ gives the tangential force from a known torque.

Theory 4: Kinetic Energy (Translational)

Equation

$$T = \frac{1}{2}mv^2$$

Description

- T is kinetic energy (J)
- m is mass (kg)
- v is velocity (m/s)

Source

Standard mechanics identity.

Derivation

Translational kinetic energy of a point mass or rigid body moving with velocity v.

Theory 5: Centrifugal Force

Equation

$$F_c = m\omega^2 r$$

Description

- F_c is centrifugal force (N)
- m is mass (kg)
- ω is angular velocity (rad/s)
- r is radius from the axis of rotation (m)

Source

Standard circular-motion identity.

Derivation

Derived from circular motion kinematics where $a_c = \omega^2 r$, then applying $F = ma$.

Theory 6: Hooke's Law (Compressional)

Equation

$$F = k\Delta x$$

Description

- F is force (N)
- k is spring constant (N/m)
- Δx is displacement (m)

Source

Standard linear-elastic constitutive identity.

Derivation

Linear spring constitutive relationship in the elastic regime.

Theory 7: Hooke's Law (Torsional)

Equation

$$F = \frac{k\Delta\theta}{r}$$

Description

- F is force (N)
- k is torsional spring constant (N m/rad)
- $\Delta\theta$ is angular deflection (rad)
- r is torque arm radius (m)

Source

Standard torsional spring identity.

Derivation

Starting from $\tau = k\Delta\theta$ and $\tau = Fr$, then $F = \frac{k\Delta\theta}{r}$.

Theory 8: Engine Torque from Power

Equation

$$\tau = \frac{P}{\omega}$$

Description

- τ is torque (N m)
- P is power (W)
- ω is angular velocity (rad/s)

Source

Standard rotational power identity.

Derivation

From the rotational power relationship $P = \tau\omega$.

Theory 9: Coulomb Friction

Equation

$$F_f = \mu_s F_n$$

Description

- F_f is friction force (N)
- μ_s is coefficient of static friction (unitless)
- F_n is normal force (N)

Source

Standard dry-friction model identity.

Derivation

Coulomb friction model for impending slip.

Theory 10: Capstan Equation

Equation

$$T_1 = T_2 e^{\mu\theta}$$

Description

- T_1 is tight-side belt tension (N)
- T_2 is slack-side belt tension (N)
- μ is belt–pulley friction coefficient (unitless)
- θ is wrap angle (rad)

Source

ScienceDirect Topics (n.d.).

Derivation

Differential force balance on an infinitesimal belt element on a pulley yields $\frac{\mathrm{d}T}{T} = \mu\ \mathrm{d}\theta$, then integrate over wrap.

Theory 11: Density

Equation

$$\rho = \frac{m}{V}$$

Description

- ρ is density (kg/m^3)
- m is mass (kg)
- V is volume (m^3)

Source

Standard definition.

Derivation

Definition of density.

Theory 12: Air Resistance (Quadratic Drag)

Equation

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

Description

- F_d is drag force (N)
- ρ is fluid density (kg/m^3)
- v is object speed relative to fluid (m/s)
- C_d is drag coefficient (unitless)
- A is reference area (m^2)

Source

Standard aerodynamic drag model identity.

Derivation

Standard empirical/derived quadratic drag model; applicable at moderate to high Reynolds number.

Theory 13: Rolling Resistance

Equation

$$F_r = C_r N$$

Description

- F_r is rolling resistance force (N)
- C_r is coefficient of rolling resistance (unitless)
- N is normal force (N)

Source

Standard rolling-resistance model identity.

Derivation

Rolling resistance arises from deformation of contact surfaces (tire and road). For a vehicle on level ground, $N = mg$ where m is vehicle mass and g is gravitational acceleration.

Theory 14: Gravitational Force

Equation

$$F_g = mg$$

Description

- F_g is gravitational force (N)
- m is mass (kg)
- g is gravitational acceleration (m/s^2)

Source

Standard near-Earth mechanics identity.

Derivation

Near Earth's surface, $g \approx 9.81$ m/s^2. On an incline with angle θ, the component parallel to the surface is $F_g \sin\theta$.

2.4.1 Model Quantity Roles and Evaluation Flow

The model distinguishes between quantities specified for a CVT configuration, quantities derived once during setup, dynamic states advanced in time, and algebraic quantities evaluated from the current state. Here, *fixed* means fixed during one simulation run. A fixed quantity may still be changed when a different CVT configuration or operating case is defined.

The physical constants and fixed material properties used by the model are listed in **??**, while the user-specified CVT, belt, and drivetrain parameters are listed in **??**. These specified quantities include the belt length, belt cross-section, sheave geometry, and endpoint pulley radii. They define the particular CVT being modeled, but they are not dynamic states.

The center-to-center distance is different. It is not prescribed independently in the present parameterization. Instead, it is derived once from the specified belt length and the known reference pulley radii, as developed in **??**. It is then retained as a fixed geometric quantity for the remainder of the run.

Table 2.1: Roles of quantities in the model evaluation.

Role	**Assigned or evaluated**	**Examples**
Physical constants and fixed properties	Before simulation	g, material densities, and other values treated as invariant within the model scope.
Specified configuration parameters	Before geometry initialization	L_b, b_{in}, h_b, sheave angle β, and endpoint pulley radii.
Derived fixed geometry	Once during geometry initialization	Center-to-center distance C.
Dynamic state variables	Advanced by the ODE integrator	ω_p, ω_s, s, $\dot{s}$, and v_b.
State-dependent algebraic quantities	At each evaluation of the system equations	$r_p(s)$, $r_s(s)$, λ, wrap angles, span length, effective radii, and other current geometry quantities.

The calculation has two levels. Geometry initialization first uses the specified configuration data to establish C. During the simulation, each evaluation of the system equations begins from the current state. The shift coordinate gives the primary radius, the fixed belt-length condition gives the compatible secondary radius, and the remaining belt-path quantities then follow directly from the current geometry:

$$\mathbf{x}(t) \longrightarrow s \longrightarrow r_{p,\text{outer}}(s) \longrightarrow r_{s,\text{outer}}(s) \longrightarrow \lambda,\ \phi_p,\ \phi_s,\ \ell \longrightarrow \text{forces, torques, and } \dot{\mathbf{x}}.$$

The secondary-radius calculation is an algebraic geometry closure within an evaluation of the system equations; it is not an additional dynamic state. Once the two radii are known, λ, the wrap angles, and the straight-span length are evaluated directly. A numerical time integrator may evaluate the system equations multiple times within one time step, so this geometry closure is performed for each such evaluation rather than only once per accepted time step.

2.5 System State Definition

The dynamic behavior of the CVT system is described using a set of *state variables*. A state variable is a quantity whose future evolution is determined by its current value and the governing differential equations of the system.

In numerical simulation, state variables are the quantities that are explicitly integrated forward in time. All other model quantities — such as torque transfer, transmission ratio, belt forces, and contact conditions — are computed algebraically from these states.

2.5.1 State Vector

State Vector **(2.5.1)**

$$\boldsymbol{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ v_b \end{bmatrix}$$

where:

- ω_p is the angular speed of the primary pulley (rad/s),
- ω_s is the angular speed of the secondary pulley (rad/s),
- s is the axial shift position of the primary sheave (m),
- $\dot{s}$ is the axial shift velocity (m/s).
- v_b is the belt transport speed (m/s).

2.5.2 Independent Dynamic Degrees of Freedom

The five state variables above are the dynamic degrees of freedom of the present state-space model.

The primary pulley can rotate independently under engine input and belt interaction. Its angular speed ω_p therefore represents one rotational degree of freedom.

The secondary pulley also rotates and interacts with both the belt and the external drivetrain load. Its angular speed ω_s represents a second independent rotational degree of freedom.

In addition to rotation, the CVT includes axial motion of the primary sheave. This axial displacement s determines the effective pulley geometry and therefore the transmission ratio. The axial motion is mechanically independent of pure rotation and must be modeled explicitly.

Because axial motion follows second-order dynamics, both the axial position s and its rate $\dot{s}$ are required to fully describe its evolution.

The quantity v_b is treated as an independent belt-transport degree of freedom. While detailed internal belt mechanics remain unresolved, v_b is modeled as a dynamic state whose evolution is given by an explicit equation of motion derived from torque, traction, and compatibility relations at the pulley–belt interfaces.

2.5.3 Time Integration Perspective

During numerical simulation, the state vector is advanced forward in time.

At each timestep Δt:

- The angular speeds ω_p and ω_s are updated according to their angular accelerations.

- The axial position s is updated using its current velocity $\dot{s}$.
- The axial velocity $\dot{s}$ is updated according to the axial acceleration.
- The belt transport speed v_b is updated according to its angular acceleration.

The angular and axial accelerations are determined from the governing equations derived in subsequent sections. Once the state vector has been updated, all other quantities in the model are computed directly from the current values of ω_p, ω_s, s, $\dot{s}$, and v_b.

Chapter 3

Derivation

3.1 Pulley Geometry

The conceptual overview introduced the CVT as two variable-radius pulleys connected by a single belt. Moving the pulley sheaves changes where the belt sits on each pulley, and the resulting pair of pulley radii determines the transmission ratio.

Both pulleys contain a movable sheave. Let x_p denote the axial position of the movable primary sheave and x_s the axial position of the movable secondary sheave. The central question of this section is

$$\text{if } x_p,\ x_s \text{ and their first and second time derivatives are known,}$$
$$\text{what are the pulley radii, and how are those radii moving?}$$

Answering this question begins locally. The axial position of each movable sheave must first be related to the belt radius on its own pulley. The two results must then be brought together, since the primary and secondary radii belong to the same belt loop. Considering the complete belt will determine which positions of the two movable sheaves can occur together.

The position-level geometry is developed first. The resulting relationships can then be differentiated to obtain the radius rates and radius accelerations from the motion of the sheaves.

The natural place to begin is the geometry of a single pulley.

3.1.1 Local Pulley Geometry

The first part of the geometric question can be answered on a single pulley: how does the axial position of its movable sheave determine the pulley radius? The same conical geometry appears on both pulleys, so the relationship can be derived once and then applied to the primary and secondary.

The axial direction and primary-sheave convention were introduced in Section 2.3.1 and illustrated in Figure 2.3. The local coordinates x_p and x_s follow the same convention on their respective pulleys.

For either pulley $j \in \{p, s\}$, $x_j = 0$ denotes the fully open configuration. Increasing x_j moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing x_j opens it. The corresponding velocity follows the same sign convention: $\dot{x}_j > 0$ denotes closing motion and $\dot{x}_j < 0$ denotes opening motion.

The primary therefore begins at $x_p = 0$ and moves in the positive direction as it closes. The secondary begins in a closed position, with $x_s > 0$, and moves toward smaller values of x_s as it opens.

Consider a belt seated between two straight sheave faces with half-angle β, as shown in Figure 3.1. Let b_{in} denote the width of the belt at its inner surface. As assumed in **??**, this width remains constant.

Suppose the movable sheave closes the pulley by a distance Δx_j. At the original belt radius, the available space between the sheaves is therefore reduced from b_{in} to

$$b_{\text{in}} - \Delta x_j.$$

The belt responds by moving outward through a radial distance Δr_j. At its new position, the angled sheave faces provide an additional axial distance d on each side.

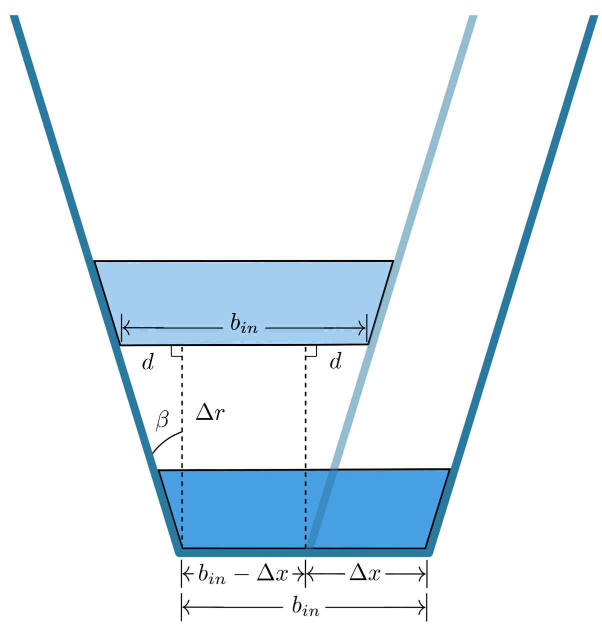


Figure 3.1: Local geometry of a belt moving outward between the sheave faces of pulley j. Closing the pulley by Δx reduces the available width at the original belt position to $b_{\text{in}} - \Delta x$. Moving outward by Δr provides an additional axial distance d at each sheave face.

At the new radius, the total space between the sheave faces must again equal the fixed belt width. Reading directly from Figure 3.1,

$$b_{\text{in}} = 2d + (b_{\text{in}} - \Delta x_j)\,. \tag{3.1.1}$$

The right triangle beside either sheave face relates d to the radial movement. For the sheave half-angle β,

$$\tan\beta = \frac{d}{\Delta r_j}, \tag{3.1.2}$$

and therefore

$$d = \Delta r_j \tan\beta. \tag{3.1.3}$$

Substituting Equation (3.1.3) into Equation (3.1.1) gives

$$b_{\text{in}} = 2\Delta r_j \tan\beta + (b_{\text{in}} - \Delta x_j)\,. \tag{3.1.4}$$

The constant belt-width terms cancel, leaving

$$\Delta r_j = \frac{\Delta x_j}{2\tan\beta}. \tag{3.1.5}$$

This is the basic axial-to-radial relationship of the pulley. Closing the pulley moves the belt outward, while opening it moves the belt inward. A smaller sheave angle produces more radial movement for the same axial motion, whereas a larger sheave angle produces less.

Taking the infinitesimal limit of Equation (3.1.5) gives

$$\frac{dr_j}{dx_j} = \frac{1}{2\tan\beta}. \tag{3.1.6}$$

Because the sheave faces are straight, this slope is constant.

Radius convention The derivation above follows the inner surface of the belt, where b_{in} is defined. The belt also has the finite radial height h_b, so its inner and outer surfaces occupy different radii. Let $r_{j,\text{inner}}$ and $r_{j,\text{outer}}$ denote those radii. They are related by

$$r_{j,\text{outer}} = r_{j,\text{inner}} + h_b. \tag{3.1.7}$$

The fixed-cross-section assumption in **??** also makes h_b constant. Consequently, the inner and outer surfaces undergo the same radial displacement, and the axial-to-radial relationship derived above applies to either one.

An absolute pulley radius must nevertheless identify which belt surface is being measured. The outer surface is used here because it provides a convenient and directly measurable reference. Throughout the geometric development,

$$r_j \equiv r_{j,\text{outer}}. \tag{3.1.8}$$

The derivative in Equation (3.1.6) describes how this radius changes, but one known configuration is needed to establish its absolute value. Let $x_{j,\mathrm{ref}}$ denote a known axial position at which the belt is seated, and let $r_{j,\mathrm{ref}}$ denote the outer pulley radius measured at that same position. The outer radius at any other seated position is then

$$r_j(x_j) = r_{j,\mathrm{ref}} + \frac{x_j - x_{j,\mathrm{ref}}}{2\tan\beta}. \tag{3.1.9}$$

The pair $(x_{j,\mathrm{ref}}, r_{j,\mathrm{ref}})$ simply anchors the linear relationship to a known physical configuration.

For the secondary pulley, a convenient reference is its known closed starting position. The secondary begins with the belt seated at a large radius and opens as the CVT shifts. Since opening corresponds to decreasing x_s, Equation (3.1.9) produces the corresponding decrease in r_s.

The primary introduces an important geometric asymmetry: it can open farther than the width of the belt. At $x_p = 0$, the distance between the primary sheaves can therefore exceed b_{in}. Initial positive motion of the movable primary sheave only removes this clearance, leaving the belt at the same radius. The primary radius begins to increase only once the movable sheave reaches the belt.

This interval is the *primary deadzone*. Let $x_{p,\mathrm{dz}}$ denote the primary position at which the movable sheave first reaches the belt. Within the deadzone,

$$0 \le x_p < x_{p,\mathrm{dz}},$$

the primary coordinate changes, but the belt remains at its minimum primary radius. At $x_p = x_{p,\mathrm{dz}}$, the clearance has been removed and the belt becomes seated between the sheave faces. Further closure then moves the belt outward according to the local geometry derived above.

The first-contact position provides the natural reference for the seated primary branch:

$$x_{p,\mathrm{ref}} = x_{p,\mathrm{dz}}, \qquad r_{p,\mathrm{ref}} = r_{p,\mathrm{min}}.$$

The complete primary radius map is therefore

$$r_p(x_p) = \begin{cases} r_{p,\mathrm{min}}, & 0 \le x_p < x_{p,\mathrm{dz}}, \\ r_{p,\mathrm{min}} + \dfrac{x_p - x_{p,\mathrm{dz}}}{2\tan\beta}, & x_{p,\mathrm{dz}} \le x_p \le x_{p,\mathrm{max}}. \end{cases} \tag{3.1.10}$$

The linear axial-to-radial relationship in Equation (3.1.9) is standard in reduced CVT geometry models **??**. It applies to both pulleys whenever the belt is seated, while Equation (3.1.10) extends the primary description through the deadzone before contact.

The local geometry now tells us the pulley radius associated with either movable-sheave position. To determine which primary and secondary positions can occur together, the two local descriptions must next be joined through the geometry of the complete belt.

3.1.2 Complete Belt-Loop Geometry

The local pulley geometry describes how axial motion changes the radius on each pulley. The belt, however, passes around both pulleys. To connect the two radii, the belt must now be followed around the complete loop.

The belt path consists of four parts: a wrapped arc on the primary pulley, a wrapped arc on the secondary pulley, and two equal straight spans. Under A 12, each wrapped portion is represented by a single radius over its full contact angle. If each straight span has length l, and the primary and secondary wrap angles are ϕ_p and ϕ_s, the total geometric length of the loop is

$$L_{\text{geo}} = r_p\phi_p + r_s\phi_s + 2l. \tag{3.1.11}$$

The quantities l, ϕ_p, and ϕ_s follow directly from the geometry shown in Figure 3.2.

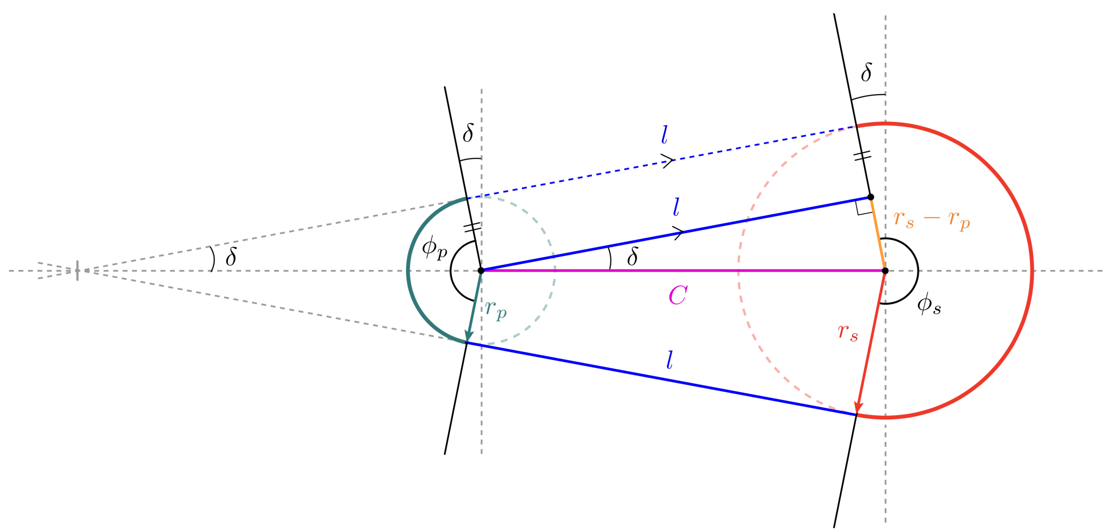


Figure 3.2: Geometry of the complete belt loop. The belt path consists of two wrapped arcs and two equal straight spans of length l. Unequal pulley radii incline the straight spans by the angle δ, producing the primary and secondary wrap angles ϕ_p and ϕ_s.

Consider the right triangle in the upper span in Figure 3.2. The line joining the pulley centres forms its hypotenuse and has length C. One side lies along the straight belt span and therefore has length l.

The other side is perpendicular to the span. The primary and secondary radii drawn to their respective contact points are also perpendicular to this span and are therefore parallel. As shown by the orange construction in Figure 3.2, the difference between their lengths gives the remaining side of the triangle, $r_s - r_p$. The triangle therefore satisfies

$$l^2 + (r_s - r_p)^2 = C^2, \tag{3.1.12}$$

and the length of either straight span is

$$l = \sqrt{C^2 - (r_s - r_p)^2}. \tag{3.1.13}$$

This expression requires

$$|r_s - r_p| < C. \tag{3.1.14}$$

This condition is already enforced by the physical separation of the pulleys. The pulley circles must remain apart, requiring $C > r_s + r_p$, and $r_s + r_p$ is necessarily greater than $|r_s - r_p|$.

The straight-span length accounts for only two parts of the belt path. The two wrapped portions still depend on how the straight spans meet the pulleys. If the pulley radii are equal, the spans are parallel to the line joining the pulley centres and the belt wraps halfway around each pulley. Both wrap angles are then equal to π.

When the radii differ, the straight spans rotate through the angle δ shown in Figure 3.2. The same right triangle gives

$$\sin\delta = \frac{r_s - r_p}{C}, \tag{3.1.15}$$

and therefore

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{3.1.16}$$

The angle δ is signed. For the configuration shown, $r_s > r_p$, so $\delta > 0$. The primary wrap loses an angle δ at each end of its contact arc, while the secondary wrap gains the same amount at each end. Their wrap angles are therefore

$$\phi_p = \pi - 2\delta, \qquad \phi_s = \pi + 2\delta. \tag{3.1.17}$$

If $r_p > r_s$, then $\delta < 0$, and the same equations automatically reverse which pulley has the larger wrap angle.

The complete belt length follows by adding the two wrapped arcs and the two straight spans. Substituting Equations (3.1.13), (3.1.16) and (3.1.17) into Equation (3.1.11) gives

Exact geometric belt length **(3.1.18)**

$$\begin{aligned} L_{\text{geo}}(r_p, r_s, C) = \pi\,(r_p + r_s) \\ + 2\,(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \\ + 2\sqrt{C^2 - (r_s - r_p)^2} \end{aligned}$$

Each term still corresponds directly to part of the belt path. The first is the combined half-circle contribution from the two pulley wraps. The second accounts for the unequal wrap angles, and the third is the combined length of the two straight spans.

For any chosen values of r_p, r_s, and C, Equation (3.1.18) tells us how much belt is required to follow that path. The installed belt cannot change its total length as the pulleys shift. Under A 10, every possible pulley position must therefore satisfy

$$L_{\text{geo}}(r_p, r_s, C) = L_b, \tag{3.1.19}$$

where L_b is the fixed belt length. This is the geometric compatibility condition for the belt loop. Once any two of r_p, r_s, and C are known, the condition determines the third.

For a particular CVT, the first quantity to determine is the centre distance C. The pulley radii change as the sheaves move, but the pulley shafts do not move with them. Once the CVT is assembled, C remains fixed throughout operation.

The fixed centre distance can be found from any installed pulley position that can be measured. At that position, the belt length and the two pulley radii are known. Denoting the measured radii by $r_{p,\text{ref}}$ and $r_{s,\text{ref}}$, the only unknown in Equation (3.1.19) is C.

In the general unequal-radius case, C appears both inside the square root and inside the inverse sine. The exact relation therefore cannot be rearranged into a useful explicit expression for C. Instead, the centre distance is found with a one-dimensional root solve.

For a trial centre distance, the difference between the belt length required by the geometry and the belt length actually available is

$$\mathcal{R}_L(r_p, r_s, C) = L_{\text{geo}}(r_p, r_s, C) - L_b. \tag{3.1.20}$$

A positive residual means that the trial geometry requires more belt than is available. A negative residual means that it requires less. The correct centre distance is the value for which the required and available lengths match:

$$\mathcal{R}_L\left(r_{p,\text{ref}}, r_{s,\text{ref}}, C\right) = 0. \tag{3.1.21}$$

The root solver varies C until this condition is satisfied.

Remark (Simplified Belt-Length Relation)

The exact root solve is required because changing pulley radii alter both the straight-span lengths and the wrap angles. When the radius difference is small relative to the centre distance,

$$\frac{|r_s - r_p|}{C} \ll 1,$$

the straight-span angle is also small:

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

A common simplification is to neglect the resulting wrap-angle correction while retaining the exact straight-span length. The compatibility relation then becomes

$$L_b \approx \pi\left(r_p + r_s\right) + 2\sqrt{C^2 - \left(r_s - r_p\right)^2}.$$

Unlike the exact relation, this approximation can be rearranged directly. For example, the centre distance is

$$C \approx \sqrt{\left(r_s - r_p\right)^2 + \frac{1}{4}\left[L_b - \pi\left(r_p + r_s\right)\right]^2}.$$

The present model retains the exact relation. The effect of using the simplified form is examined in Chapter A.

Once the initial root solve has determined C, both the centre distance and the belt length are known constants. The compatibility condition can then be used to connect the two pulley radii at every subsequent position.

Suppose the primary radius r_p is known. Substituting r_p, C, and L_b into Equation (3.1.20) leaves the secondary radius as the only unknown. Its compatible value is found from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \qquad \text{solved for } r_s. \tag{3.1.22}$$

The resulting r_s is the secondary radius required for the same fixed belt to pass around both pulleys. The calculation also works in the opposite direction. If r_s is known instead, the compatible primary radius follows from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \qquad \text{solved for } r_p. \tag{3.1.23}$$

Thus, once C has been established, either pulley radius can be used to determine the other. The two radii remain local geometric quantities, but they cannot vary independently while belonging to the same fixed-length belt loop.

The arc-and-tangent construction used above is standard in belt-drive geometry. Budynas and Nisbett (2015) presents the same underlying construction for relating pulley radii, straight spans, wrap angles, and total belt length. The exact geometry is retained here because it provides the compatibility relation between the two shifting pulleys.

The local relations $r_p(x_p)$ and $r_s(x_s)$ now describe how each pulley produces its radius, while Equation (3.1.19) describes which pairs of radii can belong to the same belt loop. Combining these results gives the relationship between the two local axial coordinates.

3.1.3 From Local Pulley Coordinates to the Shift Coordinate

The preceding geometry gives a local relation for each pulley: x_p determines $r_p(x_p)$, and x_s determines $r_s(x_s)$. A complete CVT configuration, however, cannot be formed by choosing these positions independently. Both radii must belong to the same belt loop of length L_b. The remaining problem is therefore to identify the pairs (x_p, x_s) that satisfy this shared geometry.

For any proposed pair, the local pulley relations first give the two radii. Substituting those radii into the belt-length residual gives

$$\mathcal{R}_x(x_p, x_s) \equiv \mathcal{R}_L(r_p(x_p), r_s(x_s), C) = 0. \tag{3.1.24}$$

Every pair that makes Equation (3.1.24) equal to zero is a possible position of the two pulleys. Together, these pairs form the compatibility curve.

To reveal the shape of this curve, either local coordinate could be chosen as a starting point. Take x_p simply as an arbitrary choice. For each value of x_p, the primary relation gives $r_p(x_p)$. The shaft spacing fixes C, and L_b remains fixed by A 10, so the belt-length residual can then be solved for the required secondary radius r_s. The local secondary relation finally gives the corresponding position x_s:

$$x_p \longrightarrow r_p(x_p) \longrightarrow r_s \longrightarrow x_s.$$

Repeating these steps across the primary travel produces the representative compatibility curve in Figure 3.3.

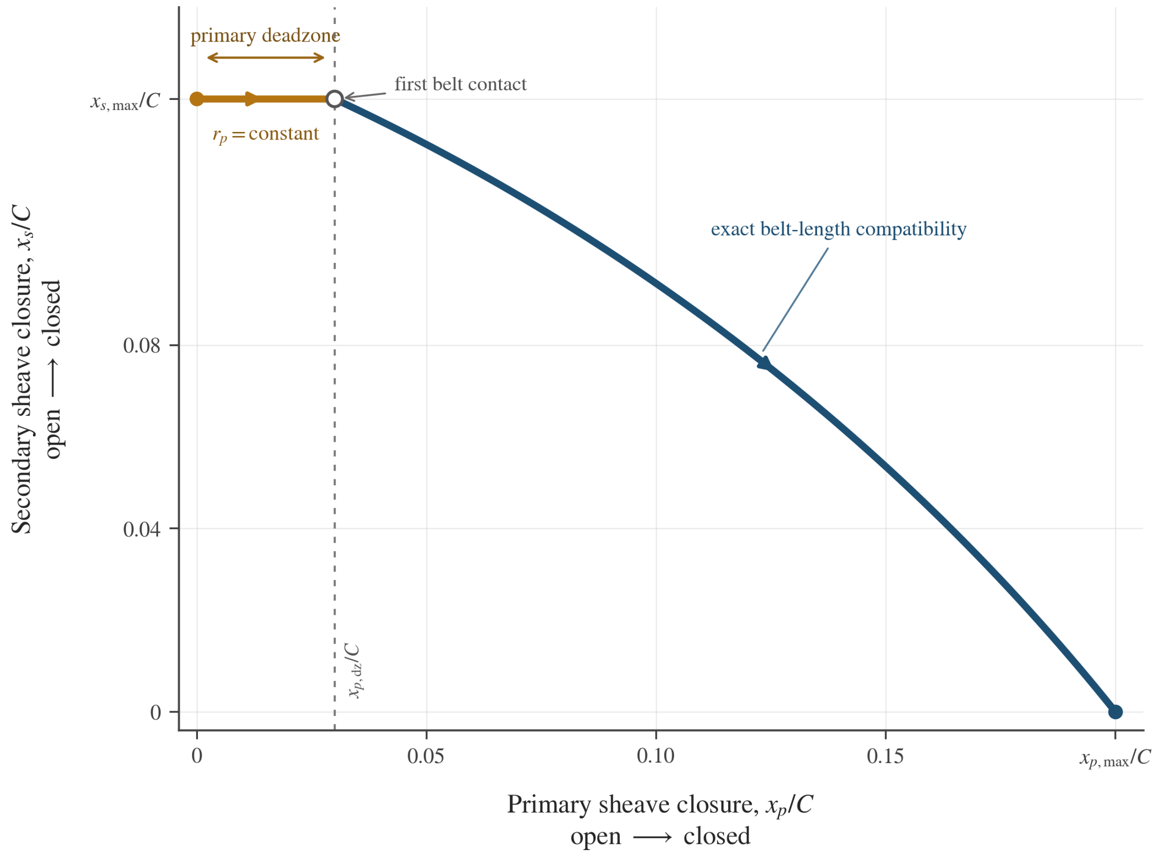


Figure 3.3: Representative compatibility curve between the local primary and secondary axial coordinates for an admissible belt–pulley geometry. Both coordinates are normalized by the same centre distance C, so their relative travel and the changing slope of the engaged branch are shown on a common scale. The dimensions are illustrative and do not represent a particular CVT.

The horizontal portion is created by the primary deadzone. While $0 \leq x_p < x_{p,\text{dz}}$, the primary sheave closes only the clearance to the belt. The primary radius remains fixed, so the compatible secondary radius and position remain fixed as well.

Once the primary reaches the belt, further closure increases r_p. The fixed belt loop then requires r_s to decrease, which means that the secondary opens and x_s decreases. The two sheaves do not move by equal amounts, and the resulting branch is not linear. Changing the radii alters both wrap lengths and both straight spans, so the compatible secondary displacement changes throughout the shift.

The curve also reveals which local coordinate can represent the complete geometry. A vertical line at any x_p meets the curve once, giving one compatible x_s. At the deadzone height, however, a horizontal line meets every x_p between 0 and $x_{p,\text{dz}}$. Thus x_s is a unique function of x_p over the full travel, whereas x_p is not a unique function of x_s.

The primary coordinate is therefore adopted as the system shift coordinate. This is the same s included in the state vector of Section 2.5.1:

System shift coordinate **(3.1.25)**

$$s \equiv x_p, \qquad x_p(s) = s, \qquad x_s = x_s(s)$$

The secondary mapping is then

$$x_s(s) = \begin{cases} x_{s,\max}, & 0 \le s < x_{p,\mathrm{dz}}, \\ \underset{0 \le x_s \le x_{s,\max}}{\mathrm{root}} \left[\mathcal{R}_x(s, x_s) = 0 \right], & x_{p,\mathrm{dz}} \le s \le x_{p,\max}. \end{cases} \tag{3.1.26}$$

The first branch holds the secondary fixed while the primary closes its clearance; the second returns the unique compatible position after belt contact. Both give $x_{s,\max}$ at $s = x_{p,\mathrm{dz}}$, so the mapping is continuous. The positions of both movable sheaves are now determined by s; the remaining kinematic question is how they move when s changes with time.

3.1.4 Pitch Radius and Geometric CVT Ratio

The two outer radii now locate the belt within both pulley grooves and define the outer belt path. This is sufficient for the belt-length geometry, including the wrap angles and straight-span length derived in **??**.

The next quantities of interest concern belt motion and torque transfer. For these purposes, the outer belt surface is not the appropriate reference. The longitudinal tension in a rubber V-belt is carried by its reinforcement cords, so the radius associated with belt transport and the tension moment arm lies at the cord layer rather than at the outer surface.

The radius of the cord layer is commonly called the *pitch radius*, or *cord radius*. Gerbert (1996) identifies the pitch radius with the cord radius and uses it to relate belt motion and transmitted torque.

Let d_{cord} denote the fixed radial distance from the outer belt surface to the centre of the cord layer, where

$$0 \le d_{\mathrm{cord}} \le h_b.$$

The primary and secondary pitch radii are therefore

$$r_{p,\mathrm{pitch}}(s) = r_{p,\mathrm{outer}}(s) - d_{\mathrm{cord}}, \tag{3.1.27}$$

$$r_{s,\mathrm{pitch}}(s) = r_{s,\mathrm{outer}}(s) - d_{\mathrm{cord}}. \tag{3.1.28}$$

Because the same belt offset is applied at both pulleys,

$$r_{s,\mathrm{pitch}}(s) - r_{p,\mathrm{pitch}}(s) = r_{s,\mathrm{outer}}(s) - r_{p,\mathrm{outer}}(s).$$

The tangency angle, wrap angles, and straight-span length are therefore unchanged by this change of reference radius. The pitch radii are introduced only to describe the belt's transport motion and its torque moment arms.

The two pitch radii also describe the effective gearing associated with the belt's current position. Define the geometric CVT ratio as

Geometric CVT Ratio **(3.1.29)**

$$R_g(s) = \frac{r_{s,\text{pitch}}(s)}{r_{p,\text{pitch}}(s)}.$$

This ratio describes the pulley leverage made available by the current belt position. A large secondary pitch radius together with a small primary pitch radius gives a large reduction ratio. As the primary sheave closes, the belt moves outward on the primary and inward on the secondary, causing $R_g(s)$ to decrease.

Unlike a pair of rigid meshed gears, however, the belt does not automatically transfer motion and torque through this exact ratio. The geometric ratio describes the gearing available from the pulley geometry; whether that gearing is realized depends on the belt maintaining sufficient traction at the pulley contacts.

3.1.5 Pulley-Radius Rates and Accelerations

The geometry above determines where the belt sits on each pulley at a given shift position s. When the primary sheave begins to move, those belt positions must move as well. The next quantities to establish are therefore the primary and secondary radial rates, $\dot{r}_p$ and $\dot{r}_s$, followed by their radial accelerations, $\ddot{r}_p$ and $\ddot{r}_s$.

The fixed belt-length geometry was written using the outer belt radii, while belt transport and the geometric CVT ratio use the pitch radii. For the rate relations, that distinction disappears. For either pulley $j \in \{p, s\}$,

$$r_{j,\text{pitch}} = r_{j,\text{outer}} - d_{\text{cord}}. \tag{3.1.30}$$

The cord offset is fixed for the selected belt. Differentiating with respect to s or time therefore removes it:

$$\frac{dr_{j,\text{pitch}}}{ds} = \frac{dr_{j,\text{outer}}}{ds}, \tag{3.1.31}$$

$$\dot{r}_{j,\text{pitch}} = \dot{r}_{j,\text{outer}}, \tag{3.1.32}$$

$$\ddot{r}_{j,\text{pitch}} = \ddot{r}_{j,\text{outer}}. \tag{3.1.33}$$

Thus, the same radial rates and accelerations apply whichever reference surface is used. To avoid carrying repeated subscripts, r_p and r_s denote the primary and secondary radii throughout the remainder of this section.

Primary Radius Rate and Acceleration

The primary radius is already known explicitly from the primary sheave geometry. Recalling **??**,

$$r_p(s) = \begin{cases} r_{p,\min}, & 0 \le s < s_{\mathrm{dz}}, \\ r_{p,\min} + \dfrac{s - s_{\mathrm{dz}}}{2\tan\beta}, & s_{\mathrm{dz}} \le s \le s_{\max}. \end{cases} \tag{3.1.34}$$

The primary radius changes in time only because the shift coordinate changes in time. Using the chain rule,

$$\dot{r}_p = \frac{dr_p}{ds}\dot{s}. \tag{3.1.35}$$

The derivative dr_p/ds comes directly from Equation (3.1.34). In the dead zone, changing s does not yet move the belt radially. Once the sheave faces engage the belt, the conical geometry gives a constant radial change for each unit of axial shift:

$$\frac{dr_p}{ds} = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ \dfrac{1}{2\tan\beta}, & s_{\mathrm{dz}} < s \le s_{\max}. \end{cases} \tag{3.1.36}$$

Substituting this result into Equation (3.1.35) gives

Primary Radius Rate **(3.1.37)**

$$\dot{r}_p = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ \dfrac{\dot{s}}{2\tan\beta}, & s_{\mathrm{dz}} < s \le s_{\max}. \end{cases}$$

To obtain the radial acceleration, differentiate Equation (3.1.35) once more. This is a product of dr_p/ds and $\dot{s}$, so use the product rule:

$$\begin{aligned} \ddot{r}_p &= \frac{d}{dt}\left(\frac{dr_p}{ds}\dot{s}\right) \\ &= \frac{d}{dt}\left(\frac{dr_p}{ds}\right)\dot{s} + \frac{dr_p}{ds}\ddot{s} \\ &= \frac{d^2 r_p}{ds^2}\dot{s}^2 + \frac{dr_p}{ds}\ddot{s}. \end{aligned} \tag{3.1.38}$$

Within either smooth branch of Equation (3.1.34), the slope dr_p/ds is constant. Hence $d^2r_p/ds^2 = 0$, leaving

Primary Radius Acceleration **(3.1.39)**

$$\ddot{r}_p = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ \dfrac{\ddot{s}}{2\tan\beta}, & s_{\mathrm{dz}} < s \le s_{\max}. \end{cases}$$

Remark (Dead-Zone Boundary)

The idealized primary-radius map changes slope at $s = s_{\mathrm{dz}}$, so its ordinary second derivative is not defined at that single point. The expressions above apply within the dead-zone and active-shift branches; the boundary is handled as a regime transition.

Secondary Radius Rate

The secondary radius is not prescribed directly by the shift coordinate. It is the radius required to keep the fixed belt compatible with the current primary radius. Its rate must therefore come from differentiating the belt-length compatibility condition itself.

Recall the residual from Equation (3.1.20):

$$\begin{aligned} \mathcal{R}_{\mathrm{belt}}(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \\ + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b = 0. \end{aligned} \tag{3.1.40}$$

Here, C and L_b remain fixed, while r_p and r_s can both vary with time. Using the multivariable chain rule,

$$\frac{\partial \mathcal{R}_{\mathrm{belt}}}{\partial r_p}\dot{r}_p + \frac{\partial \mathcal{R}_{\mathrm{belt}}}{\partial r_s}\dot{r}_s = 0. \tag{3.1.41}$$

The two partial derivatives tell us how the required belt-path length changes when one pulley radius changes while the other radius and the center distance are held fixed. We find them next.

Partial derivative with respect to r_p Start with the inverse-sine contribution. It is a product, so its derivative contains one term from differentiating $r_s - r_p$ and a second from differentiating the inverse-sine factor:

$$\begin{aligned} &\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] \\ &= \frac{\partial (r_s - r_p)}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \end{aligned} \tag{3.1.42}$$

For the first factor, r_s is held fixed:

$$\frac{\partial (r_s - r_p)}{\partial r_p} = -1. \tag{3.1.43}$$

For the inverse-sine factor, the chain rule gives

$$\begin{aligned}\frac{\partial}{\partial r_p} \sin^{-1}\left(\frac{r_s - r_p}{C}\right) &= -\frac{1/C}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \\ &= -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}.\end{aligned} \tag{3.1.44}$$

Substituting the two derivatives into Equation (3.1.42) gives

$$\begin{aligned}&\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] \\ &= -\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}.\end{aligned} \tag{3.1.45}$$

The remaining nonlinear contribution comes from the straight spans. Using the chain rule,

$$\begin{aligned}&\frac{\partial}{\partial r_p}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] \\ &= 2\left[\frac{1}{2\sqrt{C^2 - (r_s - r_p)^2}}\right][2(r_s - r_p)] \\ &= \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}.\end{aligned} \tag{3.1.46}$$

We can now assemble the derivative of the full residual. The terms involving the straight-span square root appear with equal magnitude and opposite sign:

$$\begin{aligned}\frac{\partial \mathcal{R}_{\text{belt}}}{\partial r_p} &= \pi + 2\left[-\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right] + \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \\ &= \phi_p.\end{aligned} \tag{3.1.47}$$

Partial derivative with respect to r_s The calculation with respect to r_s follows the same steps. The only difference is that increasing r_s increases the radius difference $r_s - r_p$, rather than decreasing it. Thus,

$$\frac{\partial (r_s - r_p)}{\partial r_s} = 1, \qquad \frac{\partial}{\partial r_s} \sin^{-1}\left(\frac{r_s - r_p}{C}\right) = \frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{3.1.48}$$

Using these signs in the same product-rule calculation gives

$$\frac{\partial}{\partial r_s}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{3.1.49}$$

The straight-span contribution changes sign as well:

$$\frac{\partial}{\partial r_s}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = -\frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{3.1.50}$$

Assembling the full derivative gives

$$\begin{aligned}
\frac{\partial \mathcal{R}_{\text{belt}}}{\partial r_s} &= \pi + 2\left[\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right] - \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\
&= \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \\
&= \phi_s.
\end{aligned} \tag{3.1.51}$$

Again, the square-root terms cancel. The two sensitivities are therefore the two wrap angles:

$$\frac{\partial \mathcal{R}_{\text{belt}}}{\partial r_p} = \phi_p, \qquad \frac{\partial \mathcal{R}_{\text{belt}}}{\partial r_s} = \phi_s. \tag{3.1.52}$$

Substituting these results into Equation (3.1.41) gives

$$\phi_p \dot{r}_p + \phi_s \dot{r}_s = 0. \tag{3.1.53}$$

This says that, under fixed belt length, the two radial motions are weighted by their current wrap angles. Solving for the secondary radius rate,

$$\dot{r}_s = -\frac{\phi_p}{\phi_s}\dot{r}_p. \tag{3.1.54}$$

Finally, inserting the primary radius rate from Equation (3.1.37) gives

Secondary Radius Rate **(3.1.55)**

$$\dot{r}_s = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}\dfrac{\dot{s}}{2\tan\beta}, & s_{\mathrm{dz}} < s \le s_{\max}. \end{cases}$$

The negative sign is the geometric consequence of the fixed belt length: as the belt moves outward on the primary pulley, it must move inward on the secondary pulley.

Secondary Radius Acceleration

The secondary radius rate is not constant while the CVT shifts, because the wrap angles themselves change as the belt moves. To find $\ddot{r}_s$, first find the rates of the wrap angles appearing in Equation (3.1.53).

The two wrap angles are determined by the tangency angle,

$$\lambda = \sin^{-1}\left(\frac{r_s - r_p}{C}\right).$$

Using the chain rule,

$$\dot{\lambda} = \frac{\dot{r}_s - \dot{r}_p}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{3.1.56}$$

Since $\phi_p = \pi - 2\lambda$ and $\phi_s = \pi + 2\lambda$,

$$\dot{\phi}_p = -2\dot{\lambda} = \frac{2(\dot{r}_p - \dot{r}_s)}{\sqrt{C^2 - (r_s - r_p)^2}}, \tag{3.1.57}$$

$$\dot{\phi}_s = 2\dot{\lambda} = -\frac{2(\dot{r}_p - \dot{r}_s)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{3.1.58}$$

Now differentiate the rate balance, $\phi_p\dot{r}_p + \phi_s\dot{r}_s = 0$. Each term is the product of a changing wrap angle and a changing radius rate, so use the product rule on both:

$$\begin{aligned} 0 &= \frac{d}{dt}\left(\phi_p\dot{r}_p + \phi_s\dot{r}_s\right) \\ &= \dot{\phi}_p\dot{r}_p + \phi_p\ddot{r}_p + \dot{\phi}_s\dot{r}_s + \phi_s\ddot{r}_s. \end{aligned} \tag{3.1.59}$$

Substituting the wrap-angle rates from Equations (3.1.57) and (3.1.58) gives

$$
\begin{aligned}
0 &= \frac{2(\dot{r}_p - \dot{r}_s)}{\sqrt{C^2 - (r_s - r_p)^2}}\dot{r}_p + \phi_p \ddot{r}_p - \frac{2(\dot{r}_p - \dot{r}_s)}{\sqrt{C^2 - (r_s - r_p)^2}}\dot{r}_s + \phi_s \ddot{r}_s \\
&= \phi_p \ddot{r}_p + \phi_s \ddot{r}_s + \frac{2(\dot{r}_p - \dot{r}_s)^2}{\sqrt{C^2 - (r_s - r_p)^2}}.
\end{aligned} \tag{3.1.60}
$$

The final term is present because a changing radius changes the wrap angles as well as the radius itself. Solving for the secondary radial acceleration gives

$$
\ddot{r}_s = -\frac{\phi_p}{\phi_s}\ddot{r}_p - \frac{2}{\phi_s \sqrt{C^2 - (r_s - r_p)^2}} \left(\dot{r}_p - \dot{r}_s\right)^2. \tag{3.1.61}
$$

The remaining difference of radius rates can be written using the result already obtained in Equation (3.1.54):

$$
\begin{aligned}
\dot{r}_p - \dot{r}_s &= \dot{r}_p + \frac{\phi_p}{\phi_s}\dot{r}_p \\
&= \frac{\phi_p + \phi_s}{\phi_s}\dot{r}_p \\
&= \frac{2\pi}{\phi_s}\dot{r}_p.
\end{aligned} \tag{3.1.62}
$$

Substituting this into Equation (3.1.61) gives

$$
\ddot{r}_s = -\frac{\phi_p}{\phi_s}\ddot{r}_p - \frac{8\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}}\dot{r}_p^2. \tag{3.1.63}
$$

Finally, insert the primary radius rate and acceleration from Equations (3.1.37) and (3.1.39):

Secondary Radius Acceleration **(3.1.64)**

$$
\ddot{r}_s = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\[2ex] -\dfrac{\phi_p}{\phi_s}\dfrac{\ddot{s}}{2\tan\beta} - \dfrac{2\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}\tan^2\beta}\dot{s}^2, & s_{\mathrm{dz}} < s \le s_{\max}. \end{cases}
$$

Remark (Rate of the Geometric CVT Ratio)

The individual pulley-radius rates and accelerations are the quantities needed in the subsequent development. They can also be combined to determine how quickly the geometric CVT ratio changes. This result is not required in the main derivation, but it gives a useful design-level view of how sheave angle and current pulley geometry shape the available ratio-change rate. The derivation is given in Chapter D.

Relation to prior transient models The radius-rate and radius-acceleration relations above follow from the particular global belt-path representation adopted here: fixed belt length and center-to-center distance, one nominal radius over each pulley contact arc, and straight tangent spans between the pulleys. Within this representation, the shift coordinate s determines the primary radius, while the belt-length condition determines the compatible secondary radius. Differentiating that same condition then gives the coupled radius rates and accelerations required by the remainder of the model.

Duan et al. (2016) develop a closely related chain-CVT geometry, including the open-belt arc-and-tangent path and the wedge relation between movable-sheave axial motion and nominal radius motion. Their formulation retains an additional elastic path-length degree of freedom: chain extension and its rate contribute to span tension, while pulley deformation produces a local difference between the nominal pulley radius and the radius followed by an individual chain element. The present formulation instead holds the nominal belt path at fixed length and does not resolve a spatial deformation field. Within those assumptions, it provides the compatible nominal radius pair and its rates directly from

$$s,\ \dot{s},\ \ddot{s} \quad \longrightarrow \quad r_p,\ \dot{r}_p,\ \ddot{r}_p,\ r_s,\ \dot{r}_s,\ \ddot{r}_s.$$

The corresponding tradeoff is that elastic path-length change and deformation-induced local radius variation are excluded from this geometric layer.

Rubber-belt transient models use more than one representation of radius evolution. Analytical shift-mechanics models, such as those of Sorge (2008) and Cammalleri and Sorge (2009), describe local belt migration through a pulley wrap, coupling radial motion to circumferential transport, sliding, and contact-region mechanics. Discretized rubber-belt models, such as those of Juli'o and Plante (2011) and Ballew (2015), obtain the evolving belt geometry from the motion and contact state of individual belt elements. Those approaches can resolve local belt deformation, tension redistribution, and contact motion in greater detail, but require additional spatial or nodal states and associated contact laws.

The present model does not attempt to reproduce that internal belt trajectory. It instead uses the global belt-path geometry to provide the compatible nominal pulley radii and their derivatives from the single primary shift coordinate. This lower-order closure is useful here because those quantities enter directly into the subsequent belt-transport, traction, and rotational equations.

3.1.6 Section Conclusion

The pulley geometry is now closed by a single shift coordinate. At any admissible value of s, the primary sheave geometry fixes the primary belt radius, while fixed belt length determines the compatible secondary radius. Together, these radii determine the belt path, the pitch radii, and the geometric ratio made available by the current pulley arrangement.

The same geometry also determines how these quantities move when the shift coordinate changes. A primary sheave displacement produces a corresponding primary radial motion, while the fixed belt length constrains the secondary radius to respond with it. The resulting radius rates and accelerations are therefore not separate assumptions; they follow from the same belt-path geometry that determines the instantaneous ratio.

This completes the geometric part of the model. The remaining development can treat the pulley radii, belt path, and their motion as a consistent consequence of the current shift state, rather than as independent quantities.

3.2 Pulley Axial Force Models

The geometric relations developed in the previous section define how the transmission ratio varies with axial shift. What remains is to determine *why* the shift coordinate $s(t)$ changes in time.

Axial motion of the sheaves is driven by clamping forces generated within the primary and secondary assemblies. On the primary side, centrifugal flyweights and springs produce an inward axial force that tends to close the sheaves and increase radius. On the secondary side, torque reaction and spring preload generate an opposing axial force.

The evolution of the shift coordinate is governed by the net axial force acting on the movable sheave assembly. Applying Newton's second law in the axial direction (Theory 1 (Newton's Second Law (Translation))) gives

$$m\,\ddot{s} = F_{p,\mathrm{ax}} - F_{s,\mathrm{ax}}, \tag{3.2.1}$$

where $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ denote the axial forces generated by the primary and secondary mechanisms, respectively, and m is the effective axial inertia of the moving components.

The purpose of this section is to derive expressions for $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ as functions of the current system state. Once these force models are established, the shift acceleration $\ddot{s}$ follows directly, completing the dynamic description of axial motion.

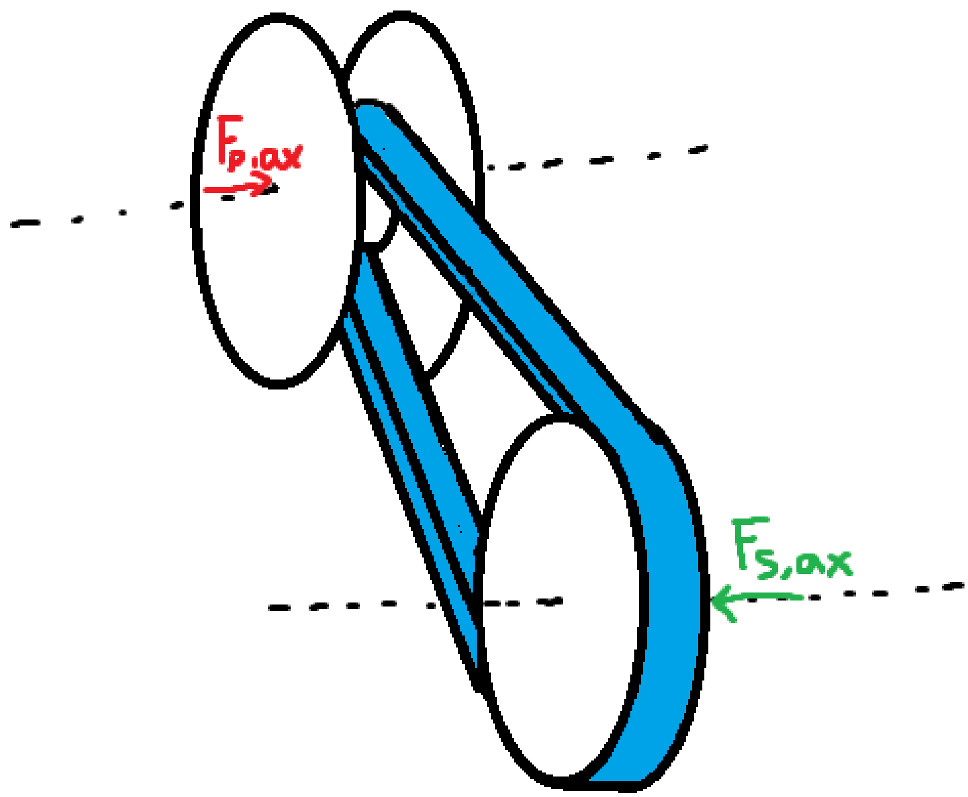


Figure 3.4: Axial force balance governing sheave motion. Primary and secondary mechanisms generate opposing axial forces. The net force determines the shift acceleration $\ddot{s}$.

Although the specific actuator mechanisms modeled here correspond to a flyweight-driven primary and a torque-reactive helix secondary, the formulation is modular: any actuator model may be substituted provided it returns the net axial force as a function of state.

3.2.1 Ramp Geometry Parameterization

Both pulley actuators redirect forces through inclined contact surfaces. As the shift coordinate s evolves, mechanical elements slide along these surfaces, converting radial or circumferential forces into axial clamping force.

To keep the formulation general and modular, the ramp geometries are parameterized directly as functions of the axial shift coordinate s. Any actuator profile may therefore be substituted provided it supplies the appropriate geometric mapping and derivative.

Primary Ramp Geometry (Axial-to-Radial Mapping) On the primary pulley, centrifugal force acting on the flyweights is redirected into axial closing force through a ramp surface (see Figure 3.5). As the movable sheave translates axially, the flyweight slides along this ramp.

We represent the ramp by an axial-to-radial profile

$$r_f = r_f(s), \tag{3.2.2}$$

where $r_f(s)$ is the effective flyweight radius.

An incremental axial displacement ds produces a radial displacement dr_f. The local slope of the ramp is therefore dr_f/ds. The instantaneous primary ramp angle is denoted by $\alpha_p(s)$ and is defined with respect to the axial direction as

$$\tan \alpha_p(s) = \frac{dr_f}{ds}, \qquad \alpha_p(s) = \tan^{-1}\left(\frac{dr_f}{ds}\right). \tag{3.2.3}$$

This definition follows directly from the geometry of the sliding motion: a steeper ramp (larger dr_f/ds) produces greater radial change per unit axial travel and therefore greater axial gain when radial force is redirected through the surface.

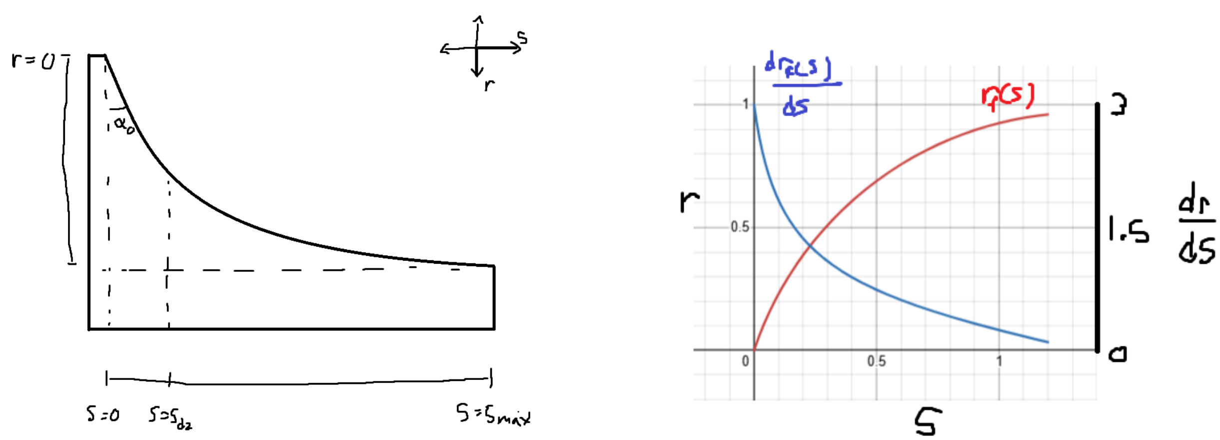


Figure 3.5: Primary ramp geometry. **Left:** As axial displacement s increases, the flyweight slides along the ramp, producing radial change dr_f. The instantaneous ramp angle satisfies $\tan \alpha_p = dr_f/ds$. **Right:** Example profiles of $r_f(s)$ and its slope dr_f/ds.

Primary Admissibility Conditions In order to be a valid primary ramp profile $r_f(s)$, it must satisfy:

1. $r_f(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{dr_f}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{dr_f}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the ramp angle $\alpha_p(s)$ is well-defined and continuous. Condition (2) guarantees that increasing axial displacement does not reduce flyweight radius. Condition (3) ensures the ramp angle remains strictly below $\pi/2$ and the axial gain $\tan \alpha_p$ remains finite.

Secondary Helix Geometry (Axial-to-Rotation Mapping) On the secondary pulley, axial travel produces relative rotation of a helical ramp mechanism, loading a torsional spring and generating torque-reactive clamping (see Figure 3.6).

We parameterize this kinematically as

$$\theta_s = \theta_s(s), \tag{3.2.4}$$

where $\theta_s(s)$ is the helix-induced rotation.

An incremental rotation $d\theta_s$ corresponds to a circumferential displacement $r_h d\theta_s$ along a cylindrical surface of effective radius r_h. The instantaneous secondary helix angle is denoted by $\alpha_s(s)$ and is defined from the local slope of the ramp surface:

$$\tan\alpha_s = \frac{\text{axial rise}}{\text{circumferential run}} = \frac{ds}{r_h d\theta_s}.$$

Rewriting in differential form gives

$$\tan\alpha_s(s) = \frac{1}{r_h \dfrac{d\theta_s}{ds}}. \tag{3.2.5}$$

Thus, a shallower helix (smaller α_s) produces greater rotation per unit axial travel, while a steeper helix produces less rotation. This inverse relationship arises naturally from the geometry of motion on a cylindrical surface.

Remark (Contrast with Primary Ramp)

The primary ramp behaves like a simple wedge: radial displacement is directly proportional to axial travel, so the ramp angle is naturally defined by $\tan\alpha_p = dr_f/ds$.
The secondary mechanism instead behaves like a power screw. Axial motion is produced through rotation around a cylindrical surface, so the relevant geometric ratio is how much rotation occurs per unit axial travel. This leads to the slope definition $ds/(r_h d\theta_s)$ and the inverse relationship in Equation (3.2.5).
In short, the primary redirects force through a wedge, while the secondary converts torque through a screw-like helix. The difference is purely geometric and reflects the distinct mechanisms by which axial motion is generated.

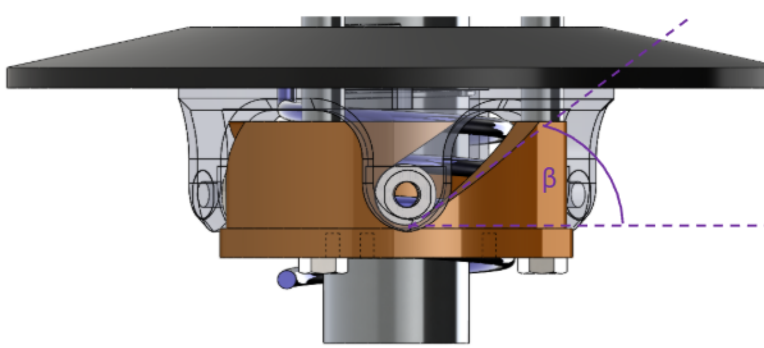


Figure 3.6: Secondary helix geometry. Axial travel s induces rotation $\theta_s(s)$ around a cylindrical surface of radius r_h. The instantaneous helix angle satisfies $\tan\alpha_s = ds/(r_h d\theta_s)$.

Secondary Admissibility Conditions The helix rotation mapping $\theta_s(s)$ must satisfy:

1. $\theta_s(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{d\theta_s}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{d\theta_s}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the torque-to-axial conversion remains continuous. Condition (2) guarantees that increasing axial displacement increases rotation. Condition (3) ensures the effective helix angle remains strictly below $\pi/2$ and the geometric gain remains finite.

Remark (Piecewise Profiles in Practice)

The C^1 requirement above is sufficient for all derivations in this document. In practice, piecewise C^1 profiles (continuous everywhere, continuously differentiable except at finitely many junction points where left and right derivatives exist) are also admissible for numerical simulation. At junction points, one-sided derivatives provide the required slope values, and the resulting force discontinuities pose no difficulty for standard ODE solvers. However, large slope discontinuities can introduce local stiffness into the system: the abrupt change in force gradients may require adaptive solvers to reduce the timestep significantly near the junction, degrading computational efficiency. In extreme cases, explicit integrators may struggle to maintain stability without impractically small step sizes, and implicit or stiff-aware solvers may be preferable.

Derivations for common ramp and helix profiles are provided in Appendix B.

3.2.2 Primary Axial Force Model

The primary axial clamping force arises from the competition between centrifugal loading of the flyweights and the restoring force of the primary compression spring.

As shown in Figure 3.7, each flyweight of mass m_f rotates with angular speed ω_p at an instantaneous radius $r_{f,0} + r_f(s)$, where $r_{f,0}$ is the flyweight radius in the reference configuration $s = 0$ and $r_f(s)$ is the additional radial displacement induced by the ramp. The resulting outward centrifugal force is redirected by the ramp geometry described in Section 3.2.1 into an axial closing force on the movable primary sheave.

Opposing this motion is the primary compression spring, characterized by stiffness k_p and preload $x_{p,0}$. As the sheaves close (s increases), the spring compression increases, producing a larger restoring force that resists further closure.

The resulting primary axial force is therefore the ramp-redirected centrifugal contribution minus the spring force. This quantity, denoted $F_{p,ax}(s, \omega_p)$, provides the primary-side contribution to the generalized axial force used in the shift equation of motion.

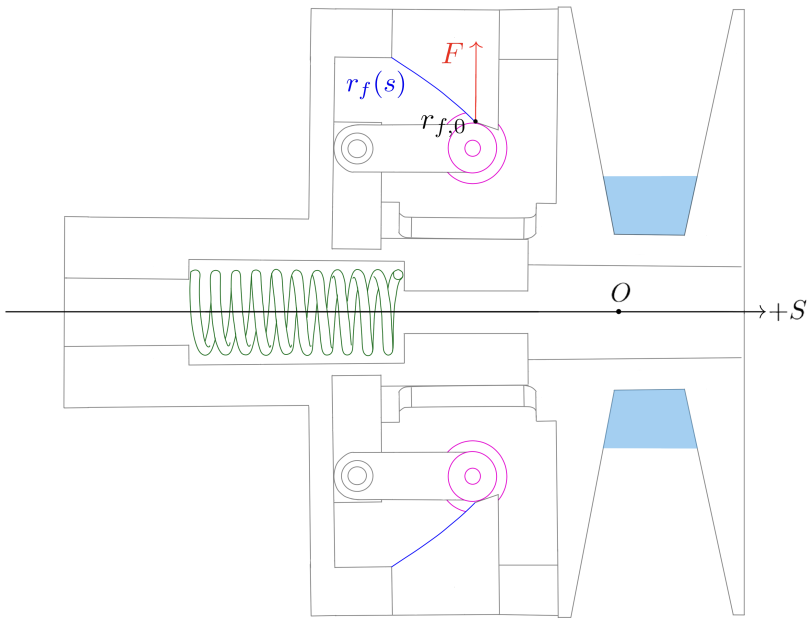


Figure 3.7: Primary CVT axial-force mechanism. Each flyweight of mass m_f rotates at angular speed ω_p and experiences an outward centrifugal force at total radius $r_{f,0} + r_f(s)$. The ramp geometry redirects this radial loading into an axial closing force on the movable primary sheave in the $+s$ direction, while the primary compression spring ($k_p,\ x_{p,0}$) provides an opposing restoring force. The lower flyweight–ramp pair is symmetric and shown unlabeled for clarity.

Centrifugal Loading of Flyweights From Theory 5 (Centrifugal Force),

$$F_c = m\omega^2 r.$$

Let

- m_f denote the flyweight mass,
- $r_{f,0}$ denote the initial flyweight radius at $s = 0$,
- $r_f(s)$ denote the incremental radial change produced by the ramp.

The total effective flyweight radius is therefore

$$r_{f,\text{tot}}(s) = r_{f,0} + r_f(s).$$

The radial centrifugal force is

$$F_{p,\text{rad}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big). \tag{3.2.6}$$

Radial-to-Axial Projection The ramp redirects radial force into axial force. As shown in Figure 3.8, the axial component of a force acting along the ramp is obtained by

$$F_{p,\text{axial component}} = F_{p,\text{rad}} \tan \alpha_p(s). \tag{3.2.7}$$

From Equation (3.2.3), the ramp angle satisfies

$$\tan \alpha_p(s) = \frac{dr_f}{ds}.$$

Substituting gives

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds}. \tag{3.2.8}$$

Thus the axial gain of centrifugal force is governed by the local ramp slope, while the centrifugal magnitude depends on the total flyweight radius.

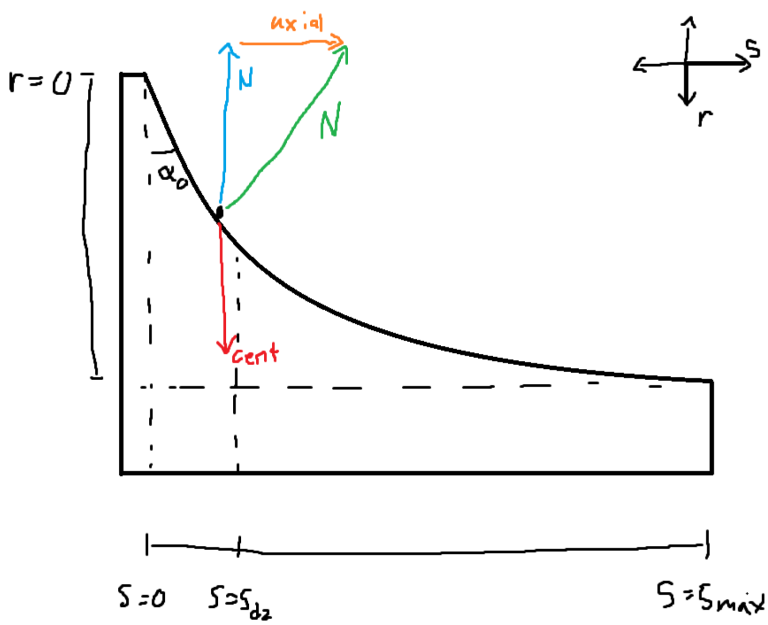


Figure 3.8: Primary ramp geometry. Radial centrifugal force is redirected into axial force through the ramp. The projection introduces a factor of $\tan \alpha_p$, which equals dr_f/ds by definition of the ramp slope.

Primary Compression Spring From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let k_p denote primary stiffness and $x_{p,0}$ preload.

$$F_{p,\text{spring}}(s) = k_p(x_{p,0} + s). \tag{3.2.9}$$

Net Primary Axial Force The net axial force generated by the primary mechanism is the difference between the centrifugal axial component and the spring force:

Primary Axial Force **(3.2.10)**

$$F_{p,\text{ax}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds} - k_p(x_{p,0} + s).$$

3.2.3 Secondary Axial Force Model

The secondary pulley generates axial clamping force in two ways. First, the secondary spring directly presses the sheaves together through axial compression. Second, the helix mechanism converts torque carried by the secondary pulley into additional axial force. The secondary is therefore torque-reactive: larger transmitted torque can increase the clamping force applied to the belt.

The goal of this section is to derive the net secondary axial force $F_{s,\text{ax}}$. The derivation follows the physical mechanism directly: determine the torque reacted through the helix, convert that reacted torque into axial force through the helix geometry, and then add the direct compression spring force.

The secondary assembly consists of a fixed sheave, a movable sheave, a helical slot between them, and a spring. The fixed sheave is attached to the secondary shaft. The movable sheave can translate axially, but the helix constrains that translation to occur with relative rotation between the movable and fixed sheaves. Under Assumption 2 (Rigid Structural Components), this coupled motion is treated as a rigid geometric constraint.

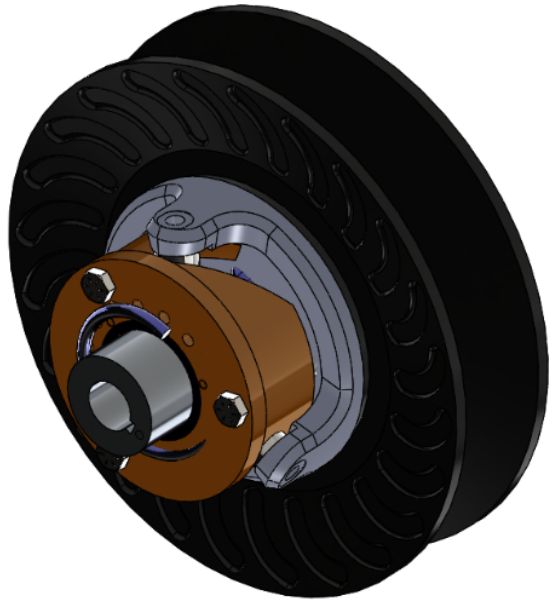


Figure 3.9: Secondary CVT assembly showing the helix mechanism and spring. The helix is modeled as a captured slot, so axial displacement of the movable sheave is coupled to relative rotation between the movable and fixed sheaves.

Physical Role of the Helix

The helix acts like a screw-like constraint between the fixed and movable secondary sheaves. A screw produces axial force only when relative rotation is resisted. The same idea applies here: torque applied to the movable sheave tends to rotate it relative to the fixed sheave. Because the movable sheave is captured by the helix, that relative rotation is tied to axial motion. The helix therefore reacts torque, and the inclined helix path converts that reacted torque into axial force.

This gives the structure of the force model. The helix contribution is not determined by secondary speed alone. It is determined by the torque that the movable sheave attempts to pass through the helix, after accounting for the torque required to accelerate the movable sheave itself.

Let τ_s denote the total signed torque transmitted from the belt to the secondary pulley. Under Assumption 8 (Equal Secondary Sheave-Face Torque Partition), this torque is treated as equally shared between the fixed and movable secondary sheave faces. The belt-applied torque acting on the movable sheave is therefore

$$\tau_{b,M} = \frac{\tau_s}{2}. \tag{3.2.11}$$

This is the portion of the belt torque that appears on the movable-sheave free body and can therefore load the helix.

Movable-Sheave Kinematics

The helix geometry was parameterized earlier by the relative rotation mapping

$$\theta_s = \theta_s(s), \tag{3.2.12}$$

as introduced in Equation (3.2.4). Here, θ_s is the relative rotation between the movable and fixed secondary sheaves.

The fixed sheave is attached to the secondary shaft, so its angular velocity is

$$\omega_F = \omega_s. \tag{3.2.13}$$

The movable sheave does not generally rotate at this same speed while the CVT is shifting. With the sign convention used here, positive helix rotation means that the movable sheave lags the fixed sheave, so

$$\dot{\theta}_s = \omega_s - \omega_M. \tag{3.2.14}$$

Since the helix angle is a function of the shift coordinate,

$$\dot{\theta}_s = \frac{d\theta_s}{ds}\dot{s}. \tag{3.2.15}$$

Combining these two relations gives the angular velocity of the movable sheave:

$$\omega_M = \omega_s - \frac{d\theta_s}{ds}\dot{s}. \tag{3.2.16}$$

Thus, secondary shift does not only move the sheave axially. It also changes the movable sheave's angular speed relative to the fixed sheave by an amount set by the helix slope.

Differentiating Equation (3.2.16) gives

$$\dot{\omega}_M = \dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2 - \frac{d\theta_s}{ds}\ddot{s}. \tag{3.2.17}$$

This acceleration is needed because torque applied to the movable sheave has two possible roles: it can be reacted through the helix, or it can angularly accelerate the movable sheave.

Torque Reacted Through the Helix

The movable sheave is now treated as a rotating rigid body. Under Assumption 13 (Lumped Inertia Representation), its inertia about the secondary shaft is represented by the scalar I_M.

The torques acting on the movable sheave are the belt-applied torque $\tau_s/2$, the torsional spring torque, and the torque applied by the helix. The secondary spring is twisted by the same relative sheave rotation that is imposed by the helix. With torsional stiffness $k_{s,\theta}$ and preload deflection $\theta_{s,0}$, the spring torque is

$$\tau_{s,\text{spring}}(s) = k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big). \tag{3.2.18}$$

Let $\tau_{\text{helix}\to M}$ denote the torque applied by the helix to the movable sheave. Applying Theory 2 (Newton's Second Law (Rotation)) to the movable sheave gives

$$\frac{\tau_s}{2} + \tau_{s,\text{spring}} + \tau_{\text{helix}\to M} = I_M \dot{\omega}_M. \tag{3.2.19}$$

The quantity needed for axial clamping is the equal and opposite torque applied by the movable sheave into the helix:

$$\tau_{M\to\text{helix}} = -\tau_{\text{helix}\to M}. \tag{3.2.20}$$

Solving the movable-sheave torque balance for this reacted torque gives

$$\tau_{M\to\text{helix}} = \frac{\tau_s}{2} + \tau_{s,\text{spring}} - I_M \dot{\omega}_M. \tag{3.2.21}$$

Substituting Equation (3.2.18) and Equation (3.2.17) gives the reacted helix torque in terms of the state and acceleration variables:

$$\tau_{M\to\text{helix}} = \frac{\tau_s}{2} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M \left[\dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2 - \frac{d\theta_s}{ds}\ddot{s}\right]. \tag{3.2.22}$$

This equation is the core of the secondary mechanism. The helix reacts the movable-face belt torque and the torsional spring torque, except for the portion of torque required to angularly accelerate the movable sheave.

Conversion from Reacted Torque to Axial Force

The remaining step is geometric. The helix relates axial displacement to relative rotation. For a virtual axial displacement δs, the corresponding relative rotation is

$$\delta\theta_s = \frac{d\theta_s}{ds}\delta s.$$

By virtual work, the axial force associated with the reacted helix torque is

$$F_{s,\text{helix}} = \tau_{M\to\text{helix}} \frac{d\theta_s}{ds}. \tag{3.2.23}$$

A steeper torque-to-shift mapping therefore produces a larger axial force from the same reacted torque.

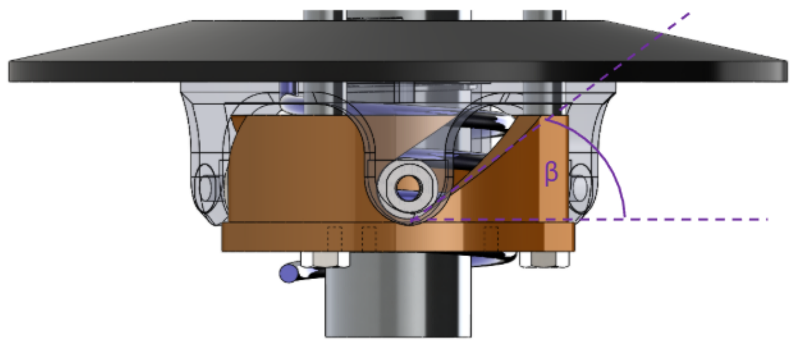


Figure 3.10: Helix free-body concept. Torque applied to the movable sheave is reacted through the helix. Because the helix is inclined, the reacted torque produces an axial force proportional to $d\theta_s/ds$.

Substituting Equation (3.2.22) into Equation (3.2.23) gives

$$F_{s,\text{helix}} = \frac{d\theta_s}{ds}\left[\frac{\tau_s}{2} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M\left(\dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2 - \frac{d\theta_s}{ds}\ddot{s}\right)\right]. \tag{3.2.24}$$

The terms inside the brackets are the torque that the helix must react. The factor multiplying the brackets is the helix slope that converts that reacted torque into axial force.

Direct Compression Spring Force

The secondary spring also contributes directly through axial compression. At $s = 0$, the spring is pre-compressed by $x_{s,0}$. As the secondary sheaves open with increasing shift, the compression increases by s. From Theory 6 (Hooke's Law (Compressional)), the direct axial spring force is

$$F_{s,\text{comp}}(s) = k_{s,x}(x_{s,0} + s). \tag{3.2.25}$$

This force exists even if no torque is being transmitted. It provides the baseline secondary clamping contribution, while the helix contribution adds the torque-reactive part.

Net Secondary Axial Force

The net secondary axial force is the sum of the direct compression-spring force and the helix-generated force:

Secondary Axial Force **(3.2.26)**

$$F_{s,\text{ax}} = k_{s,x}(x_{s,0} + s) + \frac{d\theta_s}{ds}\left[\frac{\tau_s}{2} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M\left(\dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2 - \frac{d\theta_s}{ds}\ddot{s}\right)\right].$$

This form mirrors the physical mechanism. The first term is the direct axial spring force. The second term is the helix force: the torque reacted through the movable sheave and spring is converted into axial force by the helix slope. The inertial term appears because part of the movable-sheave torque is used to accelerate the movable sheave rather than being reacted through the helix.

The secondary force therefore increases with axial spring compression, torsional spring wind-up, and transmitted secondary torque. It is also affected by the motion of the secondary sheave, because the helix couples axial shift to relative rotation. This is why the secondary is a torque-reactive clamping mechanism rather than a simple position-dependent spring.

3.2.4 Belt Centrifugal Force on the Wrap

In addition to the actuator and spring forces derived previously, the belt itself generates centrifugal loading as it rotates around each pulley. This section derives the resulting contribution to axial clamping force.

Throughout this section, let

$$j \in \{p, s\}$$

denote either pulley, where $j = p$ corresponds to the primary and $j = s$ corresponds to the secondary.

From Theory 5 (Centrifugal Force), a mass m rotating at angular velocity ω at radius r experiences centrifugal force

$$F_c = m\omega^2 r.$$

To apply this to the belt wrapped around pulley j, three quantities must be defined:

1. the radius at which the belt mass rotates,
2. the mass of belt material carried around the wrap,
3. the angular velocity of that belt mass.

Each is developed below.

1. Radius at Which the Belt Mass Rotates Centrifugal force acts at the center of mass of the rotating belt element, not at the outer or inner belt surface. The location of this center of mass therefore determines the radius that must be used in the centrifugal force calculation.

The belt cross-section is approximated as a trapezoid (see Figure 3.11) with:

- radial thickness h_b,
- outer width b_{out},
- inner width b_{in}.

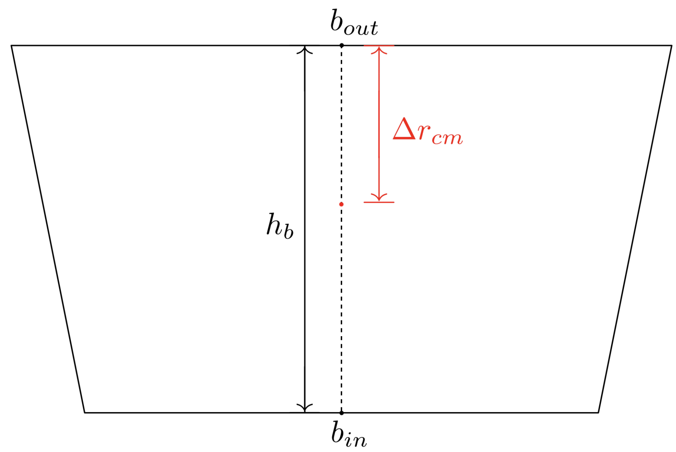


Figure 3.11: Trapezoidal belt cross-section. The outer width b_{out} corresponds to the surface away from the pulley axis; the inner width b_{in} corresponds to the surface closer to the axis. The radial thickness is h_b and the centroid is located at Δr_{cm} from the outer surface.

The cross-sectional area of the trapezoid is

$$A_b = \frac{h_b}{2}\left(b_{\text{out}} + b_{\text{in}}\right). \tag{3.2.27}$$

Let y denote radial distance measured inward from the outer surface, so that $y = 0$ at the outer surface and $y = h_b$ at the inner surface. For a trapezoid with linear sidewalls, the local width varies linearly with y:

$$b(y) = b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y. \tag{3.2.28}$$

The centroid location Δr_{cm}, measured inward from the outer surface, follows from the area-centroid definition:

$$\Delta r_{\text{cm}} = \frac{1}{A_b} \int_0^{h_b} y\, b(y)\; \mathrm{d}y. \tag{3.2.29}$$

Substituting Equation (3.2.28) into Equation (3.2.29) gives

$$
\begin{aligned}
\int_0^{h_b} y\, b(y)\ \mathrm{d}y &= \int_0^{h_b} y \left(b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y \right) \mathrm{d}y \\
&= \int_0^{h_b} b_{\text{out}} y\ \mathrm{d}y + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \int_0^{h_b} y^2\ \mathrm{d}y \\
&= \frac{h_b^2}{2} b_{\text{out}} + \frac{h_b^2}{3} \left(b_{\text{in}} - b_{\text{out}} \right) \\
&= \frac{h_b^2}{6} \left(b_{\text{out}} + 2 b_{\text{in}} \right). \qquad (3.2.30)
\end{aligned}
$$

Dividing by the area from Equation (3.2.27) yields

$$
\Delta r_{\text{cm}} = h_b \frac{b_{\text{out}} + 2 b_{\text{in}}}{3 \left(b_{\text{out}} + b_{\text{in}} \right)}. \qquad (3.2.31)
$$

For pulley j, let $r_{j,\text{out}}(s)$ denote the outer belt radius. The radius at which the belt mass rotates is therefore

$$
r_{j,\text{cm}}(s) = r_{j,\text{out}}(s) - \Delta r_{\text{cm}}. \qquad (3.2.32)
$$

2. Mass of Belt Material on the Wrap To determine the centrifugal loading on pulley j, we next determine the amount of belt mass associated with an infinitesimal portion of its wrap.

Let θ denote the angular coordinate measured from the center of the wrap, so that the wrapped region on pulley j spans

$$
\theta \in \left[-\frac{\phi_j}{2}, \frac{\phi_j}{2} \right],
$$

where $\phi_j(s)$ is the total wrap angle on pulley j (see Figure 3.12).

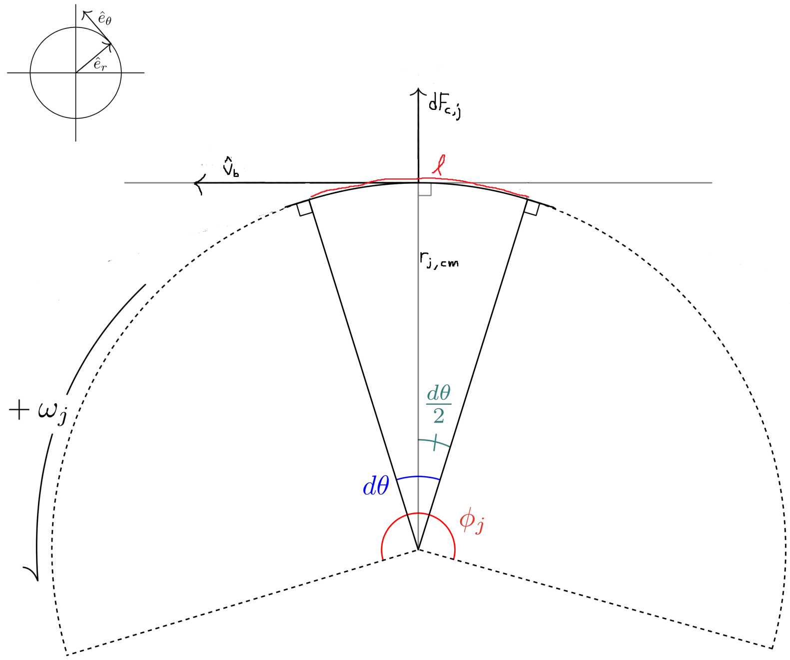


Figure 3.12: Belt wrapped around pulley j. A differential element at angle θ has arc length $\mathrm{d}\ell = r_{j,\mathrm{cm}}\,\mathrm{d}\theta$ and experiences radially outward centrifugal force.

A differential belt element subtending angle $\mathrm{d}\theta$ at the mass-centroid radius $r_{j,\mathrm{cm}}$ has arc length

$$\mathrm{d}\ell = r_{j,\mathrm{cm}}\,\mathrm{d}\theta.$$

Since the belt has cross-sectional area A_b, the differential belt element has volume

$$\mathrm{d}V_j = A_b\,\mathrm{d}\ell. \tag{3.2.33}$$

Substituting $\mathrm{d}\ell = r_{j,\mathrm{cm}}\,\mathrm{d}\theta$ gives

$$\mathrm{d}V_j = A_b\,r_{j,\mathrm{cm}}\,\mathrm{d}\theta. \tag{3.2.34}$$

Let ρ_b denote the belt material density (kg/m^3). The corresponding differential mass is then

$$\begin{aligned}\mathrm{d}m_j &= \rho_b\,\mathrm{d}V_j\\ &= \rho_b A_b\,r_{j,\mathrm{cm}}\,\mathrm{d}\theta.\end{aligned} \tag{3.2.35}$$

This gives the amount of belt material associated with an infinitesimal portion of the wrap on pulley j.

3. Angular Velocity of the Belt Mass The rotating belt mass on pulley j must also be assigned an angular speed. From Section 2.5, v_b denotes the belt transport speed. The corresponding angular velocity of the belt mass centroid on pulley j is then

$$\omega_{b,j} = \frac{v_b}{r_{j,\mathrm{cm}}(s)}. \tag{3.2.36}$$

This relation expresses the fact that the same belt transport speed corresponds to different local angular velocities depending on the radius of motion.

Distributed Centrifugal Load Along the Wrap With the radius, differential mass, and angular velocity now defined, the centrifugal force on the differential belt element follows from Theory 5 (Centrifugal Force):

$$\begin{aligned} \mathrm{d}F_{c,j} &= \mathrm{d}m_j\, \omega_{b,j}^2\, r_{j,\mathrm{cm}} \\ &= (\rho_b A_b\, r_{j,\mathrm{cm}}\, \mathrm{d}\theta) \left(\frac{v_b}{r_{j,\mathrm{cm}}}\right)^2 r_{j,\mathrm{cm}} \\ &= \rho_b A_b\, v_b^2\, \mathrm{d}\theta. \end{aligned} \tag{3.2.37}$$

This force acts radially outward at each point along the wrap.

Total Radial Force Over the Wrap Integrating Equation (3.2.37) over the full wrap angle gives the total radially-directed centrifugal force on pulley j:

$$\begin{aligned} F_{c,j,\mathrm{rad}} &= \int_{-\phi_j/2}^{\phi_j/2} \rho_b A_b\, v_b^2\, \mathrm{d}\theta \\ &= \rho_b A_b\, v_b^2 \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}\theta \\ &= \rho_b A_b\, v_b^2\, \phi_j(s). \end{aligned} \tag{3.2.38}$$

Radial-to-Axial Conversion Through the Sheave Cone The radial centrifugal force acts outward against the conical sheave faces. Because the sheaves are inclined at cone half-angle β, radial loading cannot occur without simultaneous axial motion of the sheaves. Thus, a purely radial force applied to the belt generates an axial reaction component that contributes to clamping.

From the pulley geometry derived previously (see **??**), the relationship between radial and axial displacement in the active shift region is

$$\frac{\mathrm{d}r}{\mathrm{d}s} = \frac{1}{2\tan\beta}.$$

For an infinitesimal displacement,

$$\delta W = F_{\text{rad}}\,\mathrm{d}r = F_{\text{ax}}\,\mathrm{d}s.$$

Substituting $\mathrm{d}r = \dfrac{1}{2\tan\beta}\,\mathrm{d}s$ gives

$$F_{\text{rad}}\,\frac{1}{2\tan\beta}\,\mathrm{d}s = F_{\text{ax}}\,\mathrm{d}s.$$

Canceling $\mathrm{d}s$ yields

$$F_{\text{ax}} = \frac{F_{\text{rad}}}{2\tan\beta}. \tag{3.2.39}$$

Thus, the axial clamping contribution from belt centrifugal force on pulley j is

$$F_{c,j} = \frac{\rho_b A_b\, v_b^2\, \phi_j(s)}{2\tan\beta}. \tag{3.2.40}$$

3.2.5 Axial Shift Dynamics

The previous subsections derived the axial forces produced by the primary and secondary pulley mechanisms. The remaining task is to use those forces to determine the motion of the shift coordinate $s(t)$.

The shift coordinate represents coordinated motion of several physical bodies. As s increases, the primary movable sheave closes, the secondary movable sheave opens through the belt-length constraint, and belt material migrates through the pulley grooves. The axial equation of motion is therefore written in the generalized coordinate s, with all force and inertia terms expressed in that coordinate.

From the axial force balance introduced in Equation (3.2.1),

$$m\,\ddot{s} = F_{p,\text{ax}} - F_{s,\text{ax}}, \tag{3.2.41}$$

where $F_{p,\text{ax}}$ acts in the positive s direction and $F_{s,\text{ax}}$ opposes it. This section replaces the placeholder mass m with the equivalent inertia associated with the shift coordinate and then adds the axial belt centrifugal contributions derived in Section 3.2.4.

Equivalent Inertia in the Shift Coordinate

Since s is a generalized coordinate, the inertia associated with shift is constructed from kinetic energy. Following Theory 4 (Kinetic Energy (Translational)), the translational kinetic energy associated with shift is written as

$$T_{\text{shift}} = \frac{1}{2} m_{\text{eq}} \dot{s}^2.$$

The movable primary sheave translates with speed $\dot{s}$. The movable secondary sheave translates with the same speed magnitude in the opposite physical direction, so its kinetic energy also contributes to the same generalized coordinate. Let $m_{p,m}$ and $m_{s,m}$ denote the translating masses of the primary and secondary movable sheaves.

The belt also contributes to the shift inertia. Under Assumption 10 (Constant Belt Length), the total belt mass is constant and is

$$m_b = \rho_b A_b L_b, \tag{3.2.42}$$

where A_b is the trapezoidal belt cross-sectional area from Equation (3.2.27). Under Assumption 11 (Single Belt Transport Speed), the belt is represented by a single lumped transport speed, while the simplified axial contribution associated with radial migration through the groove is represented by the effective axial speed

$$\dot{s}_b = \frac{\dot{s}}{2}. \tag{3.2.43}$$

The belt kinetic energy associated with this axial component is therefore

$$T_b = \frac{1}{2} m_b \left(\frac{\dot{s}}{2}\right)^2 = \frac{1}{2}\left(\frac{m_b}{4}\right)\dot{s}^2.$$

Thus, the base equivalent mass associated with translation in the shift coordinate is

$$m_{\text{eq}} = m_{p,m} + m_{s,m} + \frac{m_b}{4} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}. \tag{3.2.44}$$

This mass accounts for the translating sheaves and the simplified axial belt motion. The rotational inertia coupled through the secondary helix is collected separately below.

Secondary Force Split

The secondary axial force derived in Equation (3.2.26) contains the movable-sheave angular acceleration. This occurs because the helix couples axial shift to relative sheave rotation. Written as a function of the quantities that enter the shift equation,

$$F_{s,\text{ax}} = F_{s,\text{ax}}\left(s, \dot{s}, \tau_s, \dot{\omega}_s, \ddot{s}\right). \tag{3.2.45}$$

Recalling Equation (3.2.26), with the movable-sheave torque contribution written using Assumption 8 (Equal Secondary Sheave-Face Torque Partition),

$$F_{s,\text{ax}} = k_{s,x}(x_{s,0} + s) + \frac{d\theta_s}{ds}\left[\frac{\tau_s}{2} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M\left(\dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2 - \frac{d\theta_s}{ds}\ddot{s}\right)\right]. \tag{3.2.46}$$

To place this force into the shift equation, separate the part that does not multiply $\ddot{s}$ from the part that does:

$$F_{s,\text{ax}} = \widehat{F}_{s,\text{ax}} + I_M\left(\frac{d\theta_s}{ds}\right)^2 \ddot{s}. \tag{3.2.47}$$

The acceleration-independent part is

$$\widehat{F}_{s,\mathrm{ax}} = k_{s,x}(x_{s,0} + s) + \frac{d\theta_s}{ds}\left[\frac{\tau_s}{2} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M\left(\dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2\right)\right]. \tag{3.2.48}$$

This split is purely algebraic. It keeps the same secondary force model, but makes clear that the movable-sheave rotational inertia appears in the shift coordinate as the reflected inertia term

$$I_M\left(\frac{d\theta_s}{ds}\right)^2.$$

Generalized Axial Force Balance

The generalized force in the positive s direction is obtained by collecting the primary-side contributions that drive upshift and subtracting the secondary-side contributions that resist it. The forces entering this balance are:

- the primary mechanism force $F_{p,\mathrm{ax}}(s, \omega_p)$ from Equation (3.2.10),
- the secondary mechanism force $F_{s,\mathrm{ax}}$ from Equation (3.2.26),
- the primary belt centrifugal contribution $F_{c,p}(s, v_b)$ from Equation (3.2.40),
- the secondary belt centrifugal contribution $F_{c,s}(s, v_b)$ from Equation (3.2.40).

With the sign convention of s, the generalized force balance is

$$\sum F_s = \Big(F_{p,\mathrm{ax}} + F_{c,p}\Big) - \Big(F_{s,\mathrm{ax}} + F_{c,s}\Big). \tag{3.2.49}$$

Substituting the secondary force split from Equation (3.2.47) gives

$$m_{\mathrm{eq}}\ddot{s} = \Big(F_{p,\mathrm{ax}} + F_{c,p}\Big) - \left(\widehat{F}_{s,\mathrm{ax}} + I_M\left(\frac{d\theta_s}{ds}\right)^2\ddot{s} + F_{c,s}\right). \tag{3.2.50}$$

Collecting the terms proportional to $\ddot{s}$ gives

$$\left[m_{\mathrm{eq}} + I_M\left(\frac{d\theta_s}{ds}\right)^2\right]\ddot{s} = \Big(F_{p,\mathrm{ax}} + F_{c,p}\Big) - \Big(\widehat{F}_{s,\mathrm{ax}} + F_{c,s}\Big). \tag{3.2.51}$$

This form has the expected physical structure. The numerator is the net generalized force trying to move the CVT through its shift range. The bracketed factor on the left is the effective inertia associated with that motion: the translating components and the movable secondary sheave inertia reflected through the helix.

Explicit Shift Acceleration

Substituting the equivalent mass from Equation (3.2.44) into Equation (3.2.51) gives the explicit shift acceleration:

Axial Shift Acceleration **(3.2.52)**

$$\ddot{s} = \frac{\Big(F_{p,\mathrm{ax}}(s,\omega_p) + F_{c,p}(s,v_b)\Big) - \Big(\widehat{F}_{s,\mathrm{ax}}(s,\dot{s},\tau_s,\dot{\omega}_s) + F_{c,s}(s,v_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4} + I_M\left(\dfrac{d\theta_s}{ds}\right)^2}.$$

where

$$\widehat{F}_{s,\mathrm{ax}}(s,\dot{s},\tau_s,\dot{\omega}_s) = k_{s,x}(x_{s,0}+s) + \frac{d\theta_s}{ds}\left[\frac{\tau_s}{2} + k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big) - I_M\left(\dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2\right)\right]. \quad (3.2.53)$$

The shift acceleration therefore follows the same structure as any second-order force balance: net generalized force divided by effective inertia. The primary mechanism and primary belt centrifugal contribution drive positive shift, while the secondary mechanism and secondary belt centrifugal contribution oppose it. The secondary term depends on the transmitted secondary torque τ_s and secondary angular acceleration $\dot{\omega}_s$, so the shift motion remains coupled to the rotational torque-transfer dynamics.

3.2.6 Section Summary and Forward Outlook

This section derived the axial force models for the primary and secondary pulley mechanisms and assembled them into an equation of motion for the generalized shift coordinate $s(t)$.

The resulting shift equation has the expected force-balance structure: primary mechanism force and primary belt centrifugal loading drive positive shift, while secondary clamping force and secondary belt centrifugal loading oppose it. The effective inertia in the denominator includes the translating sheaves, the simplified axial contribution of belt motion, and the movable secondary sheave inertia reflected through the helix constraint.

Importantly, the axial shift relation is not a standalone subsystem. The quantity $\ddot{s}$ is written in terms of the current state, but it also depends on two quantities that are not themselves state variables: the transmitted secondary torque τ_s and the secondary angular acceleration $\dot{\omega}_s$. These appear because the secondary clamping mechanism is torque-reactive and because the helix couples axial shift to relative rotation of the secondary sheaves.

Thus, the result of this section should be interpreted as one equation in the larger coupled CVT model. It gives the axial response that must be satisfied alongside the rotational equations for the primary pulley, belt, and secondary pulley. The following section develops those torque-transfer dynamics, which provide the remaining relations needed to determine τ_s, $\dot{\omega}_s$, and the other acceleration and torque unknowns.

With the axial mechanism model established, attention now turns to the torque path through the belt and pulleys. Combining that torque path with the shift relation derived here will provide the coupled dynamic closure used by the full CVT model.

3.3 Rotational and Belt Transport Dynamics

The axial force model developed in Section 3.2 gives one equation for the shift coordinate $s(t)$. That equation depends on the transmitted secondary torque τ_s and on the secondary angular acceleration $\dot{\omega}_s$. Those quantities are determined by the torque path through the CVT.

This section derives the common equations of motion for the rotating and translating bodies in that torque path. The primary pulley, belt, and secondary pulley are treated as coupled dynamic bodies. Engine torque enters at the primary pulley, torque is exchanged between the primary pulley and the belt, the belt exchanges torque with the secondary pulley, and the secondary pulley drives the downstream drivetrain.

The equations derived here are independent of whether the belt is sticking or slipping at either pulley. They describe the bodies being accelerated; the contact state supplies the remaining closure equations later.

3.3.1 System Definition and Torque Path

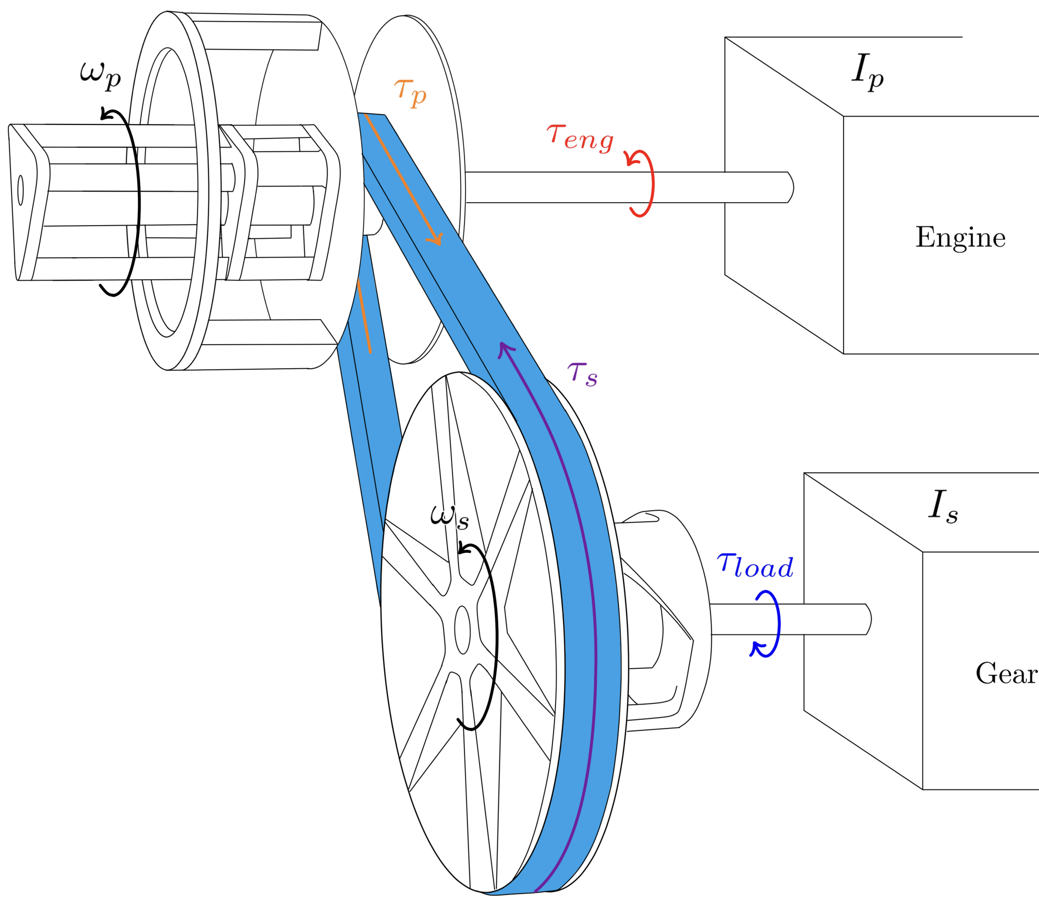


Figure 3.13: Torque path through the CVT. Engine torque τ_{eng} drives the primary pulley with angular speed ω_p. The primary pulley exchanges torque with the belt, the belt exchanges torque with the secondary pulley, and the secondary pulley drives the downstream drivetrain while opposing the reflected load torque τ_{load}.

The torque path shown in Figure 3.13 is described using three contact torques and three dynamic motions:

- the primary pulley rotates with angular speed ω_p,
- the belt translates with transport speed v_b,
- the secondary shaft and fixed sheave rotate with angular speed ω_s.

The primary pulley receives engine torque τ_{eng}. Let τ_p denote the torque exerted by the belt contact on the primary pulley. With the rotational sign convention from Section 2.3.1, τ_p opposes positive engine-driven primary rotation.

The belt receives torque from the primary contact and applies torque to the secondary contact. Let τ_s denote the total signed torque exerted by the belt on the secondary pulley. This torque acts in the positive secondary rotational direction, while the reflected drivetrain load τ_{load} opposes it.

3.3.2 Engine Torque Model

The engine provides the driving torque $\tau_{\text{eng}}(t)$ acting on the primary pulley. Under Assumption 18 (Full-Throttle Engine Model), engine torque is represented as a prescribed full-throttle function of primary angular speed:

$$\tau_{\text{eng}}(t) = \tau_{\text{eng}}\big(\omega_p(t)\big). \tag{3.3.1}$$

No specific analytic form is required. The model only requires that the mapping $\omega_p \mapsto \tau_{\text{eng}}$ be finite over the operating range used in simulation. Outside the supplied range, the torque curve is either saturated or the state is constrained to remain within the admissible engine-speed domain.

SAE Baja Restricted Kohler CH440							
Engine Speed (rpm)	3600	3400	3200	3000	2800	2600	2400
Torque Net Corrected (ft-lb)	13.5	14.5	15.4	16.6	17.4	18.1	18.5
Power Net Corrected (HP)	9.3	9.4	9.4	9.5	9.3	9.0	8.5

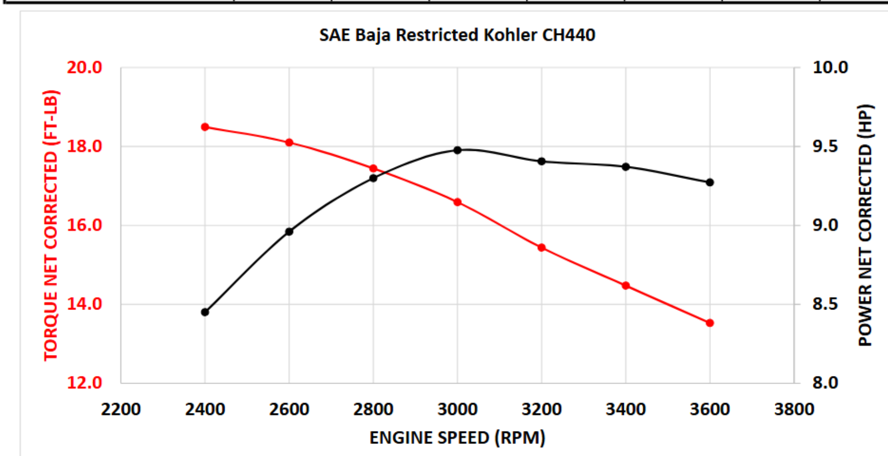


Figure 3.14: Representative torque and power curves for the Kohler CH440 engine. Torque is supplied as a function of angular velocity and serves as the input mapping in Equation (3.3.1).

In simulation, the discrete torque curve is interpolated to provide a continuous input torque over the operating range.

3.3.3 Load Torque Model

The resisting torque $\tau_{\text{load}}(t)$ represents the longitudinal vehicle load reflected back to the secondary pulley.

Let r_w denote the effective wheel rolling radius and let $G > 0$ denote the fixed gear ratio defined by

$$\omega_s = G\,\omega_w.$$

The vehicle speed is therefore

$$v(t) = r_w \omega_w(t) = \frac{r_w}{G} \omega_s(t). \tag{3.3.2}$$

The total longitudinal road force is modeled as the sum of rolling resistance, grade resistance, and aerodynamic drag:

$$F_{\text{road}} = F_{\text{rr}} + F_{\text{grade}} + F_{\text{drag}}. \tag{3.3.3}$$

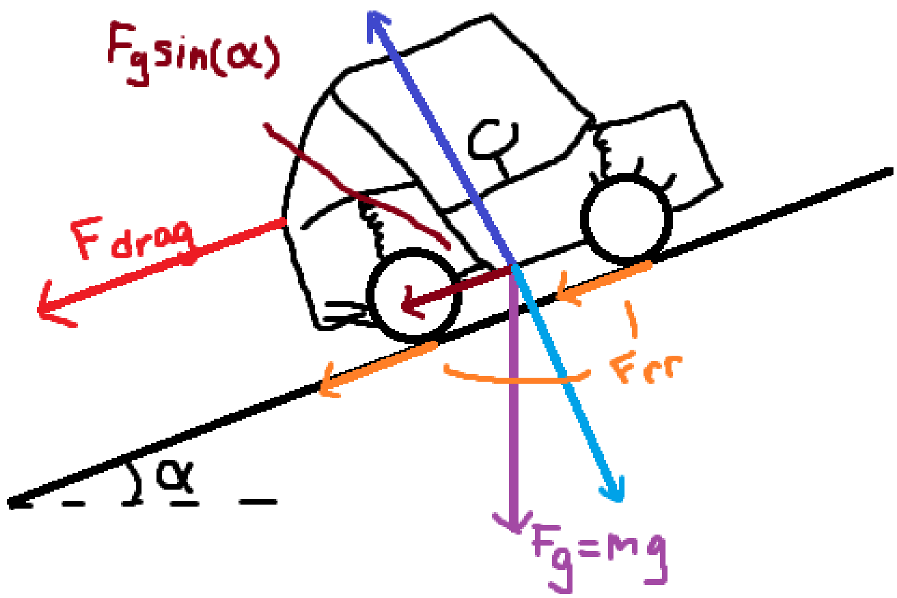


Figure 3.15: Free-body diagram of a vehicle on an incline. The weight mg resolves into $mg \sin \alpha$ along the slope and $mg \cos \alpha$ normal to the surface. Rolling resistance and aerodynamic drag oppose motion.

Using Theory 13 (Rolling Resistance), Theory 14 (Gravitational Force), and Theory 12 (Air Resistance (Quadratic Drag)), the road-load components are

$$F_{\text{rr}}(v, \alpha) = C_{\text{rr}} mg \cos \alpha \, \text{sgn}(v), \tag{3.3.4}$$

$$F_{\text{grade}}(\alpha) = mg \sin \alpha, \tag{3.3.5}$$

and

$$F_{\text{drag}}(v) = \frac{1}{2} \rho C_d A_f v^2 \, \text{sgn}(v). \tag{3.3.6}$$

Substituting these terms into Equation (3.3.3) gives

$$
\begin{aligned}
F_{\text{road}}(t) = C_{\text{rr}} mg \cos\alpha \, \text{sgn}(v(t)) + mg \sin\alpha \\
+ \frac{1}{2}\rho C_d A_f v(t)^2 \, \text{sgn}(v(t))\,.
\end{aligned}
\tag{3.3.7}
$$

The wheel torque associated with this road load is

$$
\tau_{\text{road},w}(t) = r_w F_{\text{road}}(t). \tag{3.3.8}
$$

With the downstream fixed-ratio drive treated as ideal under Assumption 16 (No Explicit CVT Loss or Thermal Model), the resisting torque reflected to the secondary pulley is

$$
\tau_{\text{load}}(t) = \frac{1}{G}\tau_{\text{road},w}(t) = \frac{r_w}{G} F_{\text{road}}(t). \tag{3.3.9}
$$

Using $v = (r_w/G)\omega_s$, this becomes

$$
\boxed{\tau_{\text{load}}(t) = \frac{r_w}{G}\left[C_{\text{rr}} mg \cos\alpha \, \text{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big) + mg\sin\alpha + \frac{1}{2}\rho C_d A_f \left(\frac{r_w}{G}\omega_s(t)\right)^2 \text{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big)\right].} \tag{3.3.10}
$$

3.3.4 Inertia Definitions

The rotational equations require inertias expressed about the primary and secondary axes. Under Assumption 13 (Lumped Inertia Representation), these inertias are treated as lumped scalar quantities about their corresponding shafts.

Primary-Side Inertia

The primary pulley is rigidly connected to the engine side. All components in this lump rotate with angular speed ω_p, so the primary-side inertia is

$$
I_p = I_{\text{eng}} + I_{\text{prim}}, \tag{3.3.11}
$$

where I_{eng} is the engine rotational inertia and I_{prim} is the primary pulley inertia.

Secondary Fixed-Side and Downstream Inertia

The secondary side requires one additional distinction. The fixed secondary sheave and downstream drivetrain rotate with ω_s. The movable secondary sheave does not generally rotate with exactly ω_s during shift, because the helix couples axial motion to relative rotation. Its rotational motion is therefore handled separately using the helix kinematics from Equation (3.2.16) and Equation (3.2.17).

Let $I_{s,F}$ denote the inertia that rotates directly with the fixed secondary sheave and shaft, excluding the movable secondary sheave inertia I_M. This fixed-side inertia includes the fixed secondary components, gearbox-side inertia, wheel inertia reflected through the fixed ratio, and the vehicle translational inertia reflected to the secondary axis.

The vehicle translational kinetic energy is

$$T_{\text{trans}} = \frac{1}{2}mv^2 = \frac{1}{2}m(r_w\omega_w)^2,$$

so the vehicle mass appears as an equivalent wheel-axis inertia mr_w^2. Since $\omega_s = G\omega_w$, wheel-axis inertias reflect to the secondary axis by division by G^2. Thus,

$$I_{s,F} = I_{\text{sec},F} + I_{\text{gb}} + \frac{I_{\text{wheel}} + mr_w^2}{G^2}. \tag{3.3.12}$$

Here $I_{\text{sec},F}$ includes the fixed secondary sheave and other secondary components that rotate directly with the secondary shaft. The movable secondary sheave inertia I_M is not included in $I_{s,F}$.

3.3.5 Common Equations of Motion

The primary pulley, belt, secondary fixed-side assembly, and shift coordinate provide the four common dynamic equations used in every contact state. The contact state will later determine the two additional closure equations.

Primary Pulley Rotational Dynamics

Applying Theory 2 (Newton's Second Law (Rotation)) to the primary pulley gives

Primary Pulley Rotational Dynamics **(3.3.13)**

$$\tau_{\text{eng}}(t) - \tau_p(t) = I_p\dot{\omega}_p(t)$$

The engine torque drives positive primary rotation, while the belt contact torque opposes it.

Belt Transport Dynamics

The belt is represented by the transport speed $v_b(t)$ introduced in Section 2.5. Under Assumption 10 (Constant Belt Length) and Assumption 11 (Single Belt Transport Speed), the belt is modeled as a single moving path with one lumped transport speed.

A contact torque produces an equivalent belt-line force when divided by the effective torque radius from ??. The primary contact contributes $\tau_p/r_{p,\text{eff}}$ to the belt-line motion, while the secondary contact contributes $\tau_s/r_{s,\text{eff}}$ in the opposing direction. Applying Theory 1 (Newton's Second Law (Translation)) to the belt gives

Belt Transport Dynamics **(3.3.14)**

$$m_b\dot{v}_b(t) = \frac{\tau_p(t)}{r_{p,\text{eff}}(s)} - \frac{\tau_s(t)}{r_{s,\text{eff}}(s)}$$

where m_b is the total belt mass from Equation (3.2.42).

Secondary Rotational Dynamics

The secondary side is not treated as a single rigid body during shift. The fixed secondary sheave and downstream drivetrain rotate with the secondary shaft speed ω_s, while the movable secondary sheave rotates with ω_M. These two angular speeds are equal only when the secondary is not shifting. When the helix is active, axial motion of the movable sheave is accompanied by relative rotation, so the movable sheave must be counted with its own angular acceleration.

The cleanest way to write the secondary rotational balance is to consider the fixed-side secondary assembly and the movable sheave together as one system. The external torques on this combined secondary system are the total belt-applied secondary torque τ_s and the reflected load torque τ_{load}. The helix and spring torques between the fixed and movable sheaves are internal to this combined system, so they do not appear in the net external torque balance.

Let $I_{s,F}$ denote the inertia of the fixed-side secondary assembly, excluding the movable secondary sheave inertia I_M. The angular momentum of the secondary system about the shaft axis is then

$$L_s = I_{s,F}\omega_s + I_M\omega_M.$$

Applying Theory 2 (Newton's Second Law (Rotation)) in angular-momentum form gives

$$\tau_s - \tau_{\text{load}} = \frac{dL_s}{dt} = I_{s,F}\dot{\omega}_s + I_M\dot{\omega}_M. \tag{3.3.15}$$

The movable-sheave acceleration is not independent. From the helix kinematics derived in Equation (3.2.17),

$$\dot{\omega}_M = \dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2 - \frac{d\theta_s}{ds}\ddot{s}.$$

Substituting this relation into Equation (3.3.15) gives the secondary rotational equation in terms of the common dynamic variables:

Secondary Rotational Dynamics **(3.3.16)**

$$\tau_s - \tau_{\text{load}} = (I_{s,F} + I_M)\,\dot{\omega}_s - I_M\frac{d^2\theta_s}{ds^2}\dot{s}^2 - I_M\frac{d\theta_s}{ds}\ddot{s}$$

This form preserves the distinction between the fixed secondary components and the movable sheave. If the CVT is not shifting, then $\dot{s} = 0$ and $\ddot{s} = 0$, so the movable sheave rotates with the fixed sheave and the secondary inertia reduces to $I_{s,F} + I_M$. During shift, the helix introduces the additional coupling terms shown above.

3.3.6 Common Dynamic Relations and Remaining Closure

At this point, the model has four dynamic relations that are used regardless of the belt contact state. These are:

- primary rotational dynamics, Equation (3.3.13),
- belt transport dynamics, Equation (3.3.14),
- secondary rotational dynamics, Equation (3.3.16),
- axial shift dynamics, Equation (3.2.52).

The first three equations describe the torque path through the primary pulley, belt, and secondary pulley. The fourth equation carries the axial force balance into the same coupled system.

Together, the common dynamics involve the six unknown quantities

$$\dot{\omega}_p, \qquad \dot{\omega}_s, \qquad \dot{v}_b, \qquad \ddot{s}, \qquad \tau_p, \qquad \tau_s.$$

The four equations above therefore describe the body dynamics, but they do not yet determine all six unknowns. Two additional relations are still needed to specify how torque is exchanged at the primary and secondary belt–pulley contacts.

This is where the belt contact state enters the model. The equations derived so far describe how the bodies respond to applied forces and torques; the contact relations determine what forces and torques can actually be exchanged between the belt and each pulley. The following section begins with the gross sticking case, where the belt is assumed to remain kinematically compatible with both pulley contacts.

3.4 Gross Stick Compatibility and No-Slip Candidate

The previous section established the common equations of motion for the primary pulley, belt, secondary assembly, and shift coordinate. These equations describe how the bodies respond to applied forces and torques, but they do not yet specify the contact state between the belt and the pulleys.

This section considers the gross sticking case. In this contact state, the belt is assumed to remain adhered to both pulleys at their effective torque radii. The resulting solution is the *no-slip candidate*: the accelerations and contact torques that would be required if both belt–pulley contacts remained stuck.

This candidate is not automatically admissible. It is first computed from the sticking constraints, then checked against the available traction limits in the following section.

3.4.1 Gross Stick Compatibility

Let $j \in \{p, s\}$ denote either the primary or secondary pulley. Under Assumption 11 (Single Belt Transport Speed), the belt is represented by a single transport speed v_b. Under Assumption 12 (Uniform Contact Radius Around Wrap), each pulley contact is represented by a single effective torque radius $r_{j,\text{eff}}$.

If the belt is grossly stuck to pulley j, the belt transport speed equals the tangential surface speed of that pulley at the effective torque radius:

$$v_b(t) = r_{j,\text{eff}}(s(t))\,\omega_j(t), \qquad j \in \{p, s\}. \tag{3.4.1}$$

Differentiating Equation (3.4.1) gives the acceleration-level stick compatibility condition:

Gross Stick Compatibility **(3.4.2)**

$$\dot{v}_b(t) = \frac{dr_{j,\text{eff}}}{ds}\dot{s}(t)\,\omega_j(t) + r_{j,\text{eff}}(s(t))\,\dot{\omega}_j(t), \qquad j \in \{p, s\}.$$

This equation is the contact closure for a stuck belt–pulley interface. It states that the belt-line acceleration must match the acceleration of the pulley surface at the effective radius, including the contribution from changing pulley radius during shift.

Solving Equation (3.4.2) for the pulley angular acceleration gives

$$\dot{\omega}_j(t) = \frac{\dot{v}_b(t) - \dfrac{dr_{j,\text{eff}}}{ds}\dot{s}(t)\omega_j(t)}{r_{j,\text{eff}}(s(t))}, \qquad j \in \{p, s\}. \tag{3.4.3}$$

For the no-slip candidate, Equation (3.4.3) is applied at both the primary and secondary contacts.

3.4.2 No-Slip Candidate Solve

For compactness, define

$$r_p = r_{p,\text{eff}}(s), \qquad r_s = r_{s,\text{eff}}(s),$$

and

$$\dot{r}_p = \frac{dr_{p,\text{eff}}}{ds}\dot{s}, \qquad \dot{r}_s = \frac{dr_{s,\text{eff}}}{ds}\dot{s}.$$

Also define the effective shift inertia from Equation (3.2.52) as

$$M_{\text{shift}} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4} + I_M \left(\frac{d\theta_s}{ds}\right)^2. \tag{3.4.4}$$

This is not a new model assumption; it is the denominator of the axial shift equation already derived.

Stick-Compatible Angular Accelerations

Applying the stick condition from Equation (3.4.3) at the primary pulley gives

$$\dot{\omega}_{p,\text{ns}} = \frac{\dot{v}_{b,\text{ns}} - \dot{r}_p\omega_p}{r_p}. \tag{3.4.5}$$

Applying the same condition at the secondary pulley gives

$$\dot{\omega}_{s,\text{ns}} = \frac{\dot{v}_{b,\text{ns}} - \dot{r}_s\omega_s}{r_s}. \tag{3.4.6}$$

Thus, once the no-slip belt acceleration $\dot{v}_{b,\text{ns}}$ is known, both pulley angular accelerations are known.

Primary No-Slip Torque Demand

The primary rotational equation is

$$\tau_{\text{eng}} - \tau_p = I_p \dot{\omega}_p.$$

Substituting Equation (3.4.5) gives the primary contact torque required to maintain no slip:

Primary No-Slip Torque Demand Before Belt Solve **(3.4.7)**

$$\tau_{p,\text{ns}} = \tau_{\text{eng}} - I_p \left(\frac{\dot{v}_{b,\text{ns}} - \dot{r}_p \omega_p}{r_p} \right)$$

This is already expressed in terms of known state quantities and the single remaining unknown $\dot{v}_{b,\text{ns}}$.

Secondary No-Slip Torque Demand

The secondary torque demand is coupled to the axial shift equation because the secondary helix couples τ_s, $\dot{\omega}_s$, and $\ddot{s}$.

From Equation (3.2.52), the shift acceleration can be written directly as

$$\ddot{s}_{\text{ns}} = \frac{F_{p,\text{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \dfrac{d\theta_s}{ds} \left[\dfrac{\tau_{s,\text{ns}}}{2} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M \left(\dot{\omega}_{s,\text{ns}} - \dfrac{d^2\theta_s}{ds^2} \dot{s}^2 \right) \right]}{M_{\text{shift}}}. \tag{3.4.8}$$

The secondary rotational equation from Equation (3.3.16) is

$$\tau_{s,\text{ns}} - \tau_{\text{load}} = (I_{s,F} + I_M)\, \dot{\omega}_{s,\text{ns}} - I_M \frac{d^2\theta_s}{ds^2} \dot{s}^2 - I_M \frac{d\theta_s}{ds} \ddot{s}_{\text{ns}}. \tag{3.4.9}$$

Substituting Equation (3.4.8) into Equation (3.4.9), then collecting terms in $\tau_{s,\text{ns}}$, gives

$$\tau_{s,\text{ns}} = \frac{\begin{array}{c} \tau_{\text{load}} + \left[I_{s,F} + I_M - \dfrac{I_M^2 \left(\dfrac{d\theta_s}{ds} \right)^2}{M_{\text{shift}}} \right] \dot{\omega}_{s,\text{ns}} - I_M \dfrac{d^2\theta_s}{ds^2} \dot{s}^2 \\ - \dfrac{I_M \dfrac{d\theta_s}{ds}}{M_{\text{shift}}} \left[F_{p,\text{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \dfrac{d\theta_s}{ds} k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M \dfrac{d\theta_s}{ds} \dfrac{d^2\theta_s}{ds^2} \dot{s}^2 \right] \end{array}}{1 - \dfrac{I_M \left(\dfrac{d\theta_s}{ds} \right)^2}{2 M_{\text{shift}}}}. \tag{3.4.10}$$

Finally, substituting the secondary stick-compatible acceleration from Equation (3.4.6) gives the secondary torque demand in terms of the belt acceleration:

$$\tau_{s,\mathrm{ns}} = \frac{\begin{array}{c}\tau_{\mathrm{load}} + \left[I_{s,F} + I_M - \dfrac{I_M^2 \left(\dfrac{d\theta_s}{ds}\right)^2}{M_{\mathrm{shift}}} \right] \left(\dfrac{\dot{v}_{b,\mathrm{ns}} - \dot{r}_s \omega_s}{r_s} \right) - I_M \dfrac{d^2\theta_s}{ds^2} \dot{s}^2 \\ - \dfrac{I_M \dfrac{d\theta_s}{ds}}{M_{\mathrm{shift}}} \left[F_{p,\mathrm{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \dfrac{d\theta_s}{ds} k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M \dfrac{d\theta_s}{ds} \dfrac{d^2\theta_s}{ds^2} \dot{s}^2 \right]\end{array}}{1 - \dfrac{I_M \left(\dfrac{d\theta_s}{ds}\right)^2}{2M_{\mathrm{shift}}}}. \tag{3.4.11}$$

This expression is longer than the primary torque demand because the secondary torque is coupled through the helix to both secondary rotation and axial shift.

Explicit No-Slip Belt Acceleration

The belt transport equation is

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}.$$

Substituting Equation (3.4.7) and Equation (3.4.11), then collecting terms in $\dot{v}_{b,\mathrm{ns}}$, gives the explicit no-slip belt acceleration:

No-Slip Belt Acceleration **(3.4.12)**

$$\dot{v}_{b,\text{ns}} = \frac{\dfrac{\tau_{\text{eng}}}{r_p} + \dfrac{I_p \dot{r}_p \omega_p}{r_p^2} - \dfrac{1}{r_s} \dfrac{\tau_{\text{load}} - \left[I_{s,F} + I_M - \dfrac{I_M^2 \left(\dfrac{d\theta_s}{ds}\right)^2}{M_{\text{shift}}} \right] \dfrac{\dot{r}_s \omega_s}{r_s} - I_M \dfrac{d^2\theta_s}{ds^2} \dot{s}^2 - \dfrac{I_M \dfrac{d\theta_s}{ds}}{M_{\text{shift}}} \left[F_{p,\text{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \dfrac{d\theta_s}{ds} k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M \dfrac{d\theta_s}{ds} \dfrac{d^2\theta_s}{ds^2} \dot{s}^2 \right]}{1 - \dfrac{I_M \left(\dfrac{d\theta_s}{ds}\right)^2}{2M_{\text{shift}}}}}{m_b + \dfrac{I_p}{r_p^2} + \dfrac{1}{r_s} \dfrac{\left[I_{s,F} + I_M - \dfrac{I_M^2 \left(\dfrac{d\theta_s}{ds}\right)^2}{M_{\text{shift}}} \right] \dfrac{1}{r_s}}{1 - \dfrac{I_M \left(\dfrac{d\theta_s}{ds}\right)^2}{2M_{\text{shift}}}}}$$

This equation contains only quantities known from the current state, evaluated geometry, force models, and inertial parameters. It therefore determines the belt acceleration required by the no-slip candidate.

Recovered No-Slip Accelerations and Candidate Torques

Once $\dot{v}_{b,\text{ns}}$ has been computed from Equation (3.4.12), the remaining no-slip quantities follow in direct order.

The pulley angular accelerations are

$$\dot{\omega}_{p,\text{ns}} = \frac{\dot{v}_{b,\text{ns}} - \dot{r}_p \omega_p}{r_p}, \tag{3.4.13}$$

and

$$\dot{\omega}_{s,\text{ns}} = \frac{\dot{v}_{b,\text{ns}} - \dot{r}_s \omega_s}{r_s}. \tag{3.4.14}$$

The primary torque candidate is

Primary No-Slip Torque Demand **(3.4.15)**

$$\tau_{p,\mathrm{ns}} = \tau_{\mathrm{eng}} - I_p \left(\frac{\dot{v}_{b,\mathrm{ns}} - \dot{r}_p \omega_p}{r_p} \right)$$

The secondary torque candidate is

Secondary No-Slip Torque Demand **(3.4.16)**

$$\tau_{s,\mathrm{ns}} = \frac{\tau_{\mathrm{load}} + \left[I_{s,F} + I_M - \frac{I_M^2 \left(\frac{d\theta_s}{ds} \right)^2}{M_{\mathrm{shift}}} \right] \left(\frac{\dot{v}_{b,\mathrm{ns}} - \dot{r}_s \omega_s}{r_s} \right) - I_M \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{I_M \frac{d\theta_s}{ds}}{M_{\mathrm{shift}}} \left[F_{p,\mathrm{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M \frac{d\theta_s}{ds} \frac{d^2\theta_s}{ds^2} \dot{s}^2 \right]}{1 - \frac{I_M \left(\frac{d\theta_s}{ds} \right)^2}{2M_{\mathrm{shift}}}}$$

Finally, the no-slip shift acceleration is obtained from the axial shift equation:

No-Slip Shift Acceleration **(3.4.17)**

$$\ddot{s}_{\mathrm{ns}} = \frac{F_{p,\mathrm{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} \left[\frac{\tau_{s,\mathrm{ns}}}{2} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M \left(\dot{\omega}_{s,\mathrm{ns}} - \frac{d^2\theta_s}{ds^2} \dot{s}^2 \right) \right]}{M_{\mathrm{shift}}}$$

Together, these equations explicitly determine the no-slip candidate:

$$\dot{v}_{b,\mathrm{ns}}, \qquad \dot{\omega}_{p,\mathrm{ns}}, \qquad \dot{\omega}_{s,\mathrm{ns}}, \qquad \tau_{p,\mathrm{ns}}, \qquad \tau_{s,\mathrm{ns}}, \qquad \ddot{s}_{\mathrm{ns}}.$$

If both belt–pulley contacts were guaranteed to remain stuck, these equations would close the CVT dynamics. In that idealized case, the belt would act as a kinematic constraint between the two pulleys while the shift coordinate changes the effective ratio.

For a frictional belt drive, however, the required no-slip torques must still be checked against the available traction capacity.

3.4.3 Transition to Traction Limits

The no-slip candidate answers one question: what accelerations and contact torques are required if the belt remains stuck to both pulleys? It does not yet answer whether the contacts can physically support those torques.

The next section derives the tangential traction limits for the primary and secondary belt–pulley contacts. The candidate torques $\tau_{p,\mathrm{ns}}$ and $\tau_{s,\mathrm{ns}}$ can then be compared with those limits. If both contact torques are admissible, the no-slip candidate is used. If either required torque exceeds the available traction capacity, the corresponding sticking constraint is replaced by a slip relation.

3.5 Traction Limits and Slip Conditions

The previous section derived the no-slip candidate, including the contact torque demands $\tau_{p,\mathrm{ns}}$ and $\tau_{s,\mathrm{ns}}$. These are the torques that the belt–pulley interfaces would have to exchange if the belt remained grossly stuck to both pulleys.

Those torque demands are not automatically attainable. Torque is transmitted through frictional traction between the belt and the sheave faces, and that traction is limited by the normal load generated by axial clamping. Therefore, each belt–pulley contact can support only a finite contact torque before gross slip must occur.

The purpose of this section is to derive the traction checks used to decide whether the no-slip candidate is admissible. The important distinction is that the dynamic variables τ_p and τ_s are treated as contact torques, not as tight–slack span-tension differences. The corresponding span tension change can still be reconstructed after the contact torque is known, but the stick/slip admissibility check is posed directly in terms of frictional contact torque.

3.5.1 Contact-Torque Capacity

The same traction argument applies to both pulleys. Let

$$j \in \{p, s\},$$

where $j = p$ denotes the primary pulley and $j = s$ denotes the secondary pulley. At pulley j, the belt–pulley contact is represented by an effective torque radius $r_{j,\mathrm{eff}}$. The signed contact torque at that pulley is denoted τ_j.

Under Assumption 4 (Idealized Belt–Pulley Friction Law) and Assumption 5 (Constant Effective Friction Coefficient), the belt–pulley interface obeys a Coulomb friction law with constant coefficient μ. Under Assumption 7 (Gross Tangential Traction Approximation), the available friction is resolved in the gross tangential torque-transfer direction. Thus, at the differential level,

$$|\,\mathrm{d}F_{t,j}| \le \mu\,\mathrm{d}N_j, \tag{3.5.1}$$

where $\mathrm{d}F_{t,j}$ is the tangential contact force and $\mathrm{d}N_j$ is the local normal load.

Integrating over the wrap gives the bound on total tangential contact force,

$$|F_{t,j}| \le \mu N_{\phi,j}, \tag{3.5.2}$$

where

$$N_{\phi,j} \equiv \int_{\mathrm{wrap}} \mathrm{d}N_j \tag{3.5.3}$$

is the total normal load carried by the belt–pulley contact on pulley j.

The corresponding contact torque magnitude is

$$|\tau_j| = r_{j,\text{eff}} |F_{t,j}|. \tag{3.5.4}$$

Therefore, for a prescribed normal load, the maximum gross contact torque that can be supported without slip is

Gross Contact-Torque Capacity **(3.5.5)**

$$|\tau_j| \leq r_{j,\text{eff}} \mu N_{\phi,j}$$

This is the contact-torque capacity used for stick/slip admissibility. It bounds the torque exchanged directly at the belt–pulley interface.

Remark (Contact Torque Versus Span Tension)

The contact torque τ_j should not be confused with the torque-equivalent of the tight–slack tension difference across the wrap. During transient motion, some contact traction may be used to accelerate belt material on the wrap, so the contact force and the span tension rise need not be equal. The present model uses contact torque as the dynamic coupling variable. A span-tension reconstruction is given in **??**, but it is not used as the primary stick/slip admissibility condition.

Remark (Other Friction Models)

Although the present simulator adopts a Coulomb friction law for simplicity and analytic transparency, richer friction descriptions are available if greater fidelity near the stick–slip transition is desired. A common next step is a Stribeck-type law, in which the effective friction level varies with relative slip speed so that breakaway friction near zero speed can exceed the kinetic friction observed at larger slip speeds. Such a model can better represent engagement and low-slip behaviour than a purely constant Coulomb coefficient, while still remaining relatively simple to implement Armstrong-H'elouvry et al. (1994); Canudas de Wit et al. (1995).
For still higher fidelity, dynamic friction models introduce internal state variables that represent the evolving contact condition. The LuGre model is a standard example and can capture presliding displacement, hysteresis, friction lag, and smoother stick–slip transitions, while related multistate models such as the generalized Maxwell–slip model can represent more detailed microslip and memory effects Canudas de Wit et al. (1995); Al-Bender et al. (2005). These models are often more physically expressive, but they also require additional parameter identification and increase numerical and modelling complexity. For the present work, Coulomb friction is therefore retained as a deliberate low-complexity approximation, while Stribeck- or state-based friction models remain natural extensions for future refinement Armstrong-H'elouvry et al. (1994).

3.5.2 Normal Force from Axial Clamping

The torque capacity in Equation (3.5.5) depends on the total normal load $N_{\phi,j}$. We now express that normal load in terms of the axial clamping force generated by pulley j.

The belt is pressed between two opposing conical sheaves. The two inclined faces apply a net radial normal load on the belt determined by the axial clamp. Using the groove geometry from Equation (3.2.39),

$$F_{\mathrm{rad},j} = 2F_{\mathrm{ax},j} \tan\beta. \tag{3.5.6}$$

The radial load in Equation (3.5.6) represents the total inward force applied by the sheave faces onto the belt. This load is carried by the distributed normal contact forces along the wrap, so

$$N_{\phi,j} = F_{\mathrm{rad},j}. \tag{3.5.7}$$

Only compressive axial clamp can create normal load. To prevent an unphysical negative traction capacity, define the compressive part of any axial force as

$$\langle F\rangle_+ = \max(F, 0). \tag{3.5.8}$$

Therefore,

Wrap-Normal Load from Axial Clamp **(3.5.9)**

$$N_{\phi,j} = 2\,\langle F_{\mathrm{ax},j}\rangle_+ \tan\beta$$

Substituting Equation (3.5.9) into Equation (3.5.5) gives the gross contact-torque capacity

Pulley Contact-Torque Capacity **(3.5.10)**

$$|\tau_j| \le 2r_{j,\mathrm{eff}}\mu\,\langle F_{\mathrm{ax},j}\rangle_+ \tan\beta$$

For a prescribed axial clamp, this bound is symmetric in positive and negative contact torque. However, the admissible torque range of a complete pulley mechanism may still be asymmetric if the axial clamp itself depends on the signed contact torque. This distinction is important for the torque-reactive secondary pulley.

Remark (Wrap Angle in the Reduced Capacity Model)

The wrap angle does not appear explicitly in Equation (3.5.10) because $N_{\phi,j}$ is the total integrated normal load imposed by the clamp. Under the averaged-contact approximation, increasing wrap angle spreads this same integrated normal load over more contact length rather than increasing the total available gross friction force. Wrap angle still matters for local contact pressure, heat, wear, pressure distribution, and span-tension reconstruction, but it is not an independent multiplier in the reduced gross contact-torque capacity.

3.5.3 Primary Pulley No-Slip Admissibility

The generic contact-torque capacity law can now be specialized to the primary pulley. At this stage, the goal is only to test whether the primary contact torque demanded by the no-slip candidate can be supported by the available primary traction.

For compactness, write

$$r_p = r_{p,\text{eff}}(s).$$

The primary axial force is known explicitly from Equation (3.2.10):

$$F_{\text{ax},p}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p(x_{p,0} + s). \tag{3.5.11}$$

Using the static friction coefficient μ_s, the primary no-slip contact is admissible if

Primary No-Slip Admissibility **(3.5.12)**

$$|\tau_{p,\text{ns}}| \le 2 r_p \mu_s \left\langle F_{\text{ax},p}(s, \omega_p) \right\rangle_+ \tan\beta$$

This condition is symmetric in positive and negative primary contact torque because the primary axial force does not depend on the signed primary torque. If Equation (3.5.12) is satisfied, the primary contact can support the torque required by the no-slip candidate. If it is not satisfied, the primary contact cannot remain grossly stuck under the demanded contact torque.

3.5.4 Secondary Pulley No-Slip Admissibility

The secondary pulley follows the same contact-torque capacity law as the primary, but with one important difference: the secondary axial clamp is torque-reactive. Through the helix, the same signed secondary contact torque being checked for admissibility also contributes to the axial force that generates the available traction.

To keep this dependence explicit, write the secondary axial force as the function

$$\begin{aligned} F_{\text{ax},s}(s, \dot{s}, \tau_s, \dot{\omega}_s, \ddot{s}) &= k_{s,x}(x_{s,0} + s) \\ &\quad + \frac{\mathrm{d}\theta_s}{\mathrm{d}s} \left[\frac{\tau_s}{2} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - I_M \left(\dot{\omega}_s - \frac{\mathrm{d}^2\theta_s}{\mathrm{d}s^2} \dot{s}^2 - \frac{\mathrm{d}\theta_s}{\mathrm{d}s} \ddot{s} \right) \right]. \end{aligned} \tag{3.5.13}$$

For compactness, write

$$r_s = r_{s,\text{eff}}(s).$$

On the no-slip candidate branch, the quantities $\tau_{s,\text{ns}}$, $\dot{\omega}_{s,\text{ns}}$, and $\ddot{s}_{\text{ns}}$ have already been determined by the no-slip solve. Therefore, the secondary clamp used in the traction check is evaluated at the candidate values:

$$F_{\mathrm{ax},s}^{\mathrm{ns}} = F_{\mathrm{ax},s}(s, \dot{s}, \tau_{s,\mathrm{ns}}, \dot{\omega}_{s,\mathrm{ns}}, \ddot{s}_{\mathrm{ns}}).$$

Using the static friction coefficient μ_s, the secondary no-slip contact is admissible if

Secondary No-Slip Admissibility **(3.5.14)**

$$|\tau_{s,\mathrm{ns}}| \le 2r_s\mu_s \left\langle F_{\mathrm{ax},s}(s, \dot{s}, \tau_{s,\mathrm{ns}}, \dot{\omega}_{s,\mathrm{ns}}, \ddot{s}_{\mathrm{ns}})\right\rangle_+ \tan\beta$$

This condition is written as a magnitude check, but it is not assuming a symmetric secondary torque range. The axial force inside the right-hand side is evaluated using the signed no-slip torque $\tau_{s,\mathrm{ns}}$. Therefore, positive and negative secondary torques of equal magnitude generally produce different clamp forces and different available traction. The condition above is a pointwise admissibility test: it asks whether the specific signed no-slip candidate torque creates enough clamp to support itself.

If Equation (3.5.14) is satisfied, the secondary contact can support the torque required by the no-slip candidate. If it is not satisfied, the secondary contact cannot remain grossly stuck under the demanded contact torque.

Remark (Optional Secondary Torque Envelope)

For branch selection, the pointwise check in Equation (3.5.14) is sufficient because the no-slip solve has already produced a specific value of $\tau_{s,\mathrm{ns}}$. However, it can also be useful to reconstruct the full secondary torque envelope at a fixed instantaneous state, for example when plotting how the torque-reactive helix changes the available torque range.
At a fixed state, the secondary stick envelope is the set of torques satisfying

$$|\tau_s| \le 2r_s\mu_s \left\langle F_{\mathrm{ax},s}(s, \dot{s}, \tau_s, \dot{\omega}_s, \ddot{s})\right\rangle_+ \tan\beta. \tag{3.5.15}$$

Because $F_{\mathrm{ax},s}$ contains the signed torque τ_s, the set defined by Equation (3.5.15) is generally not symmetric about zero. One torque direction may increase clamp through the helix, while the opposite direction may reduce clamp. This envelope is useful for diagnostic plots or design interpretation, but the branch-selection logic only needs the pointwise candidate check in Equation (3.5.14).

3.5.5 Admissibility of the No-Slip Candidate

The full no-slip candidate is admissible only if both pulley contacts satisfy their static traction checks:

$$|\tau_{p,\mathrm{ns}}| \le 2r_p\mu_s \left\langle F_{\mathrm{ax},p}(s, \omega_p)\right\rangle_+ \tan\beta, \tag{3.5.16}$$

and

$$|\tau_{s,\mathrm{ns}}| \le 2r_s\mu_s \left\langle F_{\mathrm{ax},s}(s, \dot{s}, \tau_{s,\mathrm{ns}}, \dot{\omega}_{s,\mathrm{ns}}, \ddot{s}_{\mathrm{ns}})\right\rangle_+ \tan\beta. \tag{3.5.17}$$

When both conditions hold, the no-slip candidate can be accepted as the contact state for that instant. If either condition fails, at least one contact cannot maintain gross sticking and the corresponding contact relation must be replaced by a slip relation.

3.5.6 Kinetic Traction Bound for Slipping Contacts

The same contact-torque capacity structure is used when a contact is sliding, but the kinetic friction coefficient μ_k replaces the static coefficient μ_s. Thus, the magnitude of the slipping contact torque at pulley j is

Kinetic Contact-Torque Magnitude **(3.5.18)**

$$|\tau_{j,\text{slip}}| = 2r_{j,\text{eff}}\mu_k \langle F_{\text{ax},j}\rangle_+ \tan\beta$$

The sign of the kinetic torque is chosen so that the friction force opposes the relative sliding motion between the belt and pulley surface. The exact sign convention for τ_p and τ_s is applied in the contact-state selection section, where the primary and secondary slip branches are written explicitly.

3.5.7 Section Conclusion and Transition

This section derived the traction checks needed to test the no-slip candidate from Section 3.4.2. The stick/slip admissibility check was posed directly in terms of contact torque because the rotational and belt transport equations use τ_p and τ_s as the torques exchanged at the belt–pulley contacts.

The primary check is symmetric because the primary axial force is independent of the signed primary contact torque. The secondary check is not generally symmetric in torque because the helix makes the secondary axial clamp depend on the signed secondary contact torque. For branch selection, this is handled by evaluating the secondary clamp directly on the signed no-slip candidate.

If both pulley contacts satisfy their admissibility conditions, the no-slip candidate is accepted. If one or both contacts cannot support their demanded contact torques, the corresponding contact must be described by a slipping traction law. The next section develops the contact-state selection rules and the resulting slip-branch dynamics.

3.6 Slip Branch Dynamics and Contact-State Selection

The previous section determined when the no-slip torque demands can be supported by the available belt–pulley traction. If the primary and secondary contacts both satisfy their admissibility conditions, the no-slip candidate is physically consistent and the belt remains adhered to both pulleys.

When one of these admissibility conditions fails, the no-slip candidate can no longer be used as the final torque-transfer law. The contact that exceeds its available traction cannot supply the demanded torque while remaining stuck, so relative motion begins at that belt–pulley interface. Once this occurs, the contact torque is no longer determined by the no-slip demand. Instead, it is set by the kinetic traction available at the slipping contact.

This section develops the rules used to select the active contact state and the corresponding equations of motion. The same three bodies introduced in Section 3.3.1 are retained: the primary pulley, the belt, and the secondary pulley. What changes from one branch to another is the contact condition imposed at each pulley. A stuck contact enforces belt-speed compatibility, while a slipping contact applies a saturated kinetic traction torque in the direction opposing relative motion.

The goal is therefore to complete the torque-transfer closure. We first define the relative belt–pulley speeds used to identify sliding, then state the rules for entering, maintaining, and exiting slip. The remaining subsections derive the branch dynamics for primary slip, secondary slip, and simultaneous slip at both contacts.

3.6.1 Relative Contact Speeds and Slip-State Logic

To determine whether a belt–pulley contact is sticking or slipping, the model compares the belt transport speed to the local surface speed imposed by each pulley.

For the primary pulley, define the relative contact speed as

$$v_{\mathrm{rel},p} = r_p\omega_p - v_b, \tag{3.6.1}$$

where $r_p = r_{p,\mathrm{eff}}(s)$. Thus, $v_{\mathrm{rel},p} > 0$ means the primary pulley surface is moving faster than the belt at the contact radius and tends to drive the belt forward.

For the secondary pulley, define

$$v_{\mathrm{rel},s} = v_b - r_s\omega_s, \tag{3.6.2}$$

where $r_s = r_{s,\mathrm{eff}}(s)$. Thus, $v_{\mathrm{rel},s} > 0$ means the belt is moving faster than the secondary pulley surface and tends to drive the secondary forward.

These definitions are chosen so that positive relative speed corresponds to the usual forward torque-transfer direction at each contact: primary-to-belt on the primary side, and belt-to-secondary on the secondary side.

A contact can remain stuck only if its relative contact speed is zero, or within a small numerical tolerance:

$$|v_{\mathrm{rel},j}| \le \varepsilon_v, \qquad j \in \{p, s\}. \tag{3.6.3}$$

If both contacts satisfy Equation (3.6.3), the model first tests the no-slip candidate derived in **??**. This candidate is accepted only if both pulley contacts also satisfy the traction admissibility conditions derived in Section 3.5:

$$|\tau_{p,\mathrm{ns}}| \le \tau_{p,\max}^{\mathrm{stick}}, \tag{3.6.4}$$

and

$$\tau_{s,-}^{\mathrm{stick}} \le \tau_{s,\mathrm{ns}} \le \tau_{s,+}^{\mathrm{stick}}. \tag{3.6.5}$$

If these conditions hold, the no-slip candidate becomes the active torque-transfer branch:

$$\tau_p = \tau_{p,\mathrm{ns}}, \qquad \tau_s = \tau_{s,\mathrm{ns}}, \qquad \dot{v}_b = \dot{v}_{b,\mathrm{ns}}.$$

If either contact fails its admissibility check, that contact cannot support the required no-slip torque and must enter slip. The direction of slip is determined by the direction in which the demanded torque exceeds the admissible interval. For the primary pulley,

$$\tau_{p,\mathrm{ns}} > \tau_{p,\max}^{\mathrm{stick}} \quad \Rightarrow \quad \text{positive primary slip,}$$

while

$$\tau_{p,\text{ns}} < -\tau_{p,\max}^{\text{stick}} \quad \Rightarrow \quad \text{negative primary slip.}$$

For the secondary pulley,

$$\tau_{s,\text{ns}} > \tau_{s,+}^{\text{stick}} \quad \Rightarrow \quad \text{positive secondary slip,}$$

while

$$\tau_{s,\text{ns}} < \tau_{s,-}^{\text{stick}} \quad \Rightarrow \quad \text{negative secondary slip.}$$

Once a contact is already sliding, its state is determined primarily by its relative contact speed. If

$$|v_{\text{rel},j}| > \varepsilon_v,$$

then contact j is treated as slipping, and its torque is set by kinetic traction rather than by the no-slip demand. The slip direction is recorded by

$$\sigma_j = \operatorname{sgn}(v_{\text{rel},j}), \qquad j \in \{p, s\}. \tag{3.6.6}$$

Slip persists while the relative contact speed remains nonzero. During this time, the slipping contact applies a saturated kinetic traction torque in the direction opposing relative motion. The no-slip candidate is not reaccepted until the relative speed returns to the sticking tolerance and the static admissibility conditions can again be satisfied.

This gives three possible slipping branches:

- primary slip, where the primary contact slides and the secondary contact remains stuck;
- secondary slip, where the secondary contact slides and the primary contact remains stuck;
- two-contact slip, where both pulley contacts slide simultaneously.

The following subsections derive the equations of motion for each branch.

Remark (Power Flow During Slip)

In the no-slip branch, belt-speed compatibility constrains the primary pulley, belt, and secondary pulley to move together kinematically. Once slip occurs, that compatibility is lost at the slipping contact. The contact torque is then limited by kinetic traction, while the pulley and belt speeds evolve according to their own equations of motion.

As a result, the instantaneous mechanical power at the primary contact, $\tau_p\omega_p$, does not need to match the instantaneous mechanical power at the secondary contact, $\tau_s\omega_s$. The difference is associated with changes in stored kinetic energy of the rotating components and belt, together with slip dissipation. Thus, an apparent short-term excess of output-side power does not imply energy creation; it reflects energy being drawn from inertia while the slipping contact breaks the ideal no-slip power constraint.

3.6.2 Kinetic Traction Law for a Slipping Contact

Once a belt–pulley contact is slipping, the torque at that contact is no longer determined by the no-slip torque demand. Instead, the contact applies the largest kinetic traction torque available in the direction opposing relative motion.

The generic torque-capacity expression from Equation (3.5.10) still applies, but the static coefficient μ_s is replaced by the kinetic coefficient μ_k. For a slipping contact on pulley $j \in \{p, s\}$, define

$$\tau_{j,\max}^{\text{slip}}(\dot{v}_b) = r_j \left[2\mu_k F_{\text{ax},j} \tan\beta - \rho_b A_b \phi_j \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_{j,\text{cm}}\, v_b\right)\right], \tag{3.6.7}$$

where $r_j = r_{j,\text{eff}}(s)$. The notation $\tau_{j,\max}^{\text{slip}}(\dot{v}_b)$ emphasizes that the available slipping torque is not generally a fixed number; it depends on the belt acceleration through the inertial correction term.

The direction of the slipping torque is determined by the relative contact speed defined in Section 3.6.1. Let

$$\sigma_j = \operatorname{sgn}(v_{\text{rel},j}), \qquad j \in \{p, s\}. \tag{3.6.8}$$

With the relative-speed definitions adopted in Equation (3.6.1) and Equation (3.6.2), $\sigma_j > 0$ corresponds to the usual forward torque-transfer direction at contact j. The slipping contact torque is then modeled as the saturated kinetic traction torque

$$\tau_j = \sigma_j\, \tau_{j,\max}^{\text{slip}}(\dot{v}_b). \tag{3.6.9}$$

Equation Equation (3.6.9) should not be interpreted as a completed solution for τ_j, because the right-hand side still contains $\dot{v}_b$. In a slip branch, the belt acceleration must be solved simultaneously with the saturated contact torque using the belt equation of motion Equation (3.3.14). The following subsections carry out this solve for primary slip, secondary slip, and simultaneous slip at both contacts.

3.6.3 Primary-Slip Branch

We first consider the branch where the primary contact is slipping while the secondary contact remains stuck. In this case, the primary pulley no longer enforces belt-speed compatibility. Instead, the primary contact torque is set by the saturated kinetic traction law from Section 3.6.2.

The secondary contact, however, is still adhered. Therefore, the secondary pulley remains kinematically compatible with the belt:

$$\dot{\omega}_s = \frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s}, \tag{3.6.10}$$

Using the secondary rotational equation of motion Equation (3.3.16), the secondary contact torque is then

$$\tau_s = \tau_{\text{load}} + I_s \left(\frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s}\right). \tag{3.6.11}$$

At the slipping primary contact, let

$$\sigma_p = \operatorname{sgn}(v_{\text{rel},p}).$$

Applying Equation (3.6.9) with $j = p$ gives

$$\tau_p = \sigma_p r_p \left[2\mu_k F_{\text{ax},p} \tan\beta - \rho_b A_b \phi_p \left(r_{p,\text{cm}} \dot{v}_b + \dot{r}_{p,\text{cm}} v_b\right)\right], \tag{3.6.12}$$

where $r_p = r_{p,\text{eff}}(s)$.

The remaining unknown is the belt acceleration $\dot{v}_b$. This is found by substituting Equation (3.6.11) and Equation (3.6.12) into the belt equation of motion Equation (3.3.14):

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}. \tag{3.6.13}$$

Substituting gives

$$\begin{aligned} m_b \dot{v}_b = \sigma_p \left[2\mu_k F_{\mathrm{ax},p} \tan\beta - \rho_b A_b \phi_p \left(r_{p,\mathrm{cm}} \dot{v}_b + \dot{r}_{p,\mathrm{cm}} v_b \right) \right] \\ - \frac{\tau_{\mathrm{load}}}{r_s} - \frac{I_s}{r_s} \left(\frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s} \right). \end{aligned} \tag{3.6.14}$$

Collecting the $\dot{v}_b$ terms yields the primary-slip belt acceleration:

Primary-Slip Belt Acceleration **(3.6.15)**

$$\dot{v}_{b,p-\mathrm{slip}} = \frac{\sigma_p \left[2\mu_k F_{\mathrm{ax},p} \tan\beta - \rho_b A_b \phi_p \dot{r}_{p,\mathrm{cm}} v_b \right] - \dfrac{\tau_{\mathrm{load}}}{r_s} + \dfrac{I_s \dot{r}_s \omega_s}{r_s^2}}{m_b + \sigma_p \rho_b A_b \phi_p r_{p,\mathrm{cm}} + \dfrac{I_s}{r_s^2}}$$

Once $\dot{v}_{b,p-\mathrm{slip}}$ is known, the branch torques follow by substitution into Equation (3.6.12) and Equation (3.6.11). The pulley accelerations are then

$$\dot{\omega}_p = \frac{\tau_{\mathrm{eng}} - \tau_p}{I_p}, \tag{3.6.16}$$

and

$$\dot{\omega}_s = \frac{\dot{v}_{b,p-\mathrm{slip}} - \dot{r}_s \omega_s}{r_s}. \tag{3.6.17}$$

Thus, in the primary-slip branch, the primary contact torque is set by kinetic traction, the secondary contact still enforces belt-speed compatibility, and the belt equation of motion determines the resulting belt acceleration. This branch remains valid only while the secondary contact can remain adhered. If the secondary contact also fails its sticking condition, the model must transition to the two-contact slip branch.

3.6.4 Secondary-Slip Branch

We next consider the branch where the secondary contact is slipping while the primary contact remains stuck. In this case, the secondary contact torque is set by saturated kinetic traction, while the primary contact still enforces belt-speed compatibility.

The primary contact remains adhered, so the primary pulley must satisfy

$$\dot{\omega}_p = \frac{\dot{v}_b - \dot{r}_p \omega_p}{r_p}, \tag{3.6.18}$$

where $r_p = r_{p,\text{eff}}(s)$. Using the primary rotational equation of motion Equation (3.3.13), the primary contact torque is therefore

$$\tau_p = \tau_{\text{eng}} - I_p \left(\frac{\dot{v}_b - \dot{r}_p \omega_p}{r_p} \right). \tag{3.6.19}$$

At the slipping secondary contact, let

$$\sigma_s = \text{sgn}(v_{\text{rel},s}).$$

Applying Equation (3.6.9) with $j = s$ gives

$$\tau_s = \sigma_s r_s \left[2\mu_k F_{\text{ax},s}(s, \tau_s) \tan\beta - \rho_b A_b \phi_s \left(r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b \right) \right], \tag{3.6.20}$$

where $r_s = r_{s,\text{eff}}(s)$.

Unlike the primary slip branch, this expression is implicit in τ_s because the secondary axial force is torque-reactive. Substituting the secondary axial force model from Equation (3.2.26) gives

$$\begin{aligned}\tau_s = \sigma_s r_s \Bigg[&\mu_k \tan\beta \left(\tau_s + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) \right) \frac{\mathrm{d}\theta_s}{\mathrm{d}s} \\ &+ 2\mu_k \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \left(r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b \right) \Bigg].\end{aligned} \tag{3.6.21}$$

Collecting the τ_s terms gives

$$\tau_s = \frac{\sigma_s r_s \left[\mu_k \tan\beta\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_k \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \left(r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b \right) \right]}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}. \tag{3.6.22}$$

The remaining unknown is the belt acceleration $\dot{v}_b$. Substituting Equation (3.6.19) and Equation (3.6.22) into the belt equation of motion Equation (3.3.14),

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s},$$

and collecting the $\dot{v}_b$ terms gives the secondary-slip belt acceleration:

Secondary-Slip Belt Acceleration **(3.6.23)**

$$\dot{v}_{b,s-\text{slip}} = \frac{\dfrac{\tau_{\text{eng}}}{r_p} + \dfrac{I_p \dot{r}_p \omega_p}{r_p^2} - \dfrac{\sigma_s \left[\mu_k \tan\beta\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_k \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \dot{r}_{s,\text{cm}} v_b \right]}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}}{m_b + \dfrac{I_p}{r_p^2} - \dfrac{\sigma_s \rho_b A_b \phi_s r_{s,\text{cm}}}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}}$$

Once $\dot{v}_{b,s-\text{slip}}$ is known, the branch torques follow by substitution into Equation (3.6.19) and Equation (3.6.22). The pulley accelerations are then

$$\dot{\omega}_p = \frac{\dot{v}_{b,s-\text{slip}} - \dot{r}_p \omega_p}{r_p}, \tag{3.6.24}$$

and

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}}{I_s}. \tag{3.6.25}$$

Thus, in the secondary-slip branch, the secondary contact torque is set by kinetic traction, including the torque-reactive helix feedback, while the primary contact still enforces belt-speed compatibility. This branch remains valid only while the primary contact can remain adhered. If the primary contact also fails its sticking condition, the model must transition to the two-contact slip branch.

3.6.5 Two-Contact Slip Branch

Finally, consider the branch where both belt–pulley contacts are slipping. In this case, neither pulley enforces belt-speed compatibility. Both contact torques are instead determined by saturated kinetic traction, and the belt acceleration is obtained directly from the belt equation of motion.

Let

$$\sigma_p = \operatorname{sgn}(v_{\text{rel},p}), \qquad \sigma_s = \operatorname{sgn}(v_{\text{rel},s}).$$

At the primary contact, applying Equation (3.6.9) with $j = p$ gives

$$\tau_p = \sigma_p r_p \left[2\mu_k F_{\text{ax},p} \tan\beta - \rho_b A_b \phi_p \left(r_{p,\text{cm}} \dot{v}_b + \dot{r}_{p,\text{cm}} v_b\right)\right], \tag{3.6.26}$$

where $r_p = r_{p,\text{eff}}(s)$.

At the secondary contact, applying Equation (3.6.9) with $j = s$ gives

$$\tau_s = \sigma_s r_s \left[2\mu_k F_{\text{ax},s}(s, \tau_s) \tan\beta - \rho_b A_b \phi_s \left(r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b\right)\right], \tag{3.6.27}$$

where $r_s = r_{s,\text{eff}}(s)$. As in Section 3.6.4, the secondary expression is implicit because the secondary axial force is torque-reactive. Substituting Equation (3.2.26) and collecting the τ_s terms gives

$$\tau_s = \frac{\sigma_s r_s \left[\mu_k \tan\beta \, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_k \tan\beta \, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \left(r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b\right)\right]}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}. \tag{3.6.28}$$

The belt equation of motion is still

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}.$$

Substituting Equation (3.6.26) and Equation (3.6.28), then collecting the $\dot{v}_b$ terms, gives the two-contact-slip belt acceleration:

Two-Contact-Slip Belt Acceleration **(3.6.29)**

$$\dot{v}_{b,ps-\text{slip}} = \frac{\sigma_p \left[2\mu_k F_{\text{ax},p} \tan\beta - \rho_b A_b \phi_p \dot{r}_{p,\text{cm}} v_b\right] - \dfrac{\sigma_s \left[\mu_k \tan\beta\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_k \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \dot{r}_{s,\text{cm}} v_b\right]}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}}{m_b + \sigma_p \rho_b A_b \phi_p r_{p,\text{cm}} - \dfrac{\sigma_s \rho_b A_b \phi_s r_{s,\text{cm}}}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}}$$

Once $\dot{v}_{b,ps-\text{slip}}$ is known, the branch torques follow by substitution into Equation (3.6.26) and Equation (3.6.28). Since neither contact is adhered, the pulley accelerations are obtained directly from the rotational equations of motion:

$$\dot{\omega}_p = \frac{\tau_{\text{eng}} - \tau_p}{I_p}, \tag{3.6.30}$$

and

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}}{I_s}. \tag{3.6.31}$$

Thus, in the two-contact slip branch, both pulley contacts are governed by kinetic traction, and the belt is not kinematically constrained to either pulley. The belt acceleration is determined only by the balance of the two saturated contact torques through the belt equation of motion.

3.6.6 Contact-State Selection

The preceding subsections define the possible torque-transfer closures. The active branch is selected by checking which contacts can remain adhered.

Let the primary sticking condition be

$$\mathcal{S}_p \equiv (|v_{\text{rel},p}| \le \varepsilon_v) \wedge \left(|\tau_{p,\text{ns}}| \le \tau_{p,\max}^{\text{stick}}\right),$$

and let the secondary sticking condition be

$$\mathcal{S}_s \equiv (|v_{\text{rel},s}| \le \varepsilon_v) \wedge \left(\tau_{s,-}^{\text{stick}} \le \tau_{s,\text{ns}} \le \tau_{s,+}^{\text{stick}}\right).$$

The branch closure for the belt acceleration and contact torques is then

$$(\dot{v}_b, \tau_p, \tau_s) = \begin{cases} (\dot{v}_{b,\mathrm{ns}}, \tau_{p,\mathrm{ns}}, \tau_{s,\mathrm{ns}})\,, & \mathcal{S}_p \wedge \mathcal{S}_s, \\ (\dot{v}_{b,p-\mathrm{slip}}, \tau_p \text{ from } \textit{Equation } (3.6.12), \tau_s \text{ from } \textit{Equation } (3.6.11))\,, & \neg\mathcal{S}_p \wedge \mathcal{S}_s, \\ (\dot{v}_{b,s-\mathrm{slip}}, \tau_p \text{ from } \textit{Equation } (3.6.19), \tau_s \text{ from } \textit{Equation } (3.6.22))\,, & \mathcal{S}_p \wedge \neg\mathcal{S}_s, \\ (\dot{v}_{b,ps-\mathrm{slip}}, \tau_p \text{ from } \textit{Equation } (3.6.26), \tau_s \text{ from } \textit{Equation } (3.6.28))\,, & \neg\mathcal{S}_p \wedge \neg\mathcal{S}_s. \end{cases} \tag{3.6.32}$$

The corresponding belt accelerations are given by Equation (3.4.12), Equation (3.6.15), Equation (3.6.23), and Equation (3.6.29), respectively.

Thus, when both contacts can remain adhered, the model uses the no-slip torque demands. When one or both contacts cannot remain adhered, the corresponding slip branch replaces the no-slip demand with saturated kinetic traction and solves the belt acceleration from the belt equation of motion.

3.6.7 Section Summary

This section completed the torque-transfer closure for cases where the no-slip candidate cannot be maintained. Relative contact speeds were introduced to identify whether each belt–pulley interface is adhered or sliding, and kinetic traction was used to define the torque applied by a slipping contact.

The resulting branch structure separates the possible contact states into no-slip, primary slip, secondary slip, and two-contact slip. In each slipping case, the saturated contact torque replaces the no-slip demand, and the belt equation of motion determines the corresponding belt acceleration.

Together with the no-slip demand from **??** and the traction admissibility conditions from Section 3.5, the branch selection law in Equation (3.6.32) provides the final torque-transfer rule needed by the complete CVT dynamic model.

3.7 Complete CVT Dynamic Model

This section collects the results of the preceding derivations into one closed nonlinear dynamical model. No new physics is introduced here. The purpose is to summarize, in one place,

1. the state variables,
2. the geometric and contact quantities computed from those states,
3. the torque-transfer closure,
4. and the final governing equations of motion.

3.7.1 State Definition

The state vector is

$$\mathbf{x}(t) = \big(\omega_p,\ \omega_s,\ s,\ \dot{s},\ v_b\big), \tag{3.7.1}$$

where

- ω_p is the primary pulley angular velocity,

- ω_s is the secondary pulley angular velocity,
- s is the axial shift coordinate,
- $\dot{s}$ is the axial shift velocity,
- v_b is the belt transport speed.

3.7.2 Geometry and Contact Kinematics

At each instant, the shift coordinate s determines the pulley geometry through the belt-length constraint and the conical kinematic relations derived in Section 3.1. In particular,

$$r_{p,\text{eff}}, \quad r_{s,\text{eff}}, \quad r_{p,\text{cm}}, \quad r_{s,\text{cm}}, \quad \phi_p, \quad \phi_s$$

are functions of s. For compactness in the complete model, write

$$r_p = r_{p,\text{eff}}(s), \qquad r_s = r_{s,\text{eff}}(s).$$

The geometric CVT ratio remains

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \tag{3.7.2}$$

Its time derivative is

$$\dot{R} = \dot{R}(s, \dot{s}), \tag{3.7.3}$$

as given by **??**.

The relative contact speeds at the two pulley interfaces are

$$v_{\text{rel},p} = r_p \omega_p - v_b, \qquad v_{\text{rel},s} = v_b - r_s \omega_s, \tag{3.7.4}$$

as defined in Equation (3.6.1) and Equation (3.6.2). These quantities determine whether each contact is eligible to remain adhered or must be treated as sliding.

3.7.3 Torque-Transfer Closure

The torque-transfer subsystem is closed by determining the primary contact torque τ_p, the secondary contact torque τ_s, and the belt acceleration $\dot{v}_b$. These quantities are computed algebraically from the current state and contact branch.

The no-slip candidate derived in Section 3.4.2 provides

$$\dot{v}_{b,\text{ns}}, \qquad \tau_{p,\text{ns}}, \qquad \tau_{s,\text{ns}},$$

from Equation (3.4.12), Equation (3.4.15), and Equation (3.4.16). These are the belt acceleration and contact torque demands required for adhered motion at both pulleys.

The admissibility conditions derived in Section 3.5.3 and Section 3.5.4 determine whether those no-slip demands can be supported. The primary contact is admissible when

$$|\tau_{p,\text{ns}}| \le \tau_{p,\max}^{\text{stick}}, \tag{3.7.5}$$

as in Equation (3.5.12). The secondary contact is admissible when

$$\tau_{s,-}^{\text{stick}} \le \tau_{s,\text{ns}} \le \tau_{s,+}^{\text{stick}}, \tag{3.7.6}$$

as in Equation (3.5.14).

The active contact branch is selected using the closure law from Section 3.6.6. In compact form,

$$(\dot{v}_b, \tau_p, \tau_s) = \text{the branch output selected by Equation (3.6.32).} \tag{3.7.7}$$

If both contacts remain adhered, the branch output is

$$(\dot{v}_b, \tau_p, \tau_s) = (\dot{v}_{b,\text{ns}}, \tau_{p,\text{ns}}, \tau_{s,\text{ns}}).$$

If one or both contacts slide, the corresponding slip branch replaces the no-slip demand with saturated kinetic traction and solves $\dot{v}_b$ from the belt equation of motion. The primary-slip, secondary-slip, and two-contact-slip belt accelerations are given by Equation (3.6.15), Equation (3.6.23), and Equation (3.6.29), respectively.

3.7.4 Governing ODEs

Once the branch closure has supplied $\dot{v}_b$, τ_p, and τ_s, the primary pulley equation of motion is

$$\dot{\omega}_p = \frac{\tau_{\text{eng}}(\omega_p) - \tau_p}{I_p}. \tag{3.7.8}$$

The secondary pulley equation of motion is

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}(\omega_s)}{I_s}. \tag{3.7.9}$$

The shift-position kinematics are

$$\frac{\mathrm{d}}{\mathrm{d}t} s = \dot{s}. \tag{3.7.10}$$

The axial shift acceleration is obtained from Equation (3.2.52):

$$\ddot{s} = \frac{\Big(F_{p,\text{ax}}(s, \omega_p) + F_{c,p}(s, v_b)\Big) - \Big(F_{s,\text{ax}}(s, \tau_s) + F_{c,s}(s, v_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}. \tag{3.7.11}$$

Finally, the belt transport speed evolves according to the branch-selected belt acceleration:

$$\frac{\mathrm{d}}{\mathrm{d}t} v_b = \dot{v}_b, \qquad \dot{v}_b \text{ from } \textit{Equation (3.7.7)}. \tag{3.7.12}$$

3.7.5 Model Evaluation Sequence

Given the current state

$$\mathbf{x}(t) = \big(\omega_p, \omega_s, s, \dot{s}, v_b\big),$$

the model is evaluated in the following order:

1. Compute the pulley geometry from s, including r_p, r_s, $r_{p,\text{cm}}$, $r_{s,\text{cm}}$, ϕ_p, and ϕ_s.
2. Compute the engine torque $\tau_{\text{eng}}(\omega_p)$ and the reflected load torque $\tau_{\text{load}}(\omega_s)$.

3. Compute the no-slip candidate

$$\dot{v}_{b,\mathrm{ns}}, \qquad \tau_{p,\mathrm{ns}}, \qquad \tau_{s,\mathrm{ns}},$$

using Equation (3.4.12), Equation (3.4.15), and Equation (3.4.16).

4. Compute the primary and secondary no-slip admissibility quantities from Section 3.5.3 and Section 3.5.4.

5. Compute the relative contact speeds $v_{\mathrm{rel},p}$ and $v_{\mathrm{rel},s}$ using Equation (3.7.4).

6. Select the active torque-transfer branch using Equation (3.6.32). This determines the applied values of

$$\dot{v}_b, \qquad \tau_p, \qquad \tau_s.$$

7. Evaluate the ODE right-hand sides Equation (3.7.8), Equation (3.7.9), Equation (3.7.10), Equation (3.7.11), and Equation (3.7.12).

3.7.6 Complete Nonlinear ODE System

The complete closed CVT model is therefore

$$\boxed{\begin{aligned}
\dot{\omega}_p &= \frac{\tau_{\mathrm{eng}}(\omega_p) - \tau_p}{I_p}, \\
\dot{\omega}_s &= \frac{\tau_s - \tau_{\mathrm{load}}(\omega_s)}{I_s}, \\
\frac{\mathrm{d}}{\mathrm{d}t} s &= \dot{s}, \\
\frac{\mathrm{d}}{\mathrm{d}t} \dot{s} &= \frac{\Big(F_{p,\mathrm{ax}}(s,\omega_p) + F_{c,p}(s,v_b)\Big) - \Big(F_{s,\mathrm{ax}}(s,\tau_s) + F_{c,s}(s,v_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}, \\
\frac{\mathrm{d}}{\mathrm{d}t} v_b &= \dot{v}_b.
\end{aligned}} \tag{3.7.13}$$

with the algebraic closure

$$(\dot{v}_b, \tau_p, \tau_s) = \text{the active branch output from Equation (3.6.32)}.$$

In summary, the state determines the pulley geometry and relative contact speeds. The no-slip candidate determines the torque demand required for adhered motion. The traction admissibility conditions determine whether that candidate can be maintained. If not, the appropriate slip branch supplies saturated contact torques and the resulting belt acceleration. These branch-selected quantities then feed back into the rotational, belt, and axial shift dynamics, closing the complete CVT state-space model.

Chapter 4

Results and Discussion

4.1 Simulation Results

This section compares four launch simulations: a nominal baseline case, a lighter-flyweight case, a steeper-helix case with reduced secondary torque reactivity, and a combined case with both a shallower initial primary ramp and a steeper helix. The purpose of these results is not to match experimental vehicle data directly, but to show that the model produces mechanically interpretable launch behaviour and responds to tuning changes in physically sensible ways.

The results are organized around the launch shift curves of these four cases. These curves first provide an overall picture of the launch, including engagement, low-ratio operation, active upshift, and high-ratio behaviour. The ratio terminology follows **??**: low ratio means larger R (greater speed reduction), and high ratio means smaller R. following subsections then explain those trends by examining the axial force balance across the CVT and the resulting torque-transfer and slip behaviour. In particular, the combined case is included to show a clearly slip-dominated launch, in which the model predicts loss of the low-ratio hold and sustained slip through the early portion of the launch.

Taken together, the four cases are intended to show three things: first, that the baseline tuning produces a coherent and interpretable launch trajectory; second, that changes to primary and secondary tuning shift the launch behaviour in the expected direction; and third, that when the tuning is made sufficiently unfavorable, the model captures the transition to prolonged

slipping rather than continuing to predict an unrealistic no-slip response. The implementation used to generate these simulations, along with repository, release, and data-access details, is documented in Section 4.2.

4.1.1 Simulation Cases and Phase Convention

The results section compares four full-throttle launch simulations from rest. One case is treated as the nominal baseline, while the remaining three are targeted tuning variations chosen to isolate specific changes in CVT behaviour. All cases use the same vehicle, gearbox, tire, and engine model parameters; they differ only in the clutch tuning parameters noted below.

Table 4.1: Launch cases considered in the results section. All parameters not listed remain equal to the nominal baseline.

Case	Description	Purpose in results section
D	Default tuning	Baseline reference case
F	Lighter flyweights	Shows the effect of reduced primary centrifugal clamping
H	Steeper helix	Shows the effect of reduced secondary torque reactivity
C	Shallower initial primary ramp + steeper helix	Produces a strongly slip-dominated launch

To keep the discussion consistent across the different plots, the launch is described in terms of a small number of recurring operating phases. These phases are not introduced as separate events for each case, but as common regions of behaviour used to interpret the launch trajectories and their underlying force and torque balances.

Throughout the following subsections, these phases will be indicated directly on the plots using low-opacity background shading. The same color convention is used wherever possible so that the reader can track the evolution of each case across the shift, clamping-force, and torque-transfer results.

The phase colors are

- **engagement** — light blue,
- **low-ratio operation** — light green,
- **active upshift** — light yellow, and
- **high-ratio operation** — light red.

Not every case is expected to exhibit all four phases equally clearly. The phase shading is therefore used as a visual guide to the dominant operating regime, rather than as a claim that every boundary is perfectly sharp.

4.1.2 Launch Shift Curves

The overall launch behaviour of the four tuning cases is first compared in Figure 4.1. Each trajectory is plotted in engine-speed–vehicle-speed coordinates, so that engagement, ratio change, and the final high-ratio condition can all be viewed on the same axes. All four simulations begin from the same launch initial conditions,

$$\omega_p(0) = 1800\,\mathrm{rpm}, \qquad \omega_s(0) = 0,$$

where ω_p is the primary angular speed and ω_s is the driven angular speed corresponding to zero vehicle speed. From this state, the system is released into a full-throttle forward simulation using the complete coupled model developed in the preceding sections.

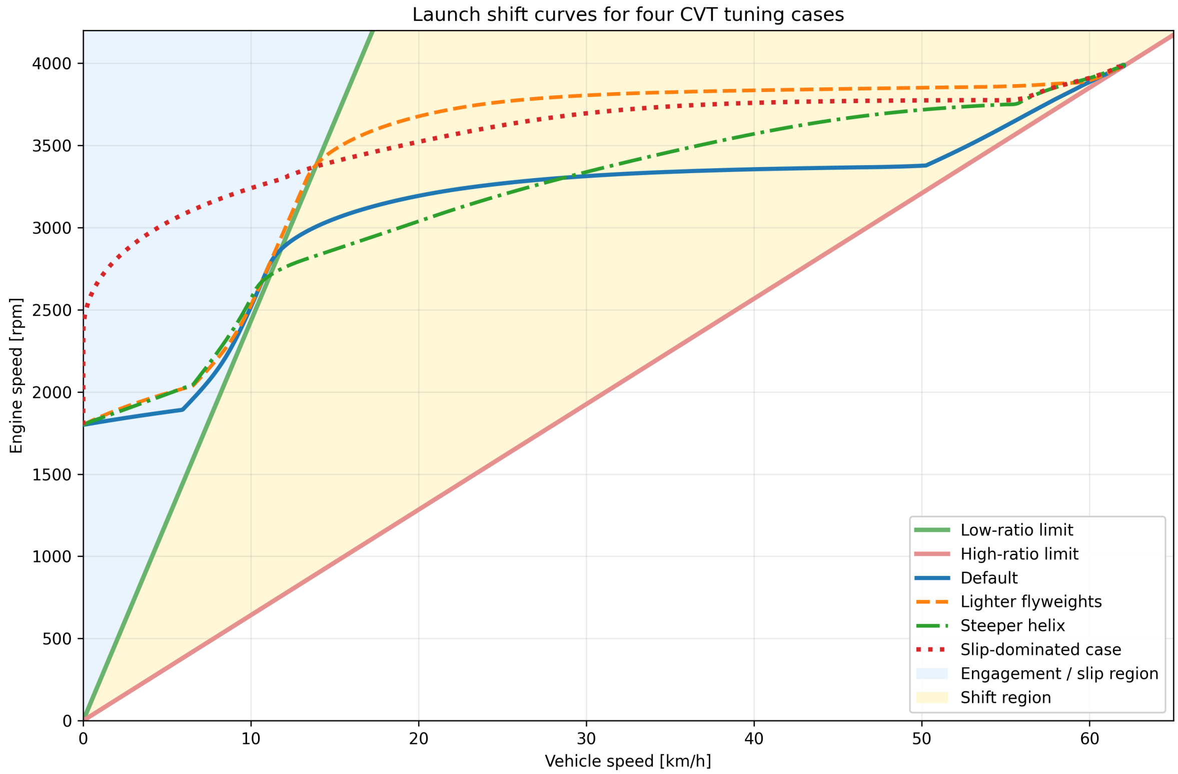


Figure 4.1: Engine-speed–vehicle-speed launch trajectories for the four tuning cases. The green line is the low-ratio limit, corresponding to a fully unshifted CVT. The red line is the high-ratio limit, corresponding to a fully shifted CVT. The blue shaded region to the left of the low-ratio line corresponds to launch engagement, where slip must be present. The yellow region between the two limit lines is the no-slip shift region.

The two reference lines are the kinematic no-slip limits of the CVT. The **low-ratio line** represents the fully unshifted transmission, where the CVT is in its torque-multiplying launch condition. The **high-ratio line** represents the fully shifted transmission, where the CVT has reached its final speed-increasing condition. Between these two limits lies the admissible shift region. A trajectory cannot move to the right of the high-ratio line, and any portion of a trajectory lying to the left of the low-ratio line must involve slip, since even the fully unshifted CVT cannot support such a low vehicle speed for the corresponding engine speed.

Read from left to right, the figure therefore shows the full launch sequence. The launch begins in the blue engagement region, where the vehicle is still too slow for no-slip operation. After crossing the low-ratio boundary, the trajectory enters the shift region. In a well-behaved launch, the CVT then tends to move approximately horizontally across this region, increasing vehicle speed while holding engine speed near a preferred operating value. This nearly horizontal portion will be referred to here as the *flat-shift region*. As the shift completes, the trajectory approaches the high-ratio line, which forms the final no-slip boundary of the launch.

With that interpretation in place, the differences between the four tuning cases are already visible. The **default** case shows the clearest baseline behaviour: a short engagement region, a brief low-ratio hold, and then a relatively flat shift through the middle of the plot before approaching the high-ratio boundary. This is the most balanced of the four trajectories, since it neither leaves low ratio too early nor rises excessively in engine speed during the main shift.

The **lighter-flyweight** case retains low ratio for longer before crossing into the shift region. Its flat-shift portion is also noticeably higher in engine speed than the default case, so the launch spends more of the main shift at elevated primary speed. The overall shape is still recognizable as a conventional launch trajectory, but the shift is delayed and carried out at a higher engine-speed level.

The **steeper-helix** case behaves differently. It leaves the low-ratio boundary much earlier, so the launch shows little sustained low-ratio hold. Its shift path is also less flat: instead of moving nearly horizontally across the shift region, it climbs upward more strongly as vehicle speed increases. This indicates that the launch does not settle into the same nearly constant engine-speed shift seen in the default case.

The **combined slip-dominated** case is the most extreme. It remains in the engagement region for much longer, showing no clear low-ratio hold before entering the shift region. Once it does cross into the no-slip range, its shift still occurs at a comparatively high engine speed, and the early launch is much more dominated by slip than by clean low-ratio operation.

At this stage, Figure 4.1 is intended to establish the overall shape of the four launches and the main qualitative differences between them. The key observations are the amount of low-ratio retention, the height and flatness of the main shift segment, and the extent of the initial slip-dominated portion of the launch. The following subsections explain these differences in terms of the underlying axial force balance and torque-transfer behaviour of the CVT.

4.1.3 Axial Force Balance Through the Launch

The shift-curve trajectories of Figure 4.1 indicate when each case remains near low ratio, when it begins to upshift, and how quickly it reaches high ratio. To explain those differences mechanically, Figure 4.2 compares the total axial force acting on the primary and secondary pulleys for each tuning case, with the same launch phases shaded in the background.

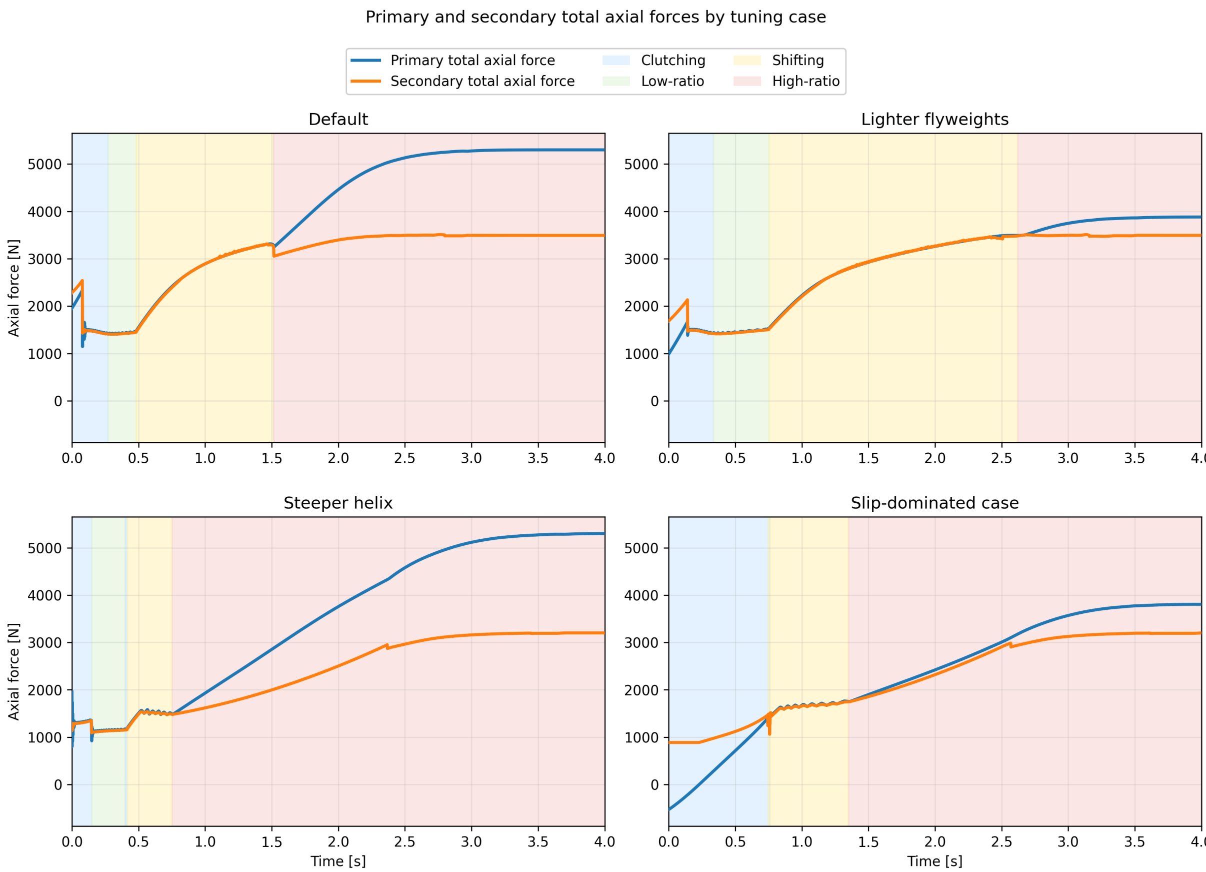


Figure 4.2: Primary and secondary total axial forces during launch for the four tuning cases. Blue curves show total primary axial force, orange curves show total secondary axial force. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as force-balance diagrams for the shift. While the secondary axial force exceeds the primary axial force, the transmission is held toward low ratio. Once the primary force overtakes the secondary, the pulley pair is driven into upshift. The timing of that crossover therefore determines how long the launch remains near low ratio and how early the transmission begins moving through its ratio range.

The **default** case shows the most balanced behaviour. During the initial clutching phase, the secondary force begins noticeably above the primary force, so the transmission is held near its low-ratio condition rather than shifting immediately. This produces the short but clear low-ratio hold seen previously in Figure 4.1. As the launch progresses, the primary axial force rises and eventually overtakes the secondary, initiating the main upshift. Importantly, the secondary force does not remain fixed while this happens. It also rises throughout the shift, so the system continues to resist ratio change even after upshift begins. This is mechanically important: if the secondary force stayed nearly constant while the primary force rose rapidly, the transmission would sweep through the shift range much more abruptly. Instead, both sides grow together, giving the default case its smoother and more controlled shift.

The **lighter-flyweight** case makes this point especially clear. Relative to the default case, the primary force is reduced in the early launch, while the secondary remains comparatively strong.

The initial force gap is therefore larger, meaning the low-ratio condition is held more firmly and for longer. This matches the longer low-ratio segment seen in the shift curves. Later in the launch, the primary still rises enough to drive upshift, but it does so later and at a higher engine-speed level than in the default case. In other words, this case does not fundamentally change the character of the shift; it mainly delays it by weakening the primary closing tendency during engagement and early launch.

The **steeper-helix** case behaves differently. Here, the initial difference between secondary and primary force is very small, and the primary overtakes the secondary much earlier. This explains why the corresponding shift curve showed little sustained low-ratio hold and entered the shifting region so quickly. We also see a differently shaped force evolution throughout the shift itself. In the default and lighter-flyweight cases, both forces continue rising together in a gradual way through the main shift region. In the steeper-helix case, both forces rise sharply early on and then level off much sooner. The two forces still remain relatively close to one another, but the overall shape they trace is different. This helps explain why the earlier shift curve did not display the same flat-shift behaviour as the default case. Rather than maintaining a long, nearly constant-speed shift, the transmission shows a more rising shift trajectory, consistent with this less gradual force evolution.

The **slip-dominated** case is the most extreme. Its early launch is characterized by a prolonged clutching phase and no meaningful low-ratio hold. This is consistent with the earlier shift-curve figure, where the trajectory spent much longer in the slip-dominated region before entering the no-slip shift region. Importantly, the beginning of geometric shift does not imply that the system has already stopped slipping. It only indicates that the shift coordinate has begun to move away from its initial value. The more important feature in this case is the low magnitude of both axial forces in the early launch. Compared with the other cases, neither pulley develops a strong axial loading during the initial portion of the event. This weak overall clamping is a key reason why this case behaves poorly, and it will be seen more directly in the following torque-transfer plots where the associated slip remains elevated for much longer.

One additional feature is visible near the beginning of each case. The early spike in secondary axial force is the torque-reactive contribution of the secondary. It appears most strongly during the initial launch when transmitted torque rises rapidly, and it provides an immediate resisting effect against premature upshift. This feature is especially useful because it shows that the secondary is responding dynamically to transmitted torque rather than acting as a static spring-only restraint. The influence of this torque reactivity will be seen more directly in the following torque-transfer results, but its signature is already visible here in the early force histories.

Taken together, Figure 4.2 provides the mechanical explanation for the differences observed in Figure 4.1. A larger initial gap between secondary and primary force corresponds to stronger low-ratio retention, as seen in the lighter-flyweight case. A very small gap corresponds to premature upshift, as seen in the steeper-helix case. The default case lies between these extremes and therefore produces the most balanced launch. Perhaps most importantly, the figure shows that the shift is not driven by the primary alone. The secondary rises with it throughout the launch, and the actual shift behaviour emerges from the evolving balance between the two rather than from either force in isolation. At the same time, this figure only explains the shift tendency and overall clamping levels; it has not yet quantified the resulting slip behaviour, which is addressed next.

4.1.4 Torque Bounds, No-Slip Demand, and Coupling Torque

The force-balance results of Figure 4.2 explain when each case tends to remain at low ratio and when it begins to shift. However, force balance alone does not determine whether the belt is actually able to transmit the torque required for sticking. To examine that question, Figure 4.3 compares four quantities during the launch: the primary and secondary traction bounds, the no-slip coupling torque $\tau_{\rm ns}$, and the actual transmitted coupling torque τ_c.

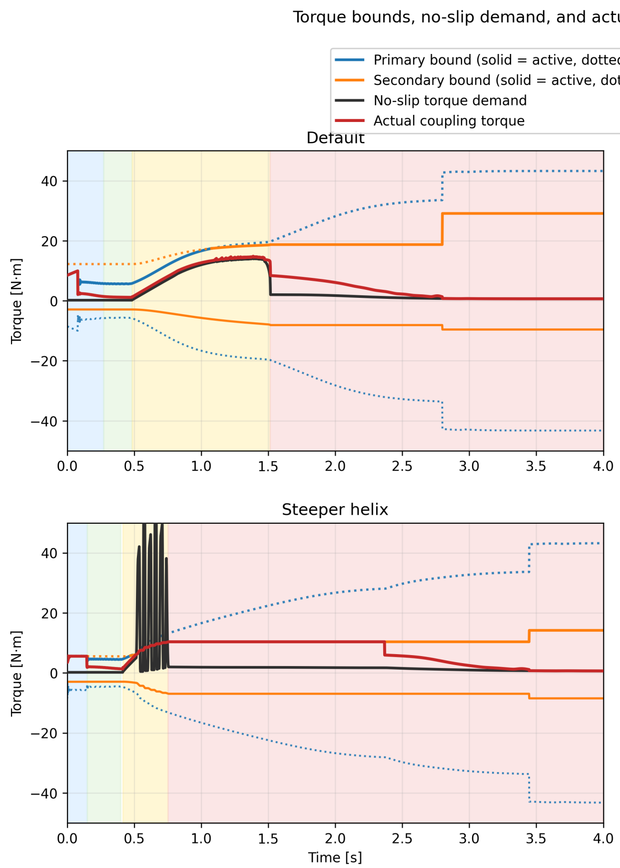


Figure 4.3: Torque bounds, no-slip demand, and actual coupling torque for the four tuning cases. Blue and orange curves show the primary and secondary bounds, with the active bound drawn solid and the inactive bound dotted. The black curve shows the no-slip torque demand $\tau_{\rm ns}$, and the red curve shows the actual transmitted coupling torque τ_c. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as traction-capacity diagrams. At any instant, the belt can only remain in stick if the required no-slip torque lies inside the admissible interval defined by the active upper and lower bounds. When $\tau_{\rm ns}$ lies inside that interval, the actual coupling torque can follow the no-slip demand closely. When $\tau_{\rm ns}$ exceeds the available bounds, sticking is no longer admissible and the transmitted torque must saturate at the corresponding active limit.

A first broad trend is immediately visible. The cases that settle out of slip early in the launch are precisely the cases with larger admissible positive torque capacity during the initial clutching and low-ratio phases. In the **default** and **lighter-flyweight** cases, the upper torque bound rises to a relatively large value early in the launch. This gives the system enough admissible coupling torque to reduce the belt-speed mismatch and recover toward sticking. By contrast, the **slip-dominated** case begins with essentially no admissible positive torque and only develops usable capacity more gradually. That lower early torque capacity directly explains why the slip-dominated case spends much longer in the clutching region.

This same distinction appears in the evolution of τ_c. In the cases that recover stick early, the actual coupling torque relaxes toward and then tracks the no-slip demand through the end of clutching and into low-ratio operation. Once low-ratio launch is established, the red and black curves remain close, indicating that the required torque stays admissible and sustained slip is not reintroduced immediately.

A second important pattern appears once the transmission begins to shift. In all four cases, the no-slip torque rises during the shift phase. This is consistent with the form of τ_{ns}, in which the ratio-rate term becomes active once the CVT begins moving away from its initial ratio. In the **default** and **lighter-flyweight** cases, this rise remains below the admissible upper torque bound, so the actual coupling torque continues to follow the no-slip demand and large renewed slip does not occur. In the **steeper-helix** and **slip-dominated** cases, however, τ_{ns} rises above the admissible upper bound during the shift. At that point, sticking is no longer feasible, the actual coupling torque saturates at the active maximum, and renewed slip appears until the state returns closer to the sticking condition.

The **steeper-helix** and **slip-dominated** cases also show visibly oscillatory behaviour in the no-slip torque during the shift onset. This is consistent with the oscillations already seen in the preceding axial-force plots. The most direct source is the ratio-rate contribution to τ_{ns}: if the shift dynamics are not smooth, then the implied no-slip torque need not evolve smoothly either. In these two poorer tuning cases, the shift motion appears more weakly damped, so the no-slip torque exhibits stronger excursions during the transition into active shifting. A plausible reason is that the present model neglects parasitic losses and other non-critical frictional effects, which in a real system would provide additional damping and suppress some of this oscillatory behaviour. This is therefore a useful limitation to note: the model captures the overall stick–slip transitions and the active limiting mechanism, but the transient oscillations in the most aggressive cases are likely amplified by the intentionally idealized loss assumptions.

Another common feature is the gradual relaxation of the actual coupling torque toward the no-slip demand rather than an instantaneous snap to it. This comes from the regularized transition law used in the simulation, discussed in Appendix C.3, which replaces a discontinuous switch with a smooth tanh-based approximation. Its purpose is numerical: it reduces chattering and improves stability near the stick–slip transition. The effect is visible here as a slight lag in the recovery of perfect sticking, but it does not change the main qualitative behaviour of the four cases. The same limiting patterns are still present, and the cases that lose admissibility remain clearly distinguishable from those that do not.

Finally, each case shows a noticeable jump in the admissible torque bound once full sticking is re-established near the end of the launch. This reflects the change in limiting behaviour used by the model after slip is removed, together with the corresponding increase in effective friction behaviour assumed in the sticking regime. The important point for the present discussion is that this jump occurs only after the system has already relaxed back toward the no-slip state; it is therefore a consequence of stick recovery rather than its cause.

Taken together, Figure 4.3 closes the interpretation begun in the previous two subsections. The shift curves showed the overall launch behaviour, and the axial-force plots explained the tendency to hold or leave low ratio. The present figure then shows whether the required coupling torque is actually admissible throughout those phases. The default and lighter-flyweight cases maintain sufficient positive torque capacity to recover stick early and remain near the no-slip demand through the main shift. The steeper-helix and slip-dominated cases do not: in both, the no-slip torque exceeds the available upper bound during the shift, forcing the actual coupling torque to saturate and reintroducing slip. In this way, the torque plots both confirm the solid patterns seen earlier and reveal one of the model's clearest shortcomings: the most weakly damped cases display exaggerated oscillatory transients due to the idealized loss assumptions.

4.1.5 Launch Performance Comparison

The preceding subsections established the mechanical differences between the four tuning cases. The shift-curve plots showed the overall launch trajectory, the axial-force plots explained the tendency to hold or leave low ratio, and the torque-bound plots showed whether the required coupling torque remained admissible throughout the launch. The final step is to compare how those differences affect overall launch performance.

Two simple metrics are used here: the time required to travel 50 m, and the time required to reach 60 km/h. These are useful because they summarize the launch in two complementary ways. The 50 m time reflects the integrated quality of the launch as a whole, while the 60 km/h time emphasizes how quickly the drivetrain can build vehicle speed through the main acceleration phase.

Table 4.2: Launch performance metrics for the four tuning cases.

Case	**Time to** 50 m [s]	**Time to** 60 km/h [s]
Default	3.93	2.27
Lighter flyweights	4.26	2.78
Steeper helix	4.19	2.82
Slip-dominated case	4.41	2.98

The default case performs best on both metrics, which is consistent with the earlier mechanism-level results. It develops enough admissible coupling torque to reduce slip early, retains low ratio long enough to launch effectively, and then transitions into a controlled shift without reintroducing large sustained slip. This is the most balanced use of the CVT: the transmission first allows a strong initial torque transfer into the driveline, and then continues accelerating the vehicle while keeping the shift progression smooth and admissible.

This point is worth emphasizing. The coupling torque plays two roles at once. It is the torque that loads the engine through the transmission, limiting how freely the primary (and therefore the engine) can accelerate, and it is also the torque delivered through the CVT that accelerates the vehicle from rest. A strong early coupling torque is therefore beneficial when it is admissible, because it quickly pulls the belt-speed mismatch down and simultaneously applies useful drive torque to the vehicle. The default case benefits from exactly this behaviour: it uses the initial slip region to establish a relatively large transmitted torque early in the launch, which helps "kick-start" vehicle acceleration before the main shift begins.

The lighter-flyweight case performs worse on both metrics, even though its shift remains comparatively well behaved. This matches the earlier interpretation that its main drawback is delayed primary loading during the early launch. By holding low ratio longer but developing less aggressive early coupling, it sacrifices initial acceleration and does not recover that loss later.

The steeper-helix case shows a different trade-off. It leaves low ratio too early and re-enters slip during the shift, which degrades the quality of the launch. However, because it still develops appreciable coupling torque early in the event, its 50 m time remains slightly better than the lighter-flyweight case. Its slower time to 60 km/h, on the other hand, is consistent with the less stable shift behaviour and the renewed slip seen in the torque-bound plots.

The slip-dominated case performs worst overall. This is the expected result from the earlier figures: its early clamping forces are weakest, its admissible torque rises too slowly, and its launch remains slip-dominated for too long. As a result, it spends too much of the start of the event establishing torque transfer rather than converting that torque cleanly into forward acceleration.

Overall, these performance metrics support the mechanism-level interpretation developed throughout the section. The best launch is not produced by simply holding low ratio as long as possible, nor by shifting as early as possible. Instead, good performance requires a balance: sufficient early admissible coupling torque to accelerate the vehicle strongly from rest, enough low-ratio retention to avoid premature upshift, and a smooth enough shift that the required no-slip torque does not repeatedly exceed the available traction limits.

4.2 Implementation and Reproducibility

The source code for the CVT simulator used in this work is available at `https://github.com/gr812b/CVT-Simulator`. This implementation includes the mathematical model developed in the main derivation sections along with the practical numerical regularization strategy described in Section C.3.

To support reproducibility, the exact version used to generate the figures and tables in this manuscript will be archived as a tagged release:

- repository release (to be finalized): `https://github.com/gr812b/CVT-Simulator/releases/tag/vX.Y.Z`,
- archival DOI landing page (placeholder): `https://doi.org/XX.XXXX/zenodo.XXXXXXX`.

Unless otherwise stated, simulation cases in Section 4.1 correspond to that archived release. The repository README and release notes describe the execution workflow, configuration files, and plotting scripts used to produce the reported results.

An interactive web deployment of the simulator is also available for demonstration purposes: `https://cvt.kaiarseneau.dev`. This live endpoint is provided for accessibility and does not replace the archived release as the canonical reproducibility artifact.

4.3 Conclusion

This work developed a unified first-principles dynamic model for a mechanically actuated dry rubber V-belt CVT and its coupling to the vehicle drivetrain. Starting from pulley geometry, belt-length closure, actuator mechanics, and rotational dynamics, the derivation produced a closed nonlinear state-space formulation in which ratio evolution, torque transmission, slip admissibility, and vehicle response are solved consistently in time.

The completed formulation brings together several mechanisms that strongly govern mechanically actuated CVT behaviour: primary flyweight actuation, secondary torque-reactive helix clamping, belt geometric closure, axial shift dynamics, traction-limited torque transfer, and longitudinal drivetrain coupling. The main outcome is a single dynamic framework in which these effects interact directly, while remaining mechanically interpretable in terms of the underlying pulley forces, ratio change, and torque limits.

The simulation results showed coherent launch behaviour across the tuning cases considered. The shift-curve results captured distinct differences in engagement, low-ratio retention, main-shift shape, and approach to high ratio. The axial-force results showed that these differences could be traced back to the evolving balance between primary and secondary clamping. The torque-bound results then showed when the no-slip coupling demand remained admissible and when renewed slip was forced by the available traction limits. Together, these results formed a consistent mechanism chain from tuning changes, to force balance, to torque admissibility, to overall launch behaviour.

An important outcome of the study is that the model preserved physical intuition across multiple levels. Changes in tuning did not simply produce different output curves; they produced changes that remained understandable in mechanical terms. Stronger or weaker low-ratio retention appeared through the initial clamping-force gap, earlier or later upshift appeared through the timing of force crossover, and reintroduced slip appeared when the no-slip torque rose beyond the admissible coupling bounds. This consistency gives confidence that the formulation is capturing the dominant interactions it was intended to represent.

The results also highlighted useful limitations of the present model. In the more aggressive cases, the no-slip torque demand exhibited visibly underdamped oscillatory transients during shift onset. These were consistent with irregular shift-rate evolution in the model and suggest that the idealized loss assumptions, particularly the neglect of parasitic losses and other non-critical frictional effects, leave the system with less damping than would be expected in real hardware. The formulation also remains limited to one-dimensional vehicle motion and does not yet include thermal effects, wear progression, or more detailed tribological contact evolution.

Overall, the present work provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behaviour during transient launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behaviour remains traceable back to clear mechanical causes. With further parameter identification, experimental validation, and extension to include more realistic damping and loss mechanisms, the framework developed here can serve as a useful basis for both engineering interpretation and future design-oriented CVT simulation.

Bibliography

Olav Aaen. *Clutch Tuning Handbook*. Aaen Performance Parts Inc., 1979.

Olav Aaen. *Clutch Tuning Handbook*. Aaen Performance, Wisconsin, USA, 2007. URL https://archive.org/details/aaen-clutch-tuning-handbook-2007/page/2/mode/2up.

Farid Al-Bender, Vincent Lampaert, and Jan Swevers. The generalized maxwell-slip model: A novel model for friction simulation and compensation. *IEEE Transactions on Automatic Control*, 50 (11):1883–1887, 2005. doi: 10.1109/TAC.2005.858676.

Ivan Arango and Sebastian Mu noz Alzate. Numerical design method for CVT supported in standard variable speed rubber V-belts. *Applied Sciences*, 10(18):6238, 2020. doi: 10.3390/app10186238.

Brian Armstrong-H'elouvry, Pierre Dupont, and Carlos Canudas de Wit. A survey of models, analysis tools and compensation methods for the control of machines with friction. *Automatica*, 30(7):1083–1138, 1994. doi: 10.1016/0005-1098(94)90209-7.

Baja SAE. Sae baja restricted kohler ch440 engine performance data. Available from the Baja SAE website under Document Resources, 2022. URL https://www.bajasae.net/cdsweb/gen/DownloadDocument.aspx?DocumentID=b250342f-ea19-46c2-95ea-6deb728ddb7a. Accessed: 2024-11-03.

Brian Gene Ballew. Investigation of a numerical algorithm for a discretized rubber-belt continuously variable transmission dynamic model and techniques for the measurement of belt material properties. Master's thesis, University of Arkansas, 2015.

Luca Bertini, Carlo Carmignani, and Francesco Frendo. Analytical model for the power losses in rubber V-belt continuously variable transmission (CVT). *Mechanism and Machine Theory*, 78: 289–306, 2014. Verify page range and DOI against final publisher version.

B. Bonsen, T. W. G. L. Klaassen, and M. Pulles. Analysis of slip in a continuously variable transmission. *Journal of Mechanical Design*, 126(5):822–829, 2004.

Richard P. Brent. *Algorithms for Minimization without Derivatives*. Prentice-Hall, Englewood Cliffs, NJ, 1973.

Richard G. Budynas and J. Keith Nisbett. *Shigley's Mechanical Engineering Design*. McGraw-Hill Education, New York, NY, 10 edition, 2015. ISBN 978-0-07-339820-4.

Marco Cammalleri. A new approach to the design of a speed-torque-controlled rubber V-belt variator. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, 219(12):1413–1427, 2005. doi: 10.1243/095440705X35080.

Marco Cammalleri and Francesco Sorge. Approximate closed-form solutions for the shift mechanics of rubber belt variators. In *Proceedings of the ASME 2009 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Volume 6: ASME Power Transmission and Gearing Conference; 3rd International Conference on Micro- and Nanosystems; 11th International Conference on Advanced Vehicle and Tire Technologies, San Diego, California, USA, 2009. ASME.

Carlos Canudas de Wit, Henrik Olsson, Karl Johan Åstr"om, and Pablo Lischinsky. A new model for control of systems with friction. *IEEE Transactions on Automatic Control*, 40(3):419–425, 1995. doi: 10.1109/9.376053.

Giovanni Carbone, Luigi Mangialardi, and Giovanni Mantriota. The influence of pulley deformations on the shifting mechanism of metal belt CVT. *Journal of Mechanical Design*, 127(1):103–113, 2005. doi: 10.1115/1.1825443.

T.-F. Chen, D.-W. Lee, and C.-K. Sung. An experimental study on transmission efficiency of a rubber V-belt CVT. *Mechanism and Machine Theory*, 33(4):351–363, 1998.

T. H. C. Childs and D. Cowburn. Power transmission losses in V-belt drives. part 2: Effects of small pulley radii. *Proceedings of the Institution of Mechanical Engineers, Part D: Transport Engineering*, 201(1):41–53, 1987. doi: 10.1243/PIME_PROC_1987_201_156_02.

J. R. Dormand and P. J. Prince. A family of embedded Runge–Kutta formulae. *Journal of Computational and Applied Mathematics*, 6(1):19–26, 1980.

Chengwu Duan, Kumar Hebbale, Fengyu Liu, and Jian Yao. Physics-based modeling of a chain continuously variable transmission. *Mechanism and Machine Theory*, 105:397–408, 2016. doi: 10.1016/j.mechmachtheory.2016.07.018.

G"oran Gerbert. Belt slip—a unified approach. *Journal of Mechanical Design*, 118(3):432–438, 1996. doi: 10.1115/1.2826904.

G. Juli'o and J.-S. Plante. An experimentally-validated model of rubber-belt CVT mechanics. *Mechanism and Machine Theory*, 46(8):1037–1053, 2011.

H. Kim, H. Lee, H. Song, and H. Kim. Steady state and transient characteristics of a rubber belt CVT with mechanical actuators. *KSME International Journal*, 16(5):646–653, 2002. Page range should be verified against the final publisher version.

K. Kim and H. Kim. Axial forces of a V-belt CVT: Part I. theoretical analysis. *KSME Journal*, 3 (1):56–61, 1989.

Lingyuan Kong and Robert G. Parker. Steady mechanics of belt-pulley systems. *Journal of Applied Mechanics*, 72(1):25–34, 2005.

Lingyuan Kong and Robert G. Parker. Mechanics and sliding friction in belt drives with pulley grooves. *Journal of Mechanical Design*, 128(2):494–502, 2006.

V. La Battaglia, A. Giogetti, S Marini, G Arcidiacono, and P. Citti. Kinematic analysis of V-belt CVT for efficient system development in motorcycle applications. *Machines*, 10(1):16, 2022. doi: 10.3390/machines10010016.

Matthew James Messick. An experimentally-validated V-belt model for axial force and efficiency in a continuously variable transmission. Master's thesis, Virginia Polytechnic Institute and State University, Blacksburg, VA, 2018.

ScienceDirect Topics. Capstan equation. Available from the ScienceDirect Topics website, n.d. URL https://www.sciencedirect.com/topics/engineering/capstan-equation. Accessed: 2026-03-28.

Sean Sebastian Skinner. Modeling and tuning of cvt systems for SAE baja vehicles. Ms thesis, West Virginia University, Statler College of Engineering and Mineral Resources, Department of Mechanical and Aerospace Engineering, Spring 2020. URL https://researchrepository.wvu.edu/etd/7590/.

Francesco Sorge. A simple model for the axial thrust in V-belt drives. *Journal of Mechanical Design*, 118(4):589–592, 1996.

Francesco Sorge. A theoretical approach to the shift mechanics of rubber belt variators. *Journal of Mechanical Design*, 130(12):122602, 2008.

Francesco Sorge and Marco Cammalleri. Helical shift mechanics of rubber V-belt variators. *Journal of Mechanical Design*, 133(4):041006, 2011.

Nilabh Srivastava and Imtiaz Haque. Dynamics of metal V-belt continuously variable transmissions. *Journal of Mechanical Design*, 128(3):616–624, 2006.

Nilabh Srivastava and Imtiaz Haque. A review on belt and chain continuously variable transmissions (CVT): Dynamics and control. *Mechanism and Machine Theory*, 44(1):19–41, 2009. doi: 10.1016/j.mechmachtheory.2008.06.007.

Ahmet Yildiz, Francesco Bottiglione, Antonio Piccininni, Osman Kopmaz, and Giuseppe Carbone. Experimental validation of the carbone–mangialardi–mantriota model of continuously variable transmissions. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, pages 1–10, 2017. doi: 10.1177/0954407017711320. Online-first publication; verify final volume, issue, and page range.

Appendix A

Belt Length Small-Angle Approximation

This appendix quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models. Using parameter values representative of the physical system considered in this work, we demonstrate that the approximation introduces substantial error in the computed center-to-center distance, then further in the ratio, which is critical for accurate dynamic modeling.

A.1 Standard Approximation

Many simplified treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this approximation, the belt-length constraint **??** reduces to

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{A.1.1}$$

A.2 Derivation of the Approximate Center Distance

To determine the center-to-center distance at the initial configuration, we substitute $r_p = r_{p,\text{outer,min}}$ and $r_s = r_{s,\text{outer,max}}$ into Equation (A.1.1) and solve for C.

Step 1: Isolate the square root. Rearranging Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) = 2\sqrt{C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}. \tag{A.2.1}$$

Step 2: Square both sides.

$$\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2 = 4\left[C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2\right]. \tag{A.2.2}$$

Step 3: Solve for C^2.

$$C^2 = \frac{\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2. \tag{A.2.3}$$

Step 4: Take the square root. Expanding the squared term in the numerator:

$$\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2 = L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2,$$

the approximate center distance becomes

$$\boxed{C_{\text{approx}} = \sqrt{\frac{L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}.} \tag{A.2.4}$$

This closed-form expression avoids the need for numerical root finding but neglects the $\sin^{-1}$ term in the exact constraint.

A.3 Error Analysis

We now evaluate the approximation error using parameter values from the physical CVT system considered in this work.

System Parameters

Parameter	**Value**	**SI Value**
Belt length L_b	37.538 in	0.9535 m
Belt height h_b	0.613 in	0.015 57 m
Groove half-angle β	11.5°	0.2007 rad
Inner primary radius	0.8125 in	0.020 637 5 m
Primary outer radius $r_{p,\text{outer,min}}$	1.4255 in	0.036 21 m
Secondary outer radius $r_{s,\text{outer,max}}$	4.0 in	0.1016 m

The primary outer radius is computed as the inner radius plus belt height: $r_{p,\text{outer,min}} = 0.8125 + 0.613 = 1.4255\,\text{in}$.

Computed Center Distances Solving the exact belt-length constraint **??** numerically yields

$$C_{\text{exact}} = 0.251\,74\,\text{m} \approx 9.911\,\text{in}.$$

Evaluating the approximate formula Equation (A.2.4) yields

$$C_{\text{approx}} = 0.268\,37\,\text{m} \approx 10.566\,\text{in}.$$

Error Metrics The absolute and relative errors are:

$$\Delta C = C_{\text{approx}} - C_{\text{exact}} = 0.016\,63\,\text{m} = 0.655\,\text{in}, \tag{A.3.1}$$

$$\varepsilon_{\text{rel}} = \frac{\Delta C}{C_{\text{exact}}} = \frac{0.01663}{0.25174} = 6.61\%. \tag{A.3.2}$$

Assessment The small-angle approximation introduces a relative error of nearly 7% in the center-to-center distance for this system. This error arises because the ratio

$$\frac{r_{s,\text{outer,max}} - r_{p,\text{outer,min}}}{C} = \frac{0.1016 - 0.03621}{0.25174} = 0.260$$

is not small. The assumption $\lambda \approx 0$ is therefore not justified for this CVT geometry.

Since the center distance C propagates into all subsequent geometric calculations—wrap angles, tangent span lengths, and derived quantities such as the transmission ratio—a 7% error in C can compound into larger errors in predicted system behavior.

A.4 Propagation to Secondary Radius and Transmission Ratio

To demonstrate how the center distance error propagates into operationally relevant quantities, we now examine the secondary radius and CVT ratio at two shift positions: zero shift and maximum shift.

A.4.1 Derivation of the Approximate Secondary Radius

Given a known primary outer radius $r_{p,\text{outer}}$ and center distance C, the secondary outer radius can be obtained by solving the belt-length constraint. Under the small-angle approximation Equation (A.1.1), this yields a closed-form expression.

Step 1: Isolate the square root. From Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer}} + r_{s,\text{outer}}) = 2\sqrt{C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2}. \tag{A.4.1}$$

Step 2: Square both sides.

$$\left[L_b - \pi r_{p,\text{outer}} - \pi r_{s,\text{outer}}\right]^2 = 4\left[C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2\right]. \tag{A.4.2}$$

Step 3: Expand and collect terms. Expanding the left side:

$$(L_b - \pi r_{p,\text{outer}})^2 - 2\pi(L_b - \pi r_{p,\text{outer}})r_{s,\text{outer}} + \pi^2 r_{s,\text{outer}}^2.$$

Expanding the right side:

$$4C^2 - 4r_{s,\text{outer}}^2 + 8r_{p,\text{outer}}r_{s,\text{outer}} - 4r_{p,\text{outer}}^2.$$

Collecting powers of $r_{s,\text{outer}}$ yields a quadratic equation:

$$(\pi^2+4)r_{s,\text{outer}}^2 - 2\left[\pi L_b - \pi^2 r_{p,\text{outer}} + 4r_{p,\text{outer}}\right]r_{s,\text{outer}} + \left[(L_b - \pi r_{p,\text{outer}})^2 + 4r_{p,\text{outer}}^2 - 4C^2\right] = 0. \tag{A.4.3}$$

Step 4: Apply the quadratic formula. The physically meaningful root (smaller secondary radius for overdrive configuration) is

$$\boxed{r_{s,\text{approx}} = \frac{\pi L_b + (4-\pi^2)r_{p,\text{outer}} - 2\sqrt{(\pi^2+4)C^2 - L_b^2 + 4\pi L_b r_{p,\text{outer}} - 4\pi^2 r_{p,\text{outer}}^2}}{\pi^2+4}.} \tag{A.4.4}$$

This closed-form expression enables direct computation of the approximate secondary radius without numerical root finding.

A.4.2 Zero Shift Configuration

At zero shift, the CVT is in its lowest ratio (maximum reduction) configuration. This is also the geometry used to compute the approximate center distance C_{approx}, so much of that initial center-distance error cancels when the same configuration is used again to recover the secondary radius.

Primary Radius at Zero Shift At $s = 0$, the primary is within the deadzone, so from **??**:

$$r_{p,\text{outer}}(0) = r_{p,\text{outer,min}} = 0.036\,21\,\text{m}.$$

Secondary Radius Comparison Solving the exact belt-length constraint with C_{exact} yields

$$r_{s,\text{exact}} = 0.101\,60\,\text{m}.$$

Evaluating the approximate formula Equation (A.4.4) with $C_{\text{approx}} = 0.268\,37\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.101\,60\,\text{m}.$$

Transmission Ratio Error Computing effective radii with $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	Exact	Approximate
$r_{p,\text{eff}}$	0.0284 m	0.0284 m
$r_{s,\text{eff}}$	0.0938 m	0.0938 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	3.300	3.300

The ratio error at zero shift is

$$\varepsilon_R = \frac{|3.300 - 3.300|}{3.300} \approx 0.000\%.$$

Thus, although C_{approx} is already 6.61% too large, the induced error has little visible effect at minimum shift because the radius calculation is being evaluated at the same operating point used to construct that approximate center distance in the first place.

Figure A.1: Scaled diagram of the CVT at zero shift ($s = 0$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The two solutions nearly overlap at this operating point, indicating negligible ratio error in this case.

A.4.3 Maximum Shift Configuration

At maximum shift, the primary and secondary pulleys are closest in size, representing a near-crossover configuration.

Primary Radius at Maximum Shift Using ?? with $s = s_{\max} = 0.75\,\text{in} = 0.019\,05\,\text{m}$ and deadzone $s_{\text{dz}} = 0.002\,489\,2\,\text{m}$:

$$\begin{aligned} r_{p,\text{outer}}(s_{\max}) &= r_{p,\text{outer,min}} + \frac{s_{\max} - s_{\text{dz}}}{2\tan\beta} \\ &= 0.03621 + \frac{0.01905 - 0.0024892}{2 \times \tan(11.5°)} \\ &= 0.03621 + \frac{0.01656}{0.4070} \\ &= 0.0769\,\text{m}. \end{aligned} \tag{A.4.5}$$

Secondary Radius Comparison Solving the exact belt-length constraint ?? numerically with $C_{\text{exact}} = 0.251\,74\,\text{m}$ yields

$$r_{s,\text{exact}} = 0.066\,73\,\text{m}.$$

Evaluating the approximate formula Equation (A.4.4) with $C_{\text{approx}} = 0.268\,37\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.056\,26\,\text{m}.$$

Transmission Ratio Error The CVT ratio is defined using effective radii, which account for belt seating depth (Equation (3.1.29)):

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}.$$

With $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	**Exact**	**Approximate**
$r_{p,\text{eff}}$	0.0691 m	0.0691 m
$r_{s,\text{eff}}$	0.0589 m	0.0485 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	0.853	0.701

Both models predict overdrive ($R < 1$), but the approximation substantially underestimates the ratio. The ratio error is

$$\varepsilon_R = \frac{|0.701 - 0.853|}{0.853} = 17.8\%.$$

Unlike the minimum-shift case, this operating point does not benefit from the same cancellation, so the center-distance error propagates into a much larger error in secondary radius and transmission ratio.

Figure A.2: Scaled diagram of the CVT at maximum shift ($s = s_{\max}$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The approximation underestimates the secondary radius, leading to a lower predicted ratio than the exact model.

A.5 Summary of Approximation Errors

Configuration	R_{exact}	R_{approx}	**Ratio Error**
Zero shift ($s = 0$)	3.300	3.300	0.000%
Maximum shift ($s = s_{\max}$)	0.853	0.701	17.8%

The small-angle approximation error is strongly operating-point dependent. At zero shift, most of the center-distance error cancels because the approximate center distance was itself constructed from that same geometry. At maximum shift, that cancellation is lost and the ratio error becomes substantial.

These errors arise from the initial 6.61% error in center distance, which propagates nonlinearly through the belt-length geometry in a state-dependent way. Since the transmission ratio directly enters the rotational equations of motion (Section 3.7), this approximation can still produce materially inaccurate dynamics near maximum shift.

This analysis justifies the use of exact numerical solutions for all geometric computations in this work.

Appendix B

Common Ramp Profile Designs

This appendix presents common ramp profile parameterizations for CVT primary pulley actuation. From Equation (3.2.8), the centrifugal axial force is

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds}, \tag{B.0.1}$$

where m_f is the flyweight mass, $r_{f,0}$ is the initial flyweight radius, and $r_f(s)$ is the radial displacement along the ramp as a function of axial shift s.

The profile $r_f(s)$ must satisfy the admissibility conditions from Section 3.2: continuously differentiable, non-negative slope, and finite slope throughout $[0, s_{\max}]$.

B.1 Linear Ramp

A linear ramp with slope k_r (radial displacement per unit axial shift):

$$r_f(s) = k_r \cdot s, \qquad \frac{dr_f}{ds} = k_r = \text{const.} \tag{B.1.1}$$

Substituting into Equation (B.0.1):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 k_r (r_{f,0} + k_r s) = m_f \omega_p^2 k_r r_{f,0} + m_f \omega_p^2 k_r^2 s. \tag{B.1.2}$$

At constant ω_p, axial force increases linearly with s because the growing radius is multiplied by a constant slope. Simple to manufacture, but may cause aggressive shifting at high ratio.

B.2 Circular Arc Ramp

A circular arc ramp is designed to reduce force variation across the shift range by achieving *partial* cancellation. The key insight is to parameterize the arc by desired slope angles α_1 and α_2 at the start and end of the segment, where $|dr_f/ds| = \tan\alpha$ specifies the force multiplier. This provides direct control over actuation behavior at engagement (high slope) versus overdrive (low slope).

For a circle of radius R_c centered at (s_c, r_c), the ramp profile satisfies:

$$(s - s_c)^2 + (r_f - r_c)^2 = R_c^2. \tag{B.2.1}$$

Solving for $r_f(s)$ on the upper arc:

$$r_f(s) = r_c + \sqrt{R_c^2 - (s - s_c)^2}. \tag{B.2.2}$$

Implicit differentiation yields:

$$\frac{dr_f}{ds} = \frac{-(s - s_c)}{\sqrt{R_c^2 - (s - s_c)^2}} = \frac{-(s - s_c)}{r_f - r_c}. \tag{B.2.3}$$

Parameterization from slope angles. At any point on the arc where the local slope angle is α (i.e. $dr_f/ds = \tan\alpha$), the circle constraint combined with Equation (B.2.3) gives the parametric form:

$$s = s_c - R_c \sin\alpha, \qquad\qquad r_f = r_c + R_c \cos\alpha. \tag{B.2.4}$$

The slope angle α decreases monotonically from α_1 (entry) to $\alpha_2 < \alpha_1$ (exit) as s increases along the ramp. Applying Equation (B.2.4) at both boundary shifts s_1 and s_2 and subtracting:

$$s_2 - s_1 = R_c(\sin\alpha_1 - \sin\alpha_2). \tag{B.2.5}$$

With $\Delta s = s_2 - s_1$ as the total axial span of the ramp segment, the arc radius and center are determined entirely by the two slope angles:

$$R_c = \frac{\Delta s}{\sin\alpha_1 - \sin\alpha_2}, \qquad s_c = s_1 + R_c \sin\alpha_1. \tag{B.2.6}$$

The boundary condition $r_f(s_1) = 0$ (flyweights at their initial radius at ramp entry) then fixes the radial center:

$$r_c = -R_c \cos\alpha_1. \tag{B.2.7}$$

Substituting into Equation (B.0.1):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f) \cdot \frac{-(s - s_c)}{r_f - r_c}. \tag{B.2.8}$$

To see the cancellation explicitly, split $r_{f,0} + r_f = (r_{f,0} + r_c) + (r_f - r_c)$:

$$\begin{aligned} F_{p,\text{cent}} &= m_f \omega_p^2 \left[\underbrace{(r_f - r_c)}_{\text{cancels}} \cdot \frac{-(s - s_c)}{r_f - r_c} + (r_{f,0} + r_c) \cdot \frac{-(s - s_c)}{r_f - r_c} \right] \\ &= -m_f \omega_p^2 \, (s - s_c) \left[1 + \frac{r_{f,0} + r_c}{r_f - r_c} \right]. \end{aligned} \tag{B.2.9}$$

This form reveals the structure clearly. The dominant factor is $-(s - s_c)$, which is linear in shift: the $(r_f - r_c)$ terms cancel exactly in the first part of the split. The bracket $\left[1 + \dfrac{r_{f,0} + r_c}{r_f - r_c}\right]$ is a nonlinear correction multiplier. Using the parametric form Equation (B.2.4) with Equation (B.2.7), the bracket evaluates to

$$1 + \frac{r_{f,0} + r_c}{r_f - r_c} = 1 + \frac{r_{f,0} - R_c \cos\alpha_1}{R_c \cos\alpha}, \tag{B.2.10}$$

where α is the local slope angle at shift s, sweeping from α_1 at entry to α_2 at exit. This makes the effective proportionality between force and $(s - s_c)$ position-dependent.

The numerator $r_{f,0} - R_c \cos\alpha_1$ is the key residual: it is controlled entirely by the choice of α_1, α_2, and Δs through $R_c = \Delta s/(\sin\alpha_1 - \sin\alpha_2)$. Increasing either α_1 or Δs raises R_c, making $R_c \cos\alpha_1$ larger and the residual smaller (or negative). The entry force scale, the exit force scale, and the degree of position-dependence between them are therefore all coupled to the same two angle choices. There is no freedom to set the overall force scale independently of the nonlinearity — changing α_1 and α_2 adjusts both simultaneously.

The condition $r_{f,0} = R_c \cos\alpha_1$ makes the residual zero, so the bracket equals 1 everywhere and force is exactly proportional to shift. This is the angle configuration at which the circular arc best approximates the hyperbolic ideal — it gives a practical design rule for tuning.

The practical consequence is important: at a fixed engine speed ω_p, the centrifugal force is approximately proportional to $(s - s_c)$, growing linearly with shift. This directly matches the linear restoring force of the primary spring (force $\propto k_s \cdot s$). A spring-loaded primary can therefore be tuned to balance the centrifugal flyweights at a target shift, and the linear character of the circular arc ensures the equilibrium is stable across the shift range rather than concentrated at a single point.

Circular arcs are widely used in practice due to ease of manufacture and sufficient compensation. The nonlinear residual is small when the ramp center offset $(r_{f,0} + r_c)$ is small relative to the arc radius R_c, but it cannot be eliminated entirely. The following section shows how to *exactly* achieve the linear target by choosing a different ramp geometry.

B.3 Hyperbolic Ramp (Perfect Cancellation)

From Equation (B.2.10), the circular arc force-to-shift proportionality constant is:

$$\frac{F_{p,\text{cent}}}{-m_f \omega_p^2 (s - s_c)} = 1 + \frac{r_{f,0} - R_c \cos\alpha_1}{R_c \cos\alpha}. \tag{B.3.1}$$

This varies with position because α changes along the arc. The natural question is: what ramp profile makes this ratio a *true constant*? That constant is what we call K, and the profile that achieves it satisfies:

$$(r_{f,0} + r_f(s)) \frac{dr_f}{ds} = Ks, \tag{B.3.2}$$

where $K > 0$. This produces centrifugal force exactly proportional to shift:

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 K s. \tag{B.3.3}$$

K is not a free parameter any more than R_c is for the circular arc: just as R_c is uniquely determined by α_1, α_2, and Δs (Equation (B.2.6)), K is uniquely determined by α_1, α_2, s_1, and s_2 (derived in the parameterization below). What differs is not the design inputs, but that the hyperbolic profile makes K exactly constant along the entire ramp, whereas the circular arc's equivalent ratio changes with position.

Deriving the profile. Equation (B.3.2) is separable. Substituting $u = r_{f,0} + r_f$, so $du/ds = dr_f/ds$:

$$u\,du = Ks\,ds. \tag{B.3.4}$$

Integrating both sides and writing the constant of integration as $\frac{1}{2}Ks_0^2$:

$$\tfrac{1}{2}u^2 = \tfrac{1}{2}K(s^2 + s_0^2). \tag{B.3.5}$$

Substituting back $u = r_{f,0} + r_f$:

$$\boxed{(r_{f,0} + r_f(s))^2 = K(s^2 + s_0^2),} \tag{B.3.6}$$

giving the explicit profile:

$$r_f(s) = \sqrt{K(s^2 + s_0^2)} - r_{f,0}. \tag{B.3.7}$$

In the (s, r_f) plane, the locus $(r_{f,0} + r_f)^2 = Ks^2$ (taking $s_0 = 0$) is a branch of a *hyperbola* — hence the name. Unlike the circular arc, where the residual multiplier in Equation (B.2.9) could not be removed, this profile forces the product $(r_{f,0} + r_f) \cdot dr_f/ds = Ks$ exactly for all s.

Local slope and its variation. To see how K relates to actual ramp angles, differentiate Equation (B.3.6) implicitly:

$$\frac{dr_f}{ds} = \frac{Ks}{r_{f,0} + r_f} = \frac{\sqrt{K}\,s}{\sqrt{s^2 + s_0^2}}. \tag{B.3.8}$$

Since $dr_f/ds = \tan\alpha(s)$ by definition of the ramp slope angle, the local angle satisfies:

$$\tan\alpha(s) = \frac{\sqrt{K}\,s}{\sqrt{s^2 + s_0^2}}. \tag{B.3.9}$$

Two observations follow. First, the slope is zero at $s = 0$ and grows monotonically toward the asymptote $\sqrt{K}$ as $s \to \infty$. This is the opposite trend from a circular arc, where the slope decreases from entry to exit. The reason is that the growing lever arm $(r_{f,0} + r_f)$ at large shift would otherwise amplify force super-linearly; a rising slope exactly compensates to keep the product Ks linear. Second, the asymptotic slope $\lim_{s\to\infty} \tan\alpha(s) = \sqrt{K}$ gives a simple interpretation: $K = \tan^2\alpha_\infty$ where α_∞ is the slope the profile approaches at large shift.

Parameterization by entry and exit angles. Following the same convention as Section B.2, specify desired slope angles α_1 and α_2 at entry shift s_1 and exit shift s_2. From Equation (B.3.9):

$$\tan^2\alpha_1 = \frac{Ks_1^2}{s_1^2 + s_0^2}, \qquad \tan^2\alpha_2 = \frac{Ks_2^2}{s_2^2 + s_0^2}. \tag{B.3.10}$$

Dividing the two equations eliminates K:

$$\frac{\tan^2\alpha_1}{\tan^2\alpha_2} = \frac{s_1^2\,(s_2^2 + s_0^2)}{s_2^2\,(s_1^2 + s_0^2)}. \tag{B.3.11}$$

Solving for s_0^2:

$$s_0^2 = \frac{s_1^2 s_2^2\,(\tan^2\alpha_2 - \tan^2\alpha_1)}{\tan^2\alpha_1\,s_2^2 - \tan^2\alpha_2\,s_1^2}. \tag{B.3.12}$$

Then K follows from either condition in Equation (B.3.10):

$$K = \frac{\tan^2\alpha_1\,(s_1^2 + s_0^2)}{s_1^2}. \tag{B.3.13}$$

For $s_0^2 > 0$ a consistent solution requires the numerator and denominator of Equation (B.3.12) to have the same sign. Since the hyperbolic slope increases with s (Eq. (B.3.8)), the physically consistent case is $\alpha_2 > \alpha_1$ (exit angle larger than entry), with $\tan^2\alpha_1 s_2^2 > \tan^2\alpha_2 s_1^2$ ensuring a positive denominator. No solution exists when the two angle conditions give a degenerate system ($\tan^2\alpha_1 s_2^2 = \tan^2\alpha_2 s_1^2$).

Practical impact. In most cases the difference between circular and hyperbolic profiles is negligible. For the specific ramps implemented in this work, the integrated force deviation across the full shift range is less than 1%. The hyperbolic derivation is nonetheless useful: it establishes the theoretical ideal and shows that the circular arc's partial cancellation is not a fundamental limitation but rather a small correction from the constant offset $(r_{f,0} + r_c)$.

Limitations. Both profiles address only the centrifugal force term. The secondary pulley torque feedback ($F_{s,\mathrm{ax}}(s, \tau_s)$ in Equation (3.7.11)) introduces load-dependent nonlinear terms that no ramp geometry alone can cancel. Circular arcs therefore remain the practical standard—simpler to manufacture, well-understood, and sufficient given these irreducible secondary-side effects.

B.4 Piecewise Ramps and Smoothing

Any of the profiles above can be combined in a piecewise fashion: multiple linear segments, multiple circular arcs with different angle pairs, or a mix of profile types joined at transition shifts $s_1, s_2, \ldots, s_{n-1}$. This allows region-specific tuning — steeper slopes at engagement, moderate for cruise, gentler near overdrive — without committing to a single global geometry.

Admissibility at junctions. The admissibility conditions from Section 3.2 require $r_f(s)$ to be continuously differentiable (C^1). A piecewise ramp satisfies C^0 by construction (the profile value is continuous if segments are joined at a common point), but slope continuity may be violated: if the left-segment slope at s_k differs from the right-segment slope, Eq. (B.0.1) gives a finite force jump at that point. No kinematic singularity occurs, but the force has a step discontinuity.

Numerical stiffness considerations. A sharp slope discontinuity is not the same as a smooth rapid transition: many real implementations use a finite but steep blend region, which the ODE solver sees as an extremely large second derivative $d^2 r_f/ds^2$ near the junction. This can create a locally stiff region and force an adaptive integrator to take very small steps near the knot, even if the discontinuity is physically mild. If simulation performance is a concern, slope transitions should be blended — a short circular arc segment tangent to both adjoining profiles is sufficient, as it restores C^1 continuity and bounds $d^2 r_f/ds^2$ throughout the ramp.

Appendix C

Numerical Methods Summary

This section documents the numerical algorithms used for belt-geometry root finding and dynamic simulation, including the specific tolerances, bracketing strategies, and solver configurations employed in the implementation.

C.1 Root Finding for Belt-Length Constraint

Given a known primary outer radius r_1 and fixed center-to-center distance C, the secondary outer radius r_2 is the unique root of the belt-length constraint. This scalar root problem is obtained by rearranging the exact belt length closure from **??** into the residual form:

$$g(r_2;\, r_1) = \pi(r_1 + r_2) + 2(r_2 - r_1)\arcsin\left(\frac{r_2 - r_1}{C}\right) + 2\sqrt{C^2 - (r_2 - r_1)^2} - L_b = 0, \tag{C.1.1}$$

where L_b is the fixed belt length. A separate invocation solves for the center-to-center distance C itself at initialization, using the same equation evaluated at the boundary radii $(r_{1,\text{min}},\, r_{2,\text{max}})$ with C as the unknown; this is exactly the initialization step implied by **??** before the ratio law Equation (3.7.2) can be evaluated over the shift range.

Bracketing strategy. The arcsin term in Equation (C.1.1) is real only when $|r_2 - r_1| \leq C$, so the natural domain for r_2 is the open interval $(r_1 - C,\, r_1 + C)$. Within this interval a sign change is guaranteed: at the lower boundary $r_2 \to r_1 - C$, the arcsin saturates to $-\pi/2$ and the square root vanishes, giving $g \to 2\pi r_1 - 2\pi C - L_b < 0$ for all physically valid parameters. At the upper boundary $r_2 \to r_1 + C$, by symmetry $g \to 2\pi r_1 + 2\pi C - L_b > 0$, satisfied whenever the belt length is less than the circumference subtended by the wider pulley pair — a requirement any real CVT must satisfy. A small safety margin $\varepsilon = 10^{-9}$ m is applied at both ends to keep arcsin away from its domain boundary where the function becomes numerically steep.

For center-to-center distance solving, the bracket is instead the interval $(\Delta r + \varepsilon,\ L_b/2)$, where $\Delta r = |r_{2,\text{eff}} - r_{1,\text{eff}}|$ is the effective-radius difference at the boundary configuration. The lower bound ensures arcsin is defined; the upper bound is a conservative geometric limit (a straight two-span belt cannot exceed L_b across a gap of $L_b/2$). Both brackets are validated by an explicit sign-change check before invoking the solver, so any failure to bracket immediately identifies an infeasible geometry rather than producing a silent wrong answer.

Brent's method. Both roots are found using `scipy.optimize.brentq`, an implementation of Brent's algorithm Brent (1973). The method combines three strategies: bisection (guaranteed bracket-halving), secant interpolation (superlinear convergence near smooth roots), and inverse quadratic interpolation (order-≈ 1.84 convergence when three previous iterates are well-conditioned). Whenever the faster interpolation steps would place the next iterate outside the current bracket, the algorithm falls back to bisection, preserving the worst-case $O(\log_2(1/\varepsilon))$ guarantee of pure bisection. Because the sign-change brackets described above are guaranteed for all valid CVT geometries, `brentq` is certain to converge. Both root solves use an absolute tolerance of 10^{-9} m, well below any physically meaningful length precision.

C.2 ODE Integration for Dynamic Simulation

The full CVT state vector $\mathbf{x} = (\omega_p, \omega_s, s, \dot{s}, v_b)$ is integrated using `scipy.integrate.solve_ivp`. The governing system is written as a first-order ODE system in Equation (3.7.13), with rotational sub-equations Equation (3.7.8) and Equation (3.7.9), axial dynamics Equation (3.7.11), belt-speed dynamics for v_b, and algebraic geometry closures Equation (3.7.2) and Equation (3.7.3). Although Equation (3.7.11) is second-order in s, introducing $(s, \dot{s})$ as separate states converts the complete model to first-order form for time integration. The time integrator is used with the `RK45` method — the Dormand–Prince explicit embedded Runge–Kutta pair Dormand and Prince (1980). The method propagates the state with a 5th-order formula and simultaneously computes a 4th-order companion estimate for local error control, using the same six function evaluations. The tableau is FSAL (first-same-as-last): the final stage of each accepted step is reused as the first stage of the next, so only five new evaluations are needed per step in steady operation.

Adaptive step size control. At each candidate step the per-component local error estimate $\hat{e}_i$ is compared to a mixed tolerance threshold. A step is accepted when

$$\max_i \frac{|\hat{e}_i|}{\mathrm{atol} + \mathrm{rtol} \cdot |y_i|} \leq 1, \tag{C.2.1}$$

and the next step size is scaled proportionally to $(\mathrm{tol}/\hat{e})^{1/5}$. The tolerances used in this work are $\mathrm{atol} = 10^{-6}$ and $\mathrm{rtol} = 10^{-4}$ (Table C.1). The absolute floor prevents spurious step rejection when a state variable passes through zero (e.g. $\dot{s} \approx 0$ at shift equilibrium or $\omega_p \approx 0$ at engine start), where a purely relative criterion would demand unreachable accuracy.

Parameter	Value	Interpretation
`atol`	10^{-6}	absolute floor: 1 µm on s, 1 µrad/s on ω
`rtol`	10^{-4}	relative tolerance: 0.01% of current state magnitude
`t_eval`	10 000 pts	uniform output grid over 30 s

Table C.1: ODE solver tolerance parameters.

Event detection. Phase transitions (shift engagement, boundary contact, back-shift onset) are located using `solve_ivp`'s event mechanism. These events correspond to qualitative changes in the dynamics around the bounded shift coordinate used in Equation (3.7.11) and the full state evolution in Equation (3.7.13). After each accepted step the solver evaluates the event functions on a degree-4 dense-output interpolant and applies a root-finding pass to detect any sign change within the step. When a crossing is found it is bracketed at sub-tolerance precision before being reported, so transitions are located with the same accuracy as the integration tolerance rather than being snapped to the coarse output grid.

Phase splitting at hard constraints. The shift bounds $s \in [0, s_{\max}]$ with $\dot{s} = 0$ at both endpoints are handled by partitioning the trajectory into separate `solve_ivp` calls rather than imposing inequality constraints inside the ODE. Each phase — normal shifting, clamped-at-maximum, and back-shifting — presents a smooth right-hand side to the integrator, allowing RK45 to operate at full 5th-order accuracy throughout. The terminal event of each phase supplies the initial condition of the next. This is numerically cleaner than directly clamping Equation (3.7.11) inside a single solve because it preserves a smooth right-hand side within each phase. Up to ten alternating full-shift / back-shift cycles are supported before the simulation terminates, preventing unbounded phase loops for unusual parameter regimes.

C.3 Slip-Law Regularization and Blend Parameter Selection

The traction law in Equation (3.6.9) is physically clear but discontinuous at $v_{\text{rel}} = 0$, which can induce step-size chatter near stick–slip transitions. A practical regularized formulation uses three ideas: a smooth saturation law, a clamped no-slip torque request, and a blending rule between those two limits.

First define a smooth saturation torque

$$\tau_{\text{sat}} = \tau_{\max} \tanh\left(\frac{v_{\text{rel}}}{v_{\text{smooth}}}\right), \tag{C.3.1}$$

where $v_{\text{smooth}} > 0$ sets the transition width in slip-speed units. For $|v_{\text{rel}}| \gg v_{\text{smooth}}$, this approaches the usual friction-limited value $\pm\tau_{\max}$, while near zero it removes the sharp sign discontinuity.

Next, clamp the no-slip torque request:

$$\tau_{\text{req}} = \operatorname{clip}(\tau_{\text{ns}},\, -\tau_{\max},\, \tau_{\max})\,. \tag{C.3.2}$$

This is the largest admissible torque consistent with the no-slip request, without allowing the requested value to exceed traction capacity.

Then define the blend factor

$$\alpha = \operatorname{clip}\left(\frac{|v_{\text{rel}}|}{v_{\text{blend}}},\, 0,\, 1\right), \tag{C.3.3}$$

with $v_{\text{blend}} > 0$ defining the width of the blending region, and use the transmitted torque

$$\tau_c = (1-\alpha)\,\tau_{\text{req}} + \alpha\,\tau_{\text{sat}}. \tag{C.3.4}$$

This has a useful interpretation. For very small mismatch, $\alpha \approx 0$, so the model stays close to the clamped no-slip demand. For larger mismatch, $\alpha \to 1$, so the law transitions to the smooth Coulomb limit. In other words, the regularized model behaves like a requested no-slip torque near stick and like a friction-limited sliding law once the mismatch becomes appreciable.

How to choose v_{smooth}. Choose the smoothing scale small enough that $\tanh(v_{\text{rel}}/v_{\text{smooth}})$ is close to a sign law through most of the slip regime, but not so small that the solver again sees an effectively discontinuous transition. A practical workflow is:

1. Start with v_{smooth} around 0.5–2% of a representative relative-speed magnitude seen during actual slip.
2. Decrease it until predicted macro outputs (ratio trajectory, launch time, peak speeds) stop changing materially.
3. Increase it slightly if event localization or adaptive step-size behavior becomes noisy near $v_{\text{rel}} \approx 0$.

The blend scale v_{blend} should generally be chosen greater than or equal to v_{smooth}, so that the handoff from clamped demand to the smooth Coulomb limit is not sharper than the smoothing law itself.

Appendix D

Rate and Acceleration of the Geometric CVT Ratio

The main derivation uses the individual pulley-radius rates and accelerations from Section 3.1.5. Those same results can be combined to show how quickly the available geometric CVT ratio changes as the sheaves move.

Recall the geometric CVT ratio from Equation (3.1.29):

$$R_g = \frac{r_s}{r_p}, \tag{D.0.1}$$

The individual primary and secondary radius rates and accelerations have already been established in Section 3.1.5. This appendix only combines those results to show how the available geometric ratio itself changes as the sheaves move.

D.0.1 Rate of Change of the Geometric CVT Ratio

Both the numerator and denominator in Equation (D.0.1) can change with time. Using the quotient rule,

$$\begin{aligned} \dot{R}_g &= \frac{d}{dt}\left(\frac{r_s}{r_p}\right) \\ &= \frac{\dot{r}_s r_p - r_s \dot{r}_p}{r_p^2} \\ &= \frac{\dot{r}_s}{r_p} - \frac{r_s}{r_p^2}\dot{r}_p. \end{aligned} \tag{D.0.2}$$

The secondary-radius result in Equation (3.1.55) supplies the coupled secondary motion. Within the active-shift branch,

$$\dot{r}_s = -\frac{\phi_p}{\phi_s}\dot{r}_p. \tag{D.0.3}$$

Substituting this into Equation (D.0.2) gives

$$\begin{aligned}\dot{R}_g &= \frac{1}{r_p}\left(-\frac{\phi_p}{\phi_s}\dot{r}_p\right) - \frac{r_s}{r_p^2}\dot{r}_p \\ &= -\frac{\dot{r}_p}{r_p}\left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p}\right) \\ &= -\frac{\dot{r}_p}{r_p}\left(\frac{\phi_p}{\phi_s} + R_g\right).\end{aligned} \tag{D.0.4}$$

Finally, Equation (3.1.37) gives $\dot{r}_p = \dot{s}/(2\tan\beta)$ in the active-shift branch. Therefore,

Rate of Change of Geometric CVT Ratio **(D.0.5)**

$$\dot{R}_g = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ -\dfrac{\dot{s}}{2\tan\beta\, r_p}\left(\dfrac{\phi_p}{\phi_s} + R_g\right), & s_{\mathrm{dz}} < s \le s_{\max}. \end{cases}$$

For an upshift, $\dot{s} > 0$, so the available geometric reduction decreases as the belt moves outward on the primary pulley.

D.0.2 Geometry Families in the Radius Plane

The preceding result gives the local rate at which the geometric ratio changes once the pulley geometry is known. To interpret that result in terms of actual CVT design choices, it is useful to first ask which pulley-radius combinations are available in the first place.

Recall once more that the geometric CVT ratio is

$$R_g = \frac{r_s}{r_p}. \tag{D.0.6}$$

Thus, every point in the (r_p, r_s) plane corresponds to one geometric ratio, and every constant-ratio contour is a line of geometrically similar pulley states. This view is helpful because it separates three different design questions that are easy to blur together:

1. What geometric ratio does a given radius pair provide?
2. Which radius pairs are compatible with a selected belt and installed geometry?
3. How does the chosen compatible path influence the way ratio changes as the CVT shifts?

The first question is answered by the constant-R_g contours. The other two come from the belt-length compatibility relation. Once one of the global geometric quantities—belt length L_b or center-to-center distance C—is fixed, the compatibility relation defines a family of allowable radius pairs in the same plane.

Figure D.1 compares these two views directly. The left panel fixes the center distance and varies the belt length, while the right panel fixes the belt length and varies the center distance. In both panels, the solid contours are constant geometric ratio. The dashed families are the geometrically compatible radius pairs selected by the remaining design choice.

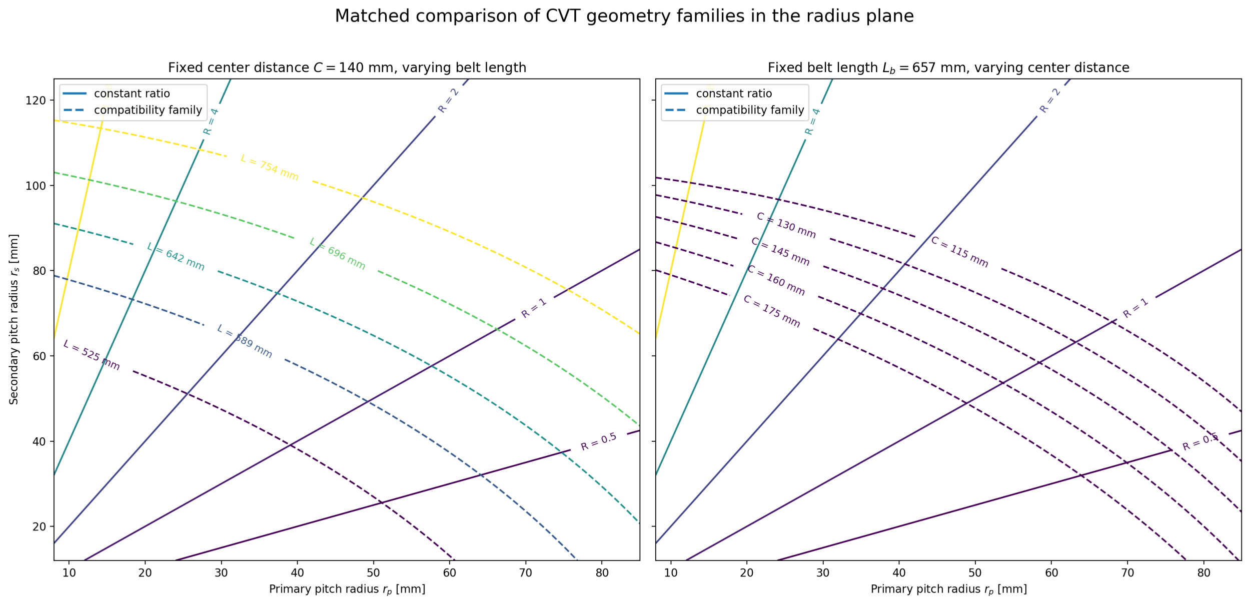


Figure D.1: Matched comparison of geometry families in the pulley-radius plane. In both panels, the solid contours are constant geometric ratio $R_g = r_s/r_p$. In (a), the center-to-center distance is fixed and the dashed families correspond to different belt lengths. In (b), the belt length is fixed and the dashed families correspond to different center-to-center distances.

This figure makes two points especially clear.

First, meeting a target ratio aperture is not only a matter of choosing two endpoint ratios. The selected belt and installed geometry determine which radius pairs are even reachable. A longer or shorter belt shifts the compatible family through the plane, and so does a change in center distance. Since the constant-ratio lines are not parallel to those compatibility families, changing L_b or C changes not only the reachable endpoints, but also the geometric route between them.

Second, the figure makes the roles of belt selection and packaging easier to distinguish. The belt length is largely a component-selection choice, while the center distance is largely a packaging choice. Both appear in the same compatibility relation, but they move the admissible family in different ways. This is why a target ratio range cannot be discussed cleanly without also specifying either the chosen belt or the available center-to-center distance.

In this sense, Figure D.1 is the static geometric view of the design problem. It shows what ratio states are available. The next question is dynamic in a geometric sense: once the CVT moves along one of those compatible paths, how much ratio change does a small increment of axial travel produce?

D.0.3 Exploring the Local Ratio-Change Map

The expression for $\dot{R}_g$ gives the instantaneous rate of change of the geometric ratio. To compare different parts of the geometry without committing to a particular time history, it is useful to divide out the shift speed and look instead at the local ratio change produced by a small increment of axial travel:

$$\frac{dR_g}{ds} = -\frac{1}{2\tan\beta\, r_p}\left(\frac{\phi_p}{\phi_s} + R_g\right). \tag{D.0.7}$$

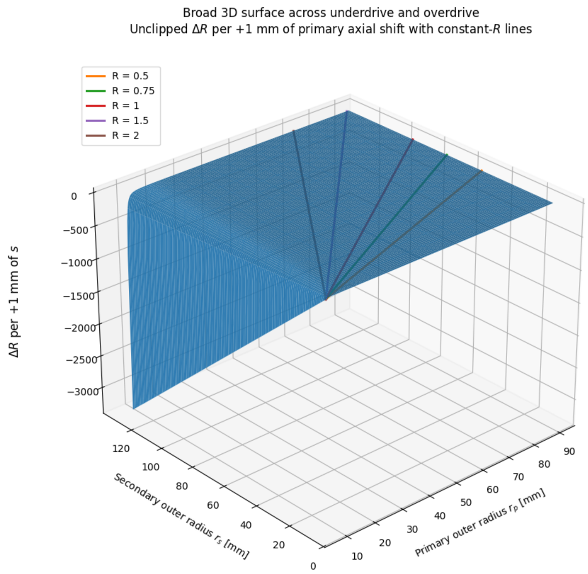
For a small positive increment Δs,

$$\Delta R_g \approx \frac{dR_g}{ds}\Delta s. \tag{D.0.8}$$

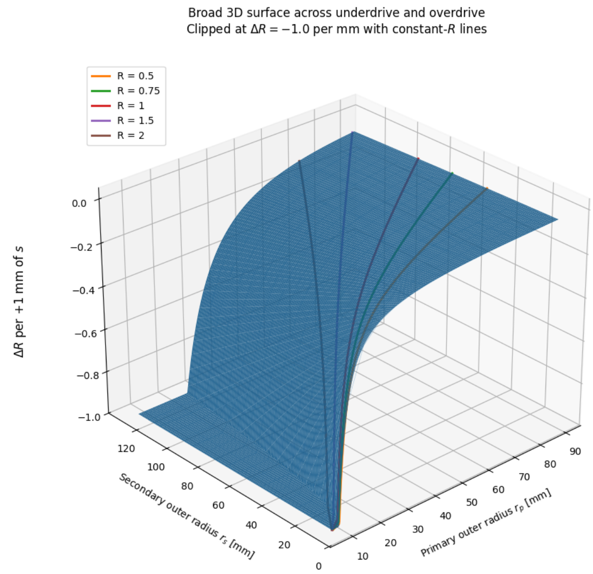
The plots below take $\Delta s = 1\,\mathrm{mm}$, so the displayed quantity may be read as the local change in geometric ratio caused by an additional 1 mm of primary closure. The negative sign simply reflects the usual upshift direction: increasing s reduces the geometric reduction ratio.

A sensitivity field over the radius plane The first view is a broad surface over the radius plane. Here, r_p and r_s are treated as independent coordinates so that every part of the geometry can be inspected. A real CVT does not wander arbitrarily across this surface; its fixed belt length and center distance constrain it to one belt-length-compatible path. Even so, the full surface is valuable because it shows where the geometry itself makes ratio change especially strong or weak.

Figure D.2 shows two versions of this surface. The unclipped view exposes the dominant trend: the local ratio change becomes much larger in magnitude as the primary radius becomes small. This follows directly from the explicit $1/r_p$ factor in Equation (D.0.7), and is reinforced by the fact that the geometric ratio $R_g = r_s/r_p$ is also largest in that same region. The clipped view hides the most extreme values so that the otherwise compressed mid-range curvature can be seen more clearly. The orange curve marks the $R_g = 1$ locus. It is a convenient reference, but not a special boundary in the local ratio-change law itself.



(a) Broad radius domain. The full scale exposes the rapidly increasing magnitude of local ratio change near small r_p.



(b) The same domain with values below -1 per 1 mm clipped, exposing the curvature through the mid-range.

Figure D.2: Local geometric-ratio change for a 1 mm increase in primary axial closure across a broad pulley-radius domain. The orange curve marks the $R_g = 1$ locus.

How belt choice and installed geometry select a path The broad surface is useful, but a particular CVT design only experiences the part of that field lying along its own compatible geometry path. This is where the earlier radius-plane view becomes directly relevant. Once a belt length and center distance are fixed, the CVT is constrained to one compatibility curve in the (r_p, r_s) plane. As the primary sheave closes, the operating point moves along that curve and samples the local ratio-change field point by point.

Figure D.3 overlays three representative belt-length-compatible paths on the local ratio-change surface. Each path corresponds to a different belt-length/reference-geometry choice. The figure shows that two designs with similar overall ratio aperture can still distribute their ratio change very differently over the available shift travel. One path may enter the steep small-r_p region early, concentrating a large amount of ratio change into a short portion of the stroke. Another may spend more of the stroke in the flatter region, distributing the same overall change more gradually.

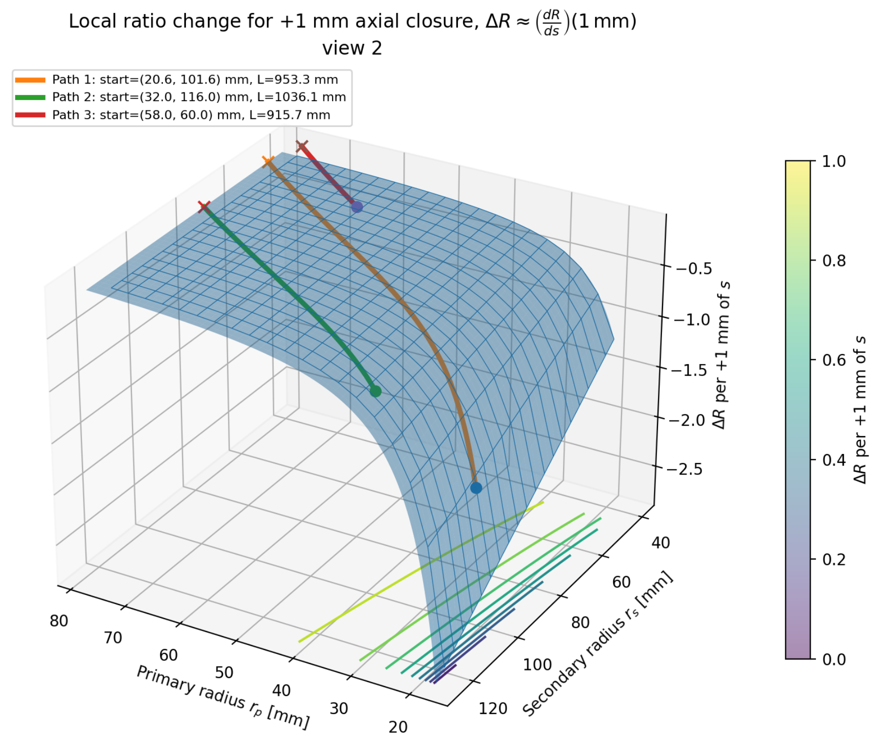
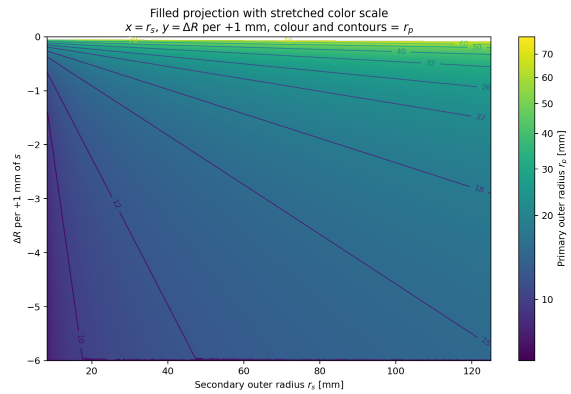


Figure D.3: Representative fixed-belt-length paths through the local ratio-change surface. Each coloured curve is a geometrically compatible pulley-radius path for a different belt-length/reference-geometry choice. The surface value gives the local change in geometric ratio associated with an additional 1 mm of primary closure at that point.

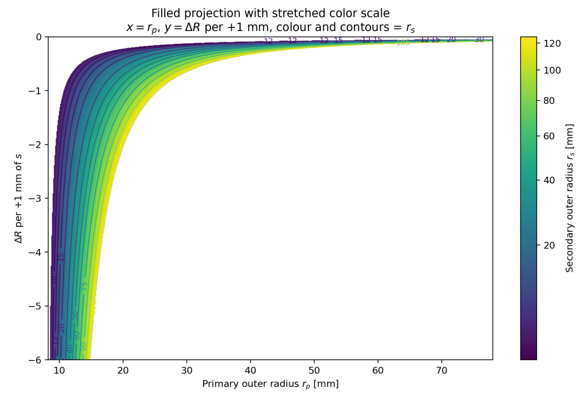
This is the key design value of the map: it does not merely say what overall ratio range is available. It shows *where along the shift stroke* that ratio change is spent.

Separating the roles of the two pulley radii The two projections in Figure D.4 help isolate the individual effects of r_p and r_s. In Figure D.4(a), the horizontal axis is the secondary radius and the contours identify the primary radius. For a fixed primary radius, increasing r_s increases the geometric ratio and makes the local ratio change more negative. The effect is visible, but the contour spacing shows that it remains secondary to the influence of the primary radius.

Figure D.4(b) makes the primary-radius effect more explicit. Every secondary-radius contour bends sharply downward as r_p approaches its lower limit, then flattens at larger radius. This is the visual form of the $1/r_p$ dependence in Equation (D.0.7). Thus, the small-primary-radius region is not only where the geometric reduction is large; it is also where the ratio becomes most sensitive to additional axial motion.



(a) Local ratio change against secondary radius. Colour and contours identify the primary radius.



(b) Local ratio change against primary radius. Colour and contours identify the secondary radius.

Figure D.4: Two projections of the local ratio-change surface for a 1 mm primary closure. The projections separate the contributions of the two radii and show the dominant sensitivity to small primary radius.

Design interpretation Taken together, Figure D.1 and Figure D.2–Figure D.4 give a more complete geometric design picture than endpoint ratios alone.

The radius-plane view answers the question of feasibility. It makes clear how a target ratio aperture depends on the selected belt length and the available center-to-center distance. A longer or shorter belt can move the admissible family enough to change both the reachable ratio range and the portion of the plane through which the CVT shifts. Likewise, a packaging change in center distance can move the same belt into a very different family of compatible radius pairs.

The ratio-change view answers the question of distribution. Once a design selects its belt and installed geometry, the compatible path through the plane determines where ratio change is concentrated. A path that reaches very small r_p can produce large ratio change in little axial travel, which may be attractive when launch or climbing performance demands strong reduction with limited shift stroke. A path that stays farther from that corner distributes ratio change more gently and may be easier to control or to package within a desired sheave travel.

This does *not* mean that the smallest possible r_p is always best. The present plots are geometric only. They show the leverage of the geometry, but not its cost. Experimental work by Chen et al. (1998) showed that, for their rubber-belt CVT, the highest transmission efficiency occurred near a 1:1 speed ratio, reminding us that the most attractive geometric reduction state is not automatically the most efficient operating state. More directly, Childs and Cowburn (1987) identified additional loss penalties at small pulley radii in V-belt drives, with belt-carcass warping and shearing proposed as important contributors. In other words, the same small-r_p region that offers strong geometric leverage is also the region where the belt is bent and seated most severely.

For that reason, these plots are best used as an early design lens rather than a final design verdict. They help screen combinations of belt length, center distance, and pulley radii by showing not only how much ratio range is available, but how that range is distributed over the stroke. Later traction, loss, and dynamic analyses can then be used to judge whether the geometrically attractive part of the design space is physically supportable (Gerbert, 1996; Bertini et al., 2014; Messick, 2018).

D.0.4 Acceleration of the Geometric CVT Ratio

The ratio rate can change for two reasons. The primary sheave may accelerate, and the pulley geometry continues to change as the belt moves. Differentiate Equation (D.0.2) term by term to capture both effects.

For the first term, $\dot{r}_s/r_p$, use the quotient rule:

$$\frac{d}{dt}\left(\frac{\dot{r}_s}{r_p}\right) = \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s\dot{r}_p}{r_p^2}. \tag{D.0.9}$$

For the second term, $r_s\dot{r}_p/r_p^2$, all three factors may change. Using the product rule,

$$\frac{d}{dt}\left(\frac{r_s\dot{r}_p}{r_p^2}\right) = \frac{\dot{r}_s\dot{r}_p}{r_p^2} + \frac{r_s\ddot{r}_p}{r_p^2} - \frac{2r_s\dot{r}_p^2}{r_p^3}. \tag{D.0.10}$$

Subtracting Equation (D.0.10) from Equation (D.0.9) gives

$$\begin{aligned}\ddot{R}_g &= \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s\dot{r}_p}{r_p^2} - \left[\frac{\dot{r}_s\dot{r}_p}{r_p^2} + \frac{r_s\ddot{r}_p}{r_p^2} - \frac{2r_s\dot{r}_p^2}{r_p^3}\right] \\ &= \frac{\ddot{r}_s}{r_p} - \frac{2\dot{r}_s\dot{r}_p}{r_p^2} - \frac{r_s\ddot{r}_p}{r_p^2} + \frac{2r_s\dot{r}_p^2}{r_p^3}. \end{aligned} \tag{D.0.11}$$

The main derivation gives the required coupled secondary motion in Equations (3.1.54) and (3.1.63):

$$\dot{r}_s = -\frac{\phi_p}{\phi_s}\dot{r}_p, \tag{D.0.12}$$

$$\ddot{r}_s = -\frac{\phi_p}{\phi_s}\ddot{r}_p - \frac{8\pi^2}{\phi_s^3\sqrt{C^2-(r_s-r_p)^2}}\dot{r}_p^2. \tag{D.0.13}$$

Substituting these into Equation (D.0.11) gives

$$\begin{aligned}\ddot{R}_g &= \frac{1}{r_p}\left[-\frac{\phi_p}{\phi_s}\ddot{r}_p - \frac{8\pi^2}{\phi_s^3\sqrt{C^2-(r_s-r_p)^2}}\dot{r}_p^2\right] - \frac{2}{r_p^2}\left[-\frac{\phi_p}{\phi_s}\dot{r}_p\right]\dot{r}_p \\ &\quad - \frac{r_s}{r_p^2}\ddot{r}_p + \frac{2r_s}{r_p^3}\dot{r}_p^2 \\ &= -\frac{1}{r_p}\left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p}\right)\ddot{r}_p + \left[\frac{2}{r_p^2}\left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p}\right) - \frac{8\pi^2}{\phi_s^3 r_p\sqrt{C^2-(r_s-r_p)^2}}\right]\dot{r}_p^2 \\ &= -\frac{1}{r_p}\left(\frac{\phi_p}{\phi_s} + R_g\right)\ddot{r}_p + \left[\frac{2}{r_p^2}\left(\frac{\phi_p}{\phi_s} + R_g\right) - \frac{8\pi^2}{\phi_s^3 r_p\sqrt{C^2-(r_s-r_p)^2}}\right]\dot{r}_p^2. \end{aligned} \tag{D.0.14}$$

Finally, use the primary radius rate and acceleration from Equations (3.1.37) and (3.1.39):

Acceleration of Geometric CVT Ratio **(D.0.15)**

$$\ddot{R}_g = \begin{cases} 0, & 0 \leq s < s_{\mathrm{dz}}, \\ -\dfrac{\ddot{s}}{2\tan\beta\, r_p}\left(\dfrac{\phi_p}{\phi_s} + R_g\right) + \dfrac{\dot{s}^2}{2\tan^2\beta\, r_p^2}\left(\dfrac{\phi_p}{\phi_s} + R_g\right) - \dfrac{2\pi^2\dot{s}^2}{\phi_s^3 r_p \tan^2\beta\sqrt{C^2 - (r_s - r_p)^2}}, & s_{\mathrm{dz}} < s \leq s_{\mathrm{max}}. \end{cases}$$

The ratio acceleration has two sources. The term proportional to $\ddot{s}$ is the direct conversion of primary-sheave acceleration into ratio acceleration. Its magnitude is scaled by the local ratio sensitivity dR_g/ds, so the same axial acceleration produces a larger ratio response where the geometric ratio changes rapidly with shift.

The terms proportional to $\dot{s}^2$ arise even when the primary sheave moves at constant axial speed. They reflect the curvature of the geometric relation $R_g(s)$: as the belt moves through the sheaves, the ratio change produced by each additional increment of axial travel need not remain constant. In the usual upshift direction, the large ratio sensitivity associated with small primary radius progressively weakens as the primary radius grows. The ratio therefore often changes rapidly at the beginning of the shift and then eases off, even without a change in axial shift speed.